(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-06-840-007-G00

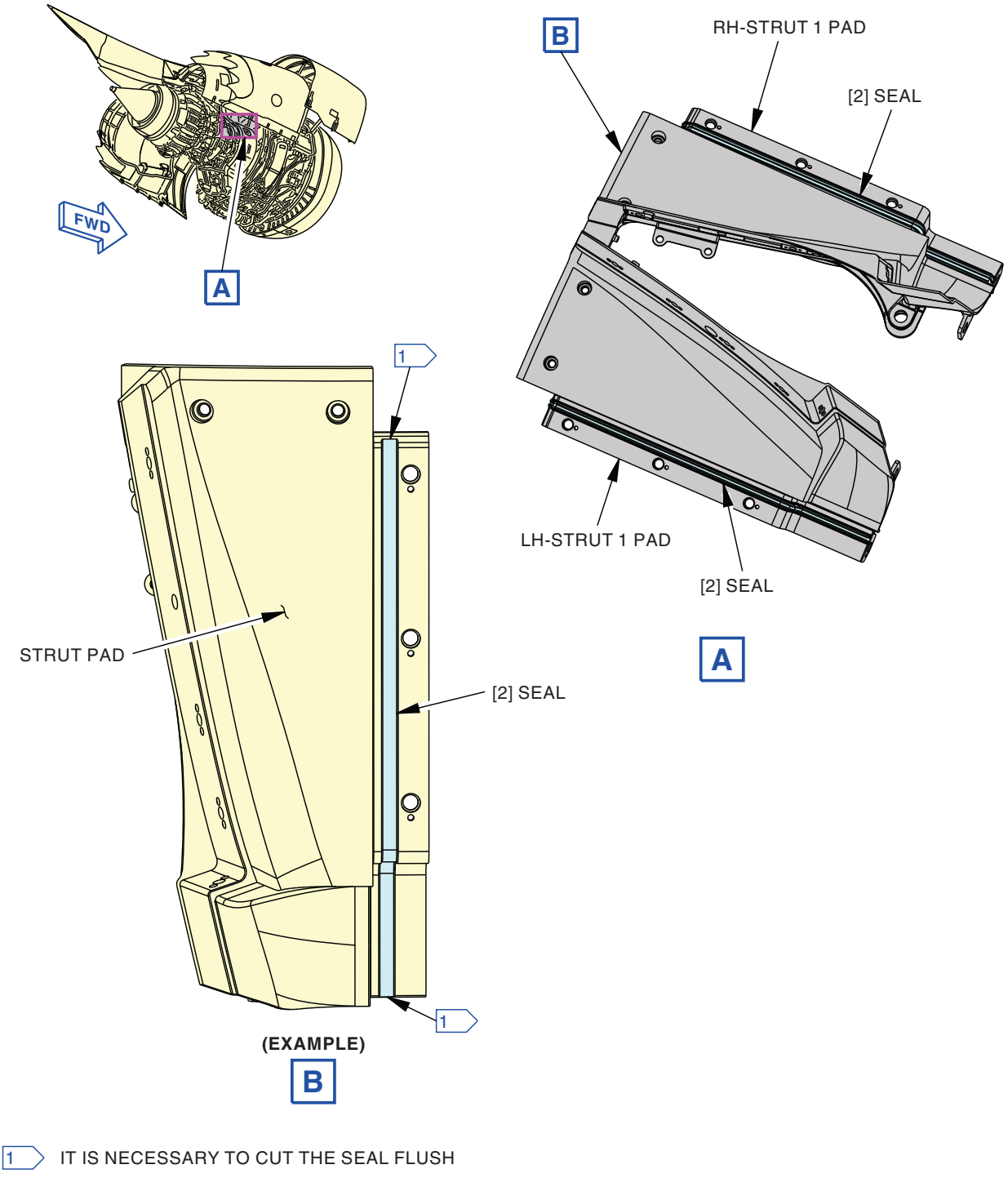
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

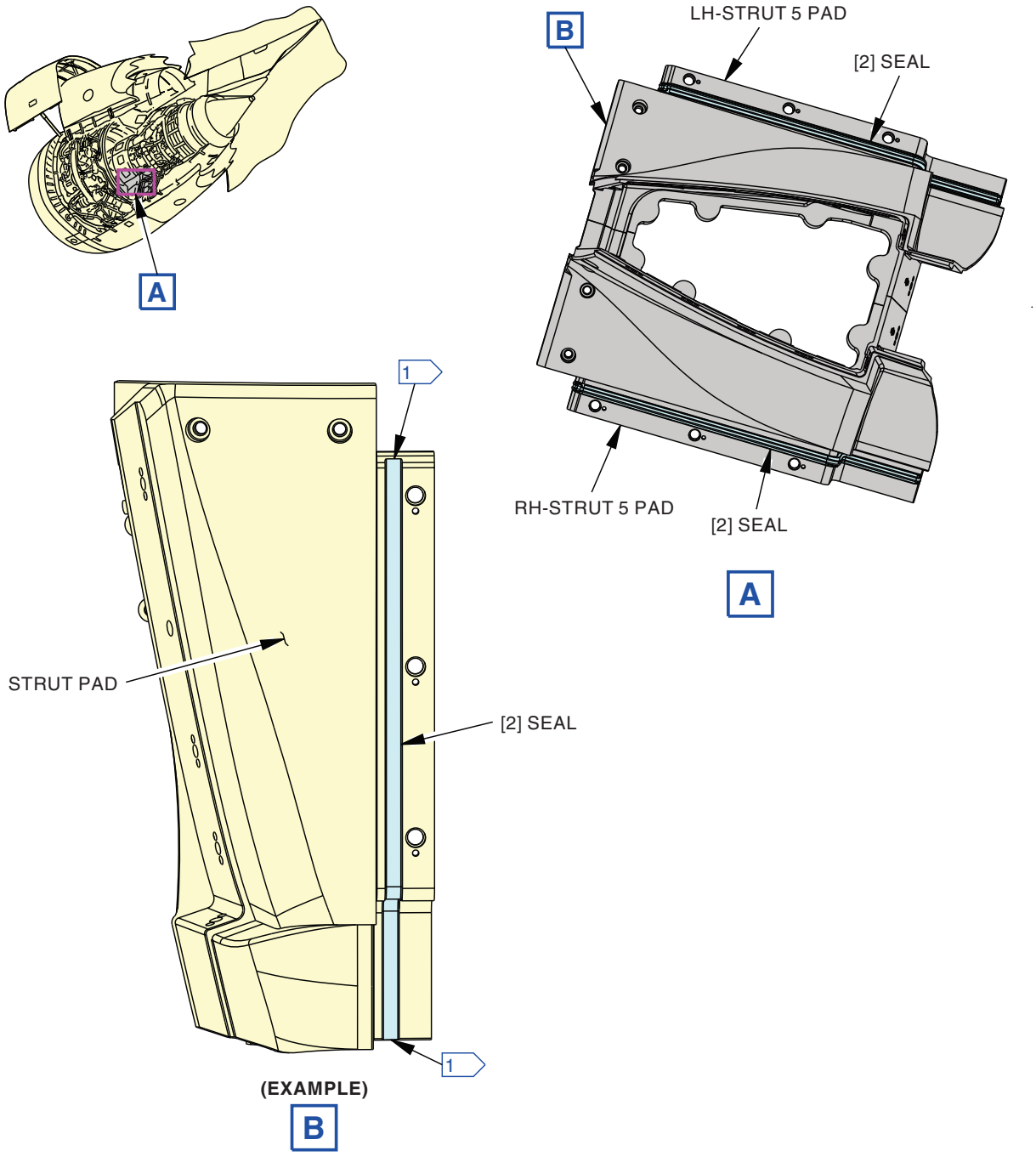


1 IT IS NECESSARY TO CUT THE SEAL FLUSH

2980033 S0000758027_V1

Fan Frame Strut Pad (RH- and LH-Strut 1 Pad) Seal Replacement
Figure 804/72-23-06-990-828-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

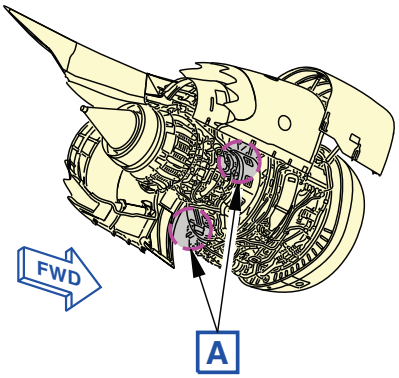


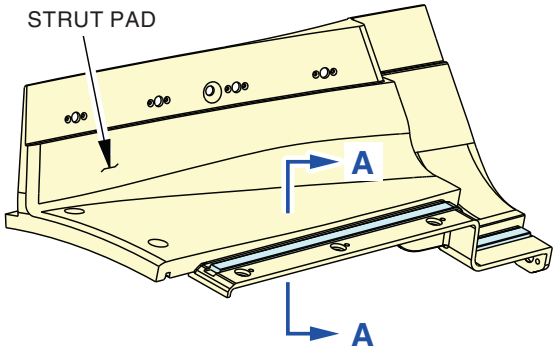
1 IT IS NECESSARY TO CUT THE SEAL FLUSH

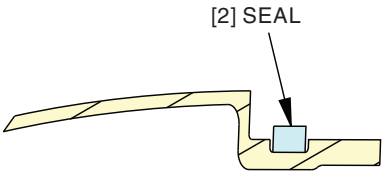
2980045 S0000758028_V1

Fan Frame Strut Pad (RH- and LH-Strut 5 Pad) Seal Replacement
Figure 805/72-23-06-990-829-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



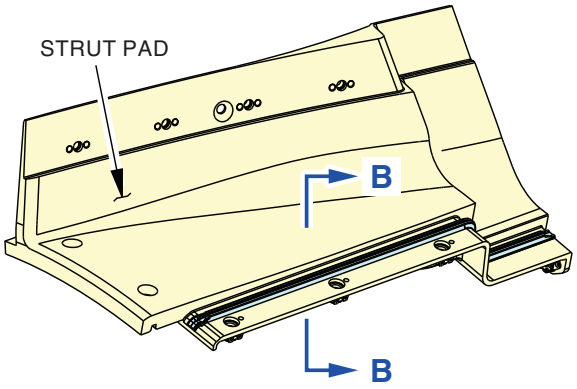


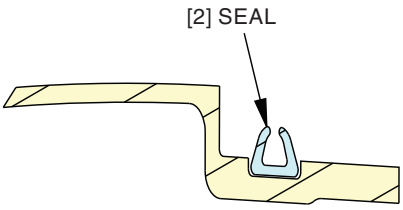


A-A

RECTANGLE SHAPE SEAL
(EXAMPLE)

A





B-B

U SHAPE SEAL
(EXAMPLE)

A

2980058 S0000758033_V1

Fan Frame Strut Pad Seal Types
Figure 806/72-23-06-990-830-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-06-300-801-G00

4. Fan Frame Strut 1 and 5 Fairings Blending

(Figure 807, Figure 808, Figure 809, Figure 810).

A. General

(1) This task gives instructions to blend the right and left strut 1 fairings and right and left strut 5 fairings.

(2) The subsequent table gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 805/72-23-06-993-806-G00

Part number	Service Bulletin	Designation
364-031-801-0	-	RH STRUT 1 FAIRING
364-031-901-0	-	LH STRUT 1 FAIRING
364-033-101-0	-	RH STRUT 5 FAIRING
364-033-201-0	-	LH STRUT 5 FAIRING

B. References

Reference	Title
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-06-200-801-G00	Fan Frame Strut Panels Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G51485 [S1093]	Etchant - Titanium Alloy	

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Blending

SUBTASK 72-23-06-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-06-010-004-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

G. Fan Frame Strut 1 and 5 Fairings Blending

SUBTASK 72-23-06-211-015-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(1) Do a visual inspection of the part.

(a) Maximum of three blending are permitted.

(b) Make sure that the indication to be blended is on available surface and is not more than DIM. AB (Figure 807, Figure 808, Figure 809).

1) Make sure that the depth DIM. AB is less or equal to 0.023 in. (0.6 mm).

2) To locate the available areas, use the following dimensions:

- DIA A is equal to 0.71 in. (18 mm)
- DIM. B is equal to 1.78 in. (45 mm)
- DIM. C is equal to 2.52 in. (64 mm)
- DIM. D is equal to 0.24 in. (6 mm)
- DIA E is equal to 0.67 in. (17 mm)
- DIA F is equal to 0.63 in. (16 mm)
- DIM. G is equal to 2.64 in. (67 mm)
- DIM. H is equal to 1.38 in. (35 mm)
- DIM. J is equal to 1.97 in. (50 mm)
- DIM. K is equal to 1.54 in. (39 mm)
- DIM. L is equal to 3.00 in. (76 mm)
- DIM. M is equal to 0.12 in. (3 mm)
- DIM. N is equal to 2.13 in. (54 mm)
- DIM. P is equal to 1.74 in. (44 mm)
- DIM. Q is equal to 3.39 in. (86 mm)
- DIA R is equal to 0.67 in. (17 mm)
- DIM. S is equal to 4.30 in. (109 mm)
- DIM. T is equal to 2.05 in. (52 mm)
- DIM. U is equal to 2.25 in. (57 mm)
- DIM. V is equal to 2.96 in. (75 mm)
- DIM. W is equal to 0.32 in. (8 mm)
- DIM. X is equal to 2.88 in. (73 mm)
- DIM. Y is equal to 3.12 in. (79 mm)
- DIM. Z is equal to 1.07 in. (27 mm)
- DIM. AA is equal to 2.17 in. (55 mm).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-23-06-320-001-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

WARNING DO NOT LET TITANIUM DUST COLLECT. TITANIUM DUST IS VERY FLAMMABLE. DISCARD ALL TITANIUM WASTE IN AN APPROVED CONTAINER. TITANIUM DUST CAN CAUSE INJURY TO PERSONNEL.

CAUTION REMOVE MINIMUM MATERIAL MORE THAN THE DEPTH OF DEVIATION WHEN YOU BLEND. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

CAUTION DO NOT USE TOOLS ON TITANIUM THAT WERE USED ON OTHER MATERIALS. IT WILL CAUSE CONTAMINATION OF THE TITANIUM. KEEP THE BLENDING ABRASIVE TOOL USED FOR THE TITANIUM ISOLATED FROM TOOLS USED FOR OTHER METALS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do the blend of the strut fairings.

NOTE: Blended areas must make a smooth contour into the adjacent areas to keep correct airflow.

(a) For the Method I, do this step:

1) Remove the indication by hand blending (TASK 70-43-42-300-802-G00).

(b) For the Method II, do this step:

CAUTION MAKE SURE THAT POWER BLENDING DOES NOT SUPPLY TOO MUCH HEAT AND THERMAL STRESS. IF YOU DO NOT OBEY, TOO MUCH HEAT CAN CAUSE DAMAGE TO THE MATERIAL.

1) Remove the indication by power blending (TASK 70-43-42-300-803-G00).

(c) If the minimum acceptable distance between two repaired areas is less than 40 times the DIM. AB, mix them to get only one repaired area.

SUBTASK 72-23-06-110-001-G00

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(3) Clean the part using alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-23-06-350-001-G00

(4) Do a local swab etching of the repaired areas with titanium alloy etchant, G51485 [S1093] (TASK 70-43-24-100-802-G00).

SUBTASK 72-23-06-230-001-G00

(5) Do a fluorescent penetrant inspection of the repaired areas (TASK 70-11-06-230-803-G00).

(a) Use sensitivity level 2.

(b) Standard unit surface: 0.98 in. (25 mm) x 0.98 in. (25 mm) inspection field.

(c) Indications less than 0.0118 in. (0.30 mm) are not to be taken into account.

(d) Indications greater than 0.059 in. (1.5 mm) are not permitted.

(e) Dimension of the satisfactory indications to count: from 0.040 in. (1.0 mm) to 0.059 in. (1.5 mm). Maximum number of satisfactory indications to count: 10.

(f) Make sure that the minimum spacing between two indications are greater than or equal to three times the size of the largest indication.

NOTE: Otherwise, they are considered as forming only one indication.

1) If there are more than two aligned indications, to determine the spacing, take the largest real dimension of the indications bounding this interval.

(g) Make sure that the minimum spacing between 2 H level clusters are greater than or equal to 0.79 in. (20 mm).

SUBTASK 72-23-06-211-016-G00

(6) Do a visual inspection of the parts (TASK 72-23-06-200-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-06-410-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-23-06-860-006-G00

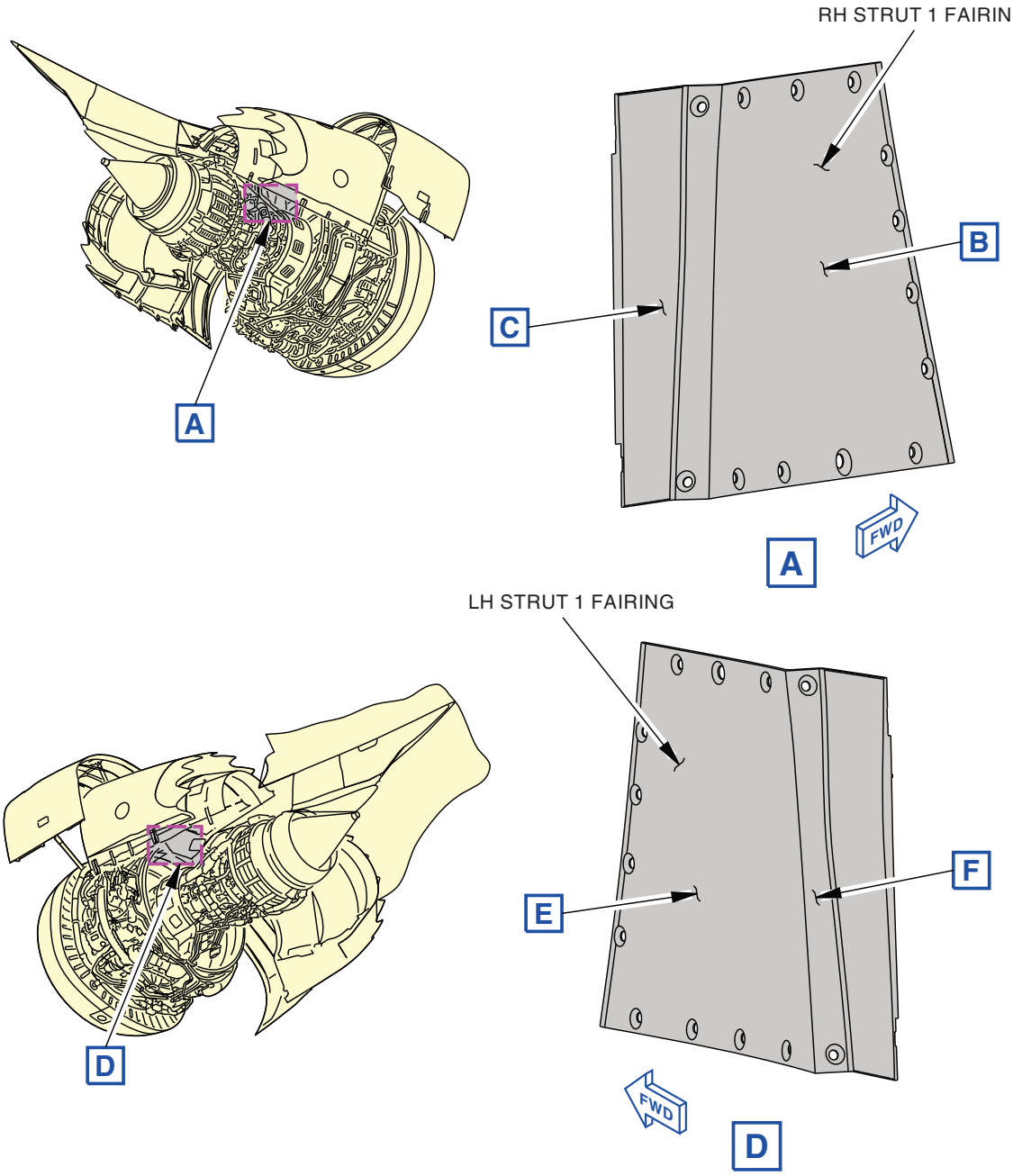
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



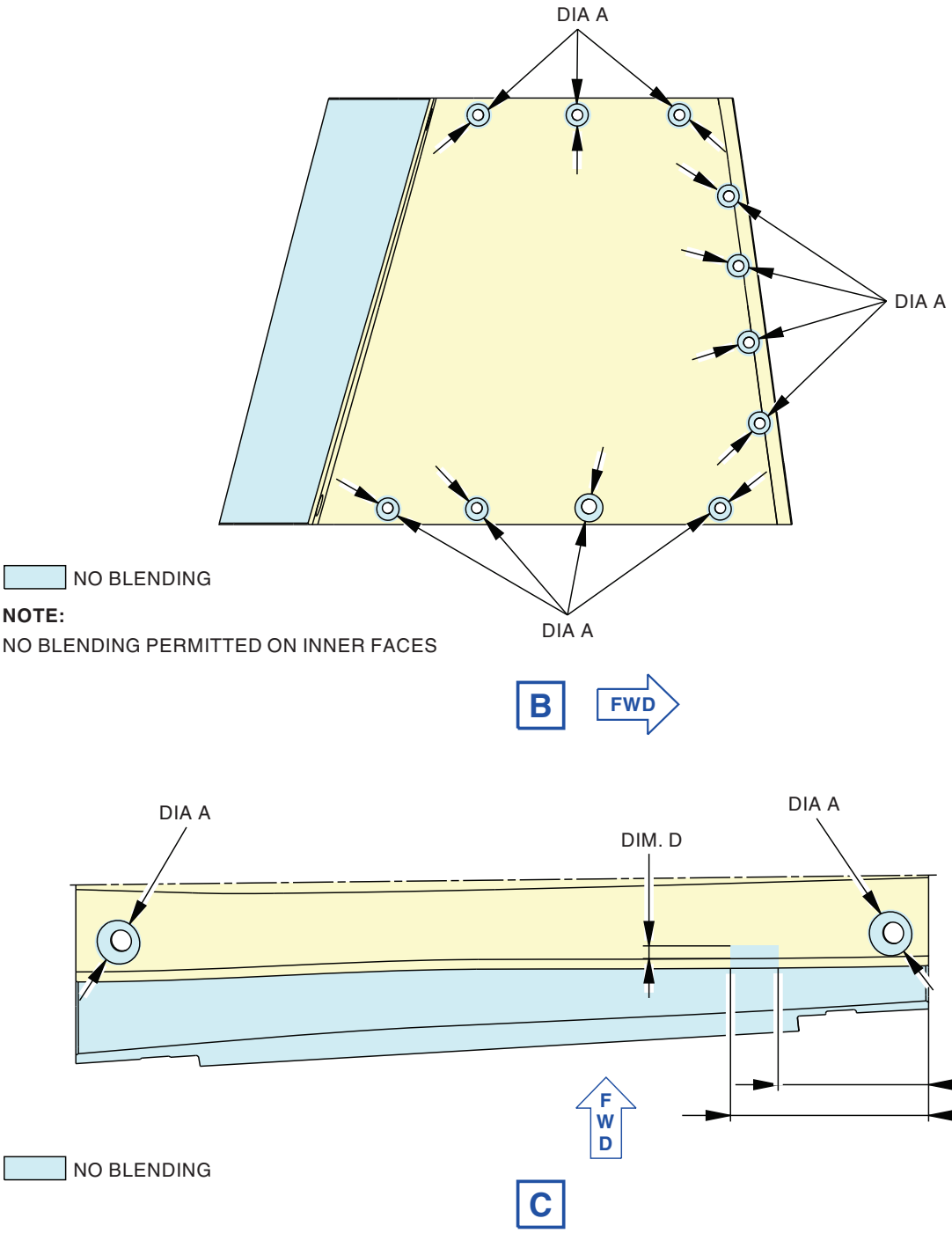
NOTE:

MATERIAL FOR ALL PARTS : 6.4 (TA6V)

2991593 S0000768337_V1

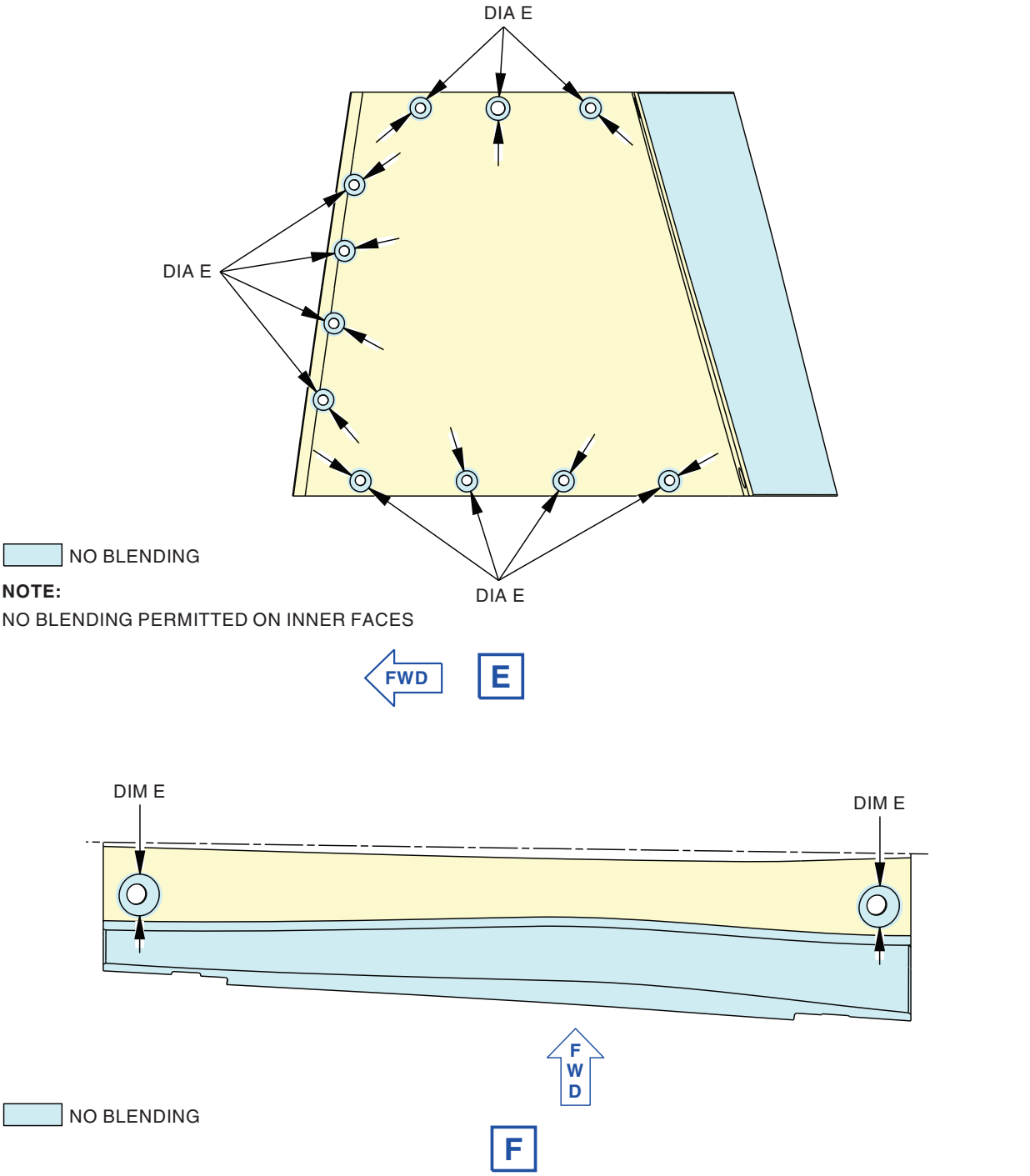
Fan Frame Strut 1 Fairings Blending
Figure 807/72-23-06-990-838-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



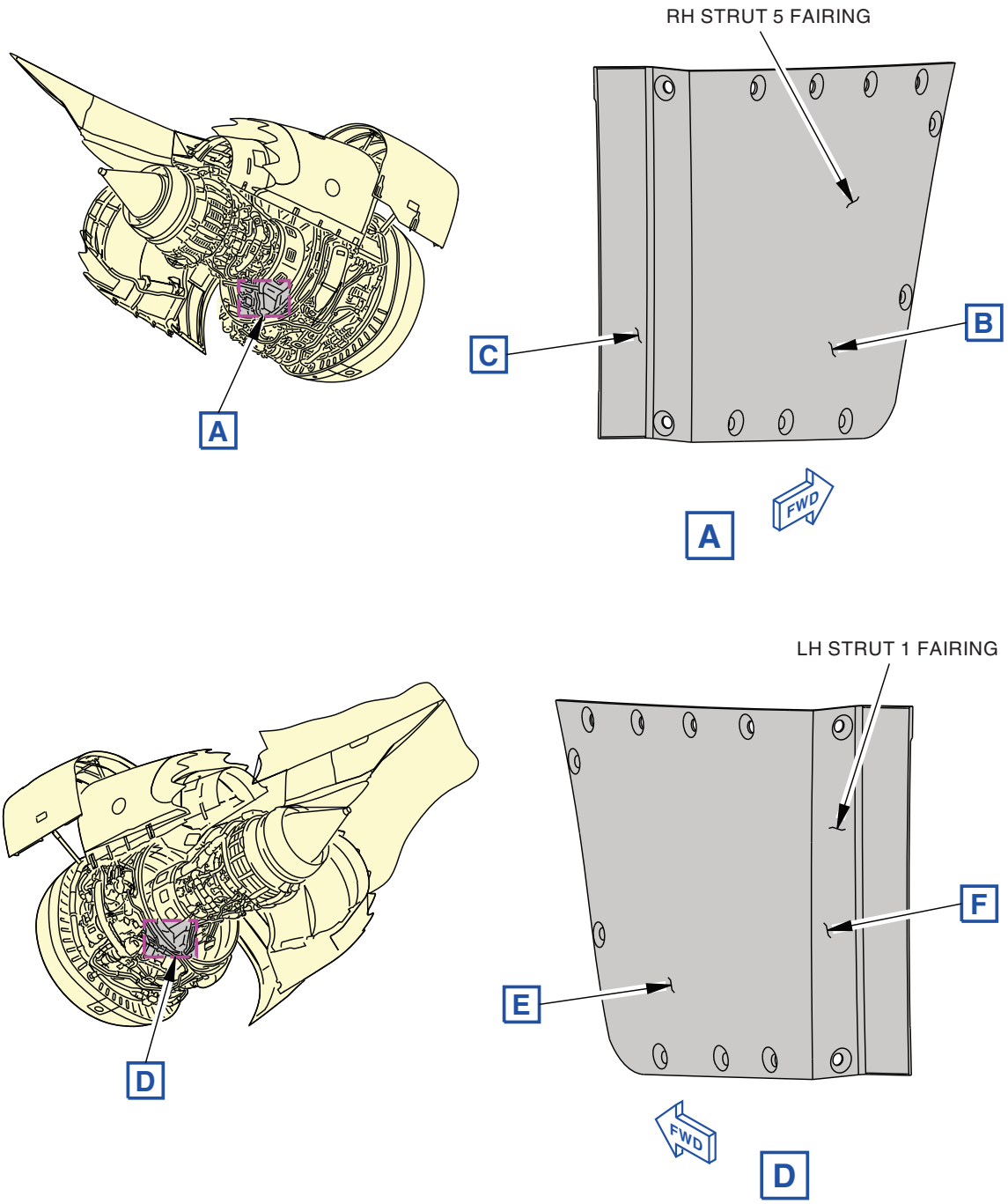
Fan Frame Strut 1 Fairings Blending
Figure 807/72-23-06-990-838-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut 1 Fairings Blending
Figure 807/72-23-06-990-838-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



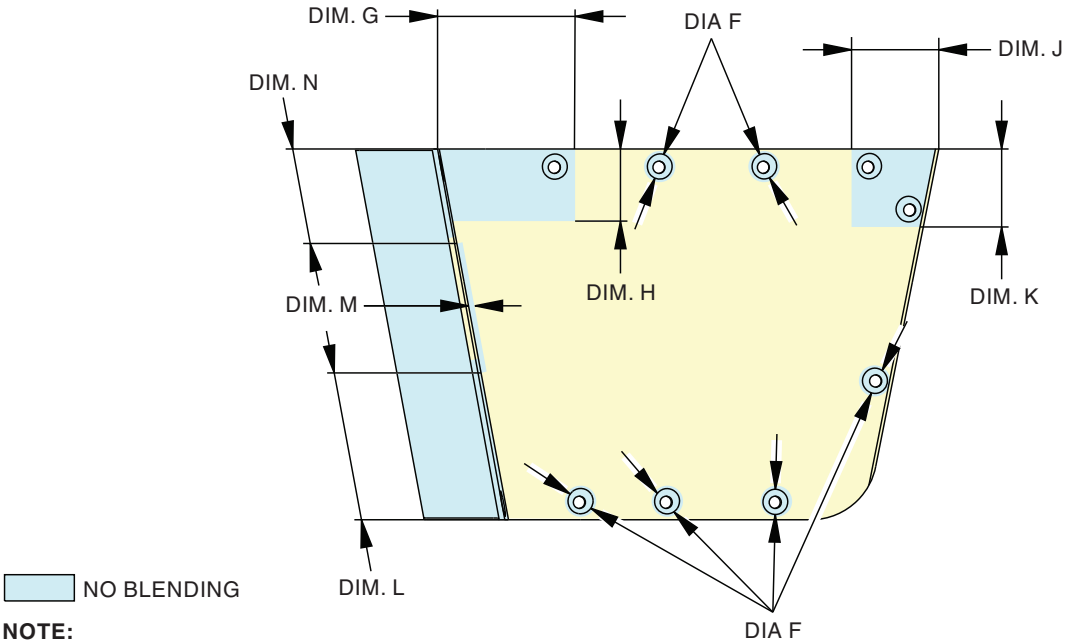
NOTE:

MATERIAL FOR ALL PARTS : 6.4 (TA6V)

2991615 S0000768339_V1

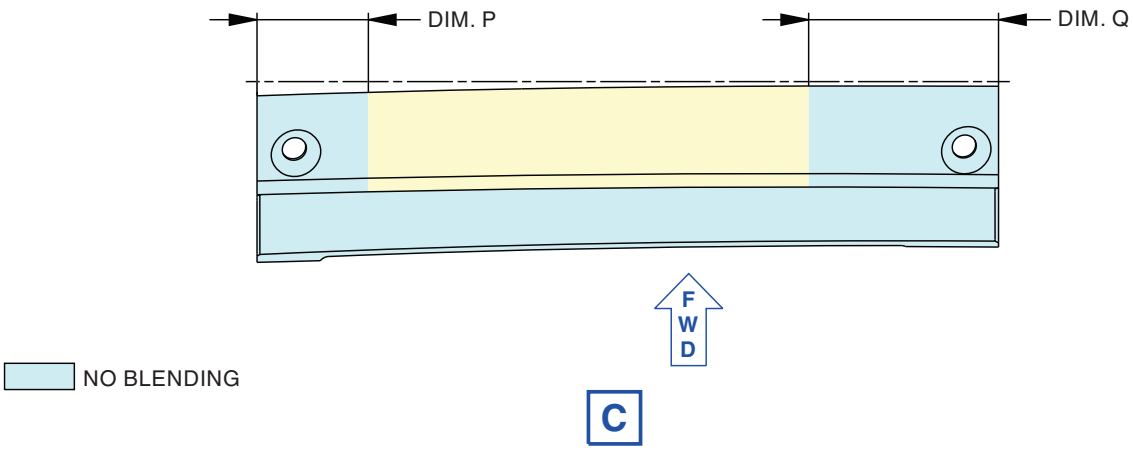
Fan Frame Strut 5 Fairings Blending
Figure 808/72-23-06-990-841-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:
NO BLENDING PERMITTED ON INNER FACES

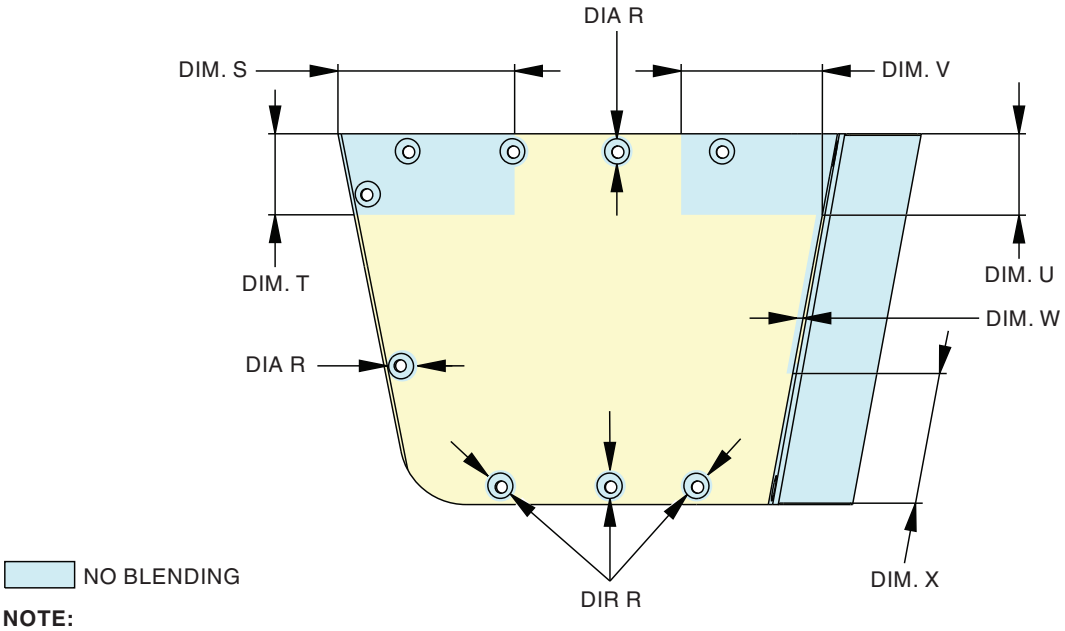




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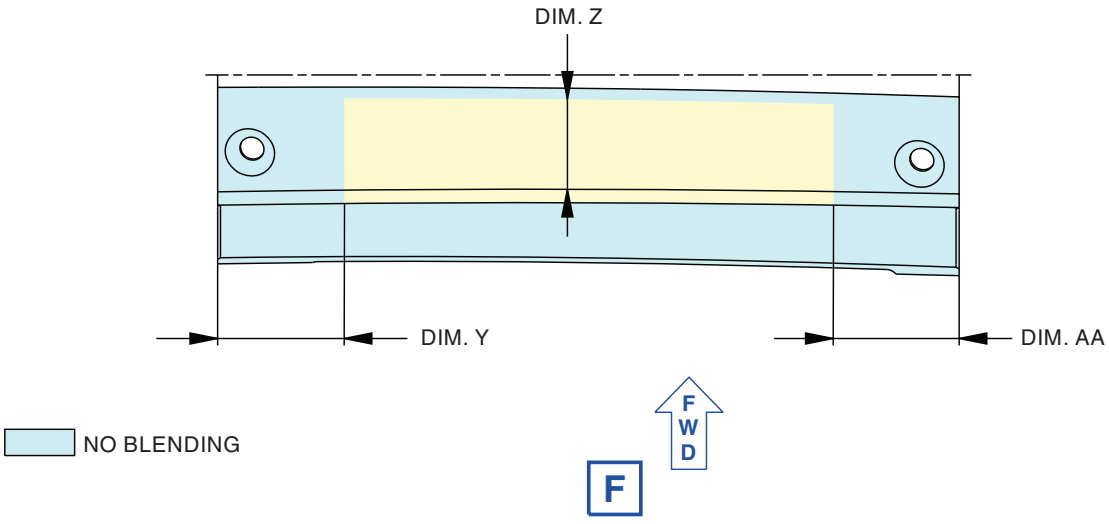
Fan Frame Strut 5 Fairings Blending
Figure 808/72-23-06-990-841-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:
NO BLENDING PERMITTED ON INNER FACES

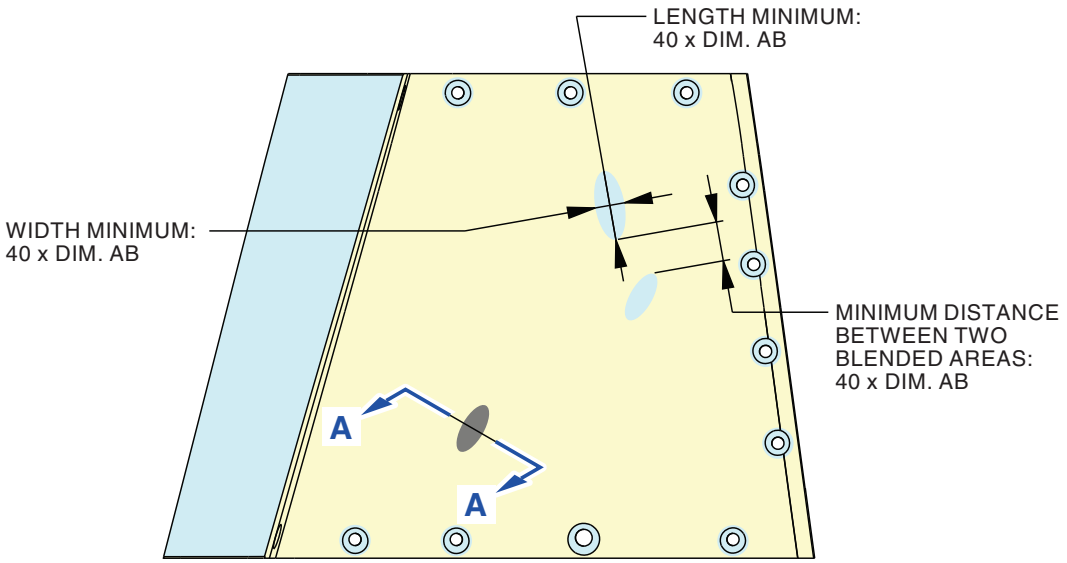




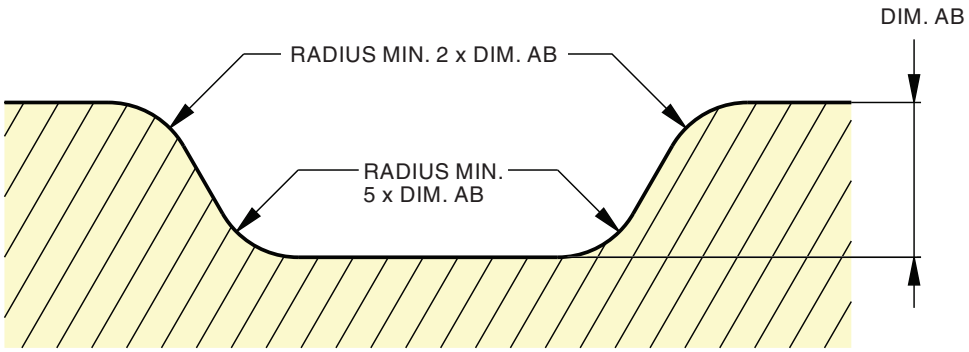
2991790 S0000768449_V1

Fan Frame Strut 5 Fairings Blending
Figure 808/72-23-06-990-841-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RH STRUT 1 FAIRING
(EXAMPLE)



ACCEPTABLE BLENDING PROFILE

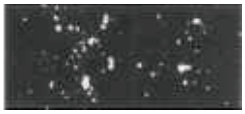
A-A

NO BLENDING

2991793 S0000768455_V1

Dimensional Information
Figure 809/72-23-06-990-844-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LEVELS	PLATES
G	NO DEFECT
M	
N	
H	
I	
J	
K	
L	

2991794 S0000768459_V1

Clustered Indications Levels
Figure 810/72-23-06-990-845-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-06-300-802-G00

5. Fan Frame Strut Pad Blending

(Figure 811, Figure 812, and Figure 813)

A. General

(1) This task gives instructions to do the blend repair on the right Strut 1 Pad, on the left Strut 1 Pad, on the right Strut 5 Pad, and on the left Strut 5 Pad (Figure 811 and Figure 812).

(2) The table below gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 806/72-23-06-993-805-G00

Part number	Service Bulletin	Designation
364-080-510-0	-	RH Strut 1 Pad
364-082-810-0	-	LH Strut 1 Pad
364-082-910-0	-	RH Strut 5 Pad
364-083-010-0	-	LH Strut 5 Pad

B. References

Reference	Title
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-06-200-801-G00	Fan Frame Strut Panels Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G51485 [S1093]	Etchant - Titanium Alloy	

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Blending

SUBTASK 72-23-06-840-008-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-06-010-005-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

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(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

G. Fan Frame Strut Pad Blending

SUBTASK 72-23-06-211-017-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(1) Do a visual inspection of the part.

(a) Maximum three blending are permitted on each strut pad.

(b) Make sure that the indication to be blended is on available surface and is not more than DIM. AF (View D and View E, Figure 811 and View D and View E, Figure 812).

1) Make sure that the DIM. AF is less or equal to 0.023 in. (0.6 mm).

2) Locate the available areas, use the following dimensions:

- DIM. A is equal to 0.99 in. (25 mm)
- DIM. B is equal to 0.60 in. (15 mm)
- DIM. C is equal to 1.86 in. (47 mm)
- DIM. D is equal to 1.70 in. (43 mm)
- DIM. E is equal to 2.33 in. (59 mm)
- DIM. F is equal to 1.89 in. (48 mm)
- DIM. G is equal to 1.38 in. (35 mm)
- DIM. H is equal to 1.46 in. (37 mm)
- DIM. J is equal to 0.48 in. (12 mm)
- DIM. K is equal to 2.01 in. (51 mm)
- DIM. L is equal to 1.50 in. (38 mm)
- DIM. M is equal to 1.42 in. (36 mm)
- DIM. N is equal to 2.21 in. (56 mm)
- DIM. P is equal to 1.66 in. (42 mm)
- DIM. Q is equal to 2.56 in. (65 mm)
- DIM. R is equal to 4.22 in. (107 mm)
- DIM. S is equal to 1.11 in. (28 mm)
- DIM. T is equal to 2.45 in. (62 mm)
- DIM. U is equal to 5.20 in. (132 mm)
- DIM. V is equal to 1.97 in. (50 mm)
- DIM. W is equal to 2.52 in. (64 mm)
- DIM. X is equal to 4.26 in. (108 mm)
- DIM. Y is equal to 2.60 in. (66 mm)
- DIM. Z is equal to 3.51 in. (89 mm)
- DIM. AA is equal to 4.81 in. (122 mm)

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- DIM. AB is equal to 4.06 in. (103 mm)
- DIM. AC is equal to 1.03 in. (26 mm)
- DIM. AD is equal to 1.58 in. (40 mm)
- DIM. AE is equal to 2.37 in. (60 mm).

SUBTASK 72-23-06-320-002-G00

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

WARNING DO NOT LET TITANIUM DUST COLLECT. TITANIUM DUST IS VERY FLAMMABLE. DISCARD ALL TITANIUM WASTE IN AN APPROVED CONTAINER. TITANIUM DUST CAN CAUSE INJURY TO PERSONNEL.

CAUTION REMOVE MINIMUM MATERIAL MORE THAN THE DEPTH OF DEVIATION WHEN YOU BLEND. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

CAUTION BLENDED AREAS MUST MAKE A SMOOTH CONTOUR INTO THE ADJACENT AREAS TO KEEP CORRECT AIRFLOW. IF YOU DO NOT OBEY, DAMAGE TO PARTS CAN OCCUR.

(2) Do the blend of the struts pads as follows (View D and View E, Figure 811, View D and View E, Figure 812, and Figure 813):

(a) For the Method I, do this step:

1) Remove the indication by hand blending (TASK 70-43-42-300-802-G00).

(b) For the Method II, do this step:

1) Remove the indication by power blending (TASK 70-43-42-300-803-G00).

(c) If the minimum acceptable distance between two repaired areas is less than 40 times the DIM. AF, mix them to get only one repaired area.

SUBTASK 72-23-06-110-002-G00

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(3) Clean the part using alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-23-06-350-002-G00

(4) Do a local swab etching of the repaired areas with titanium alloy etchant, G51485 [S1093] (TASK 70-43-24-100-802-G00).

SUBTASK 72-23-06-230-002-G00

(5) Do a fluorescent penetrant inspection of the repaired areas (TASK 70-11-06-230-803-G00).

(a) Use sensitivity level 2.

(b) Indications more than 0.0118 in. (0.30 mm) are not permitted.

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(c) Linear indications are not permitted.

NOTE: Linear indications are indications that are at least four times longer than they are wide.

SUBTASK 72-23-06-211-018-G00

(6) Do a visual inspection of the parts (TASK 72-23-06-200-801-G00).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-06-410-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-06-840-009-G00

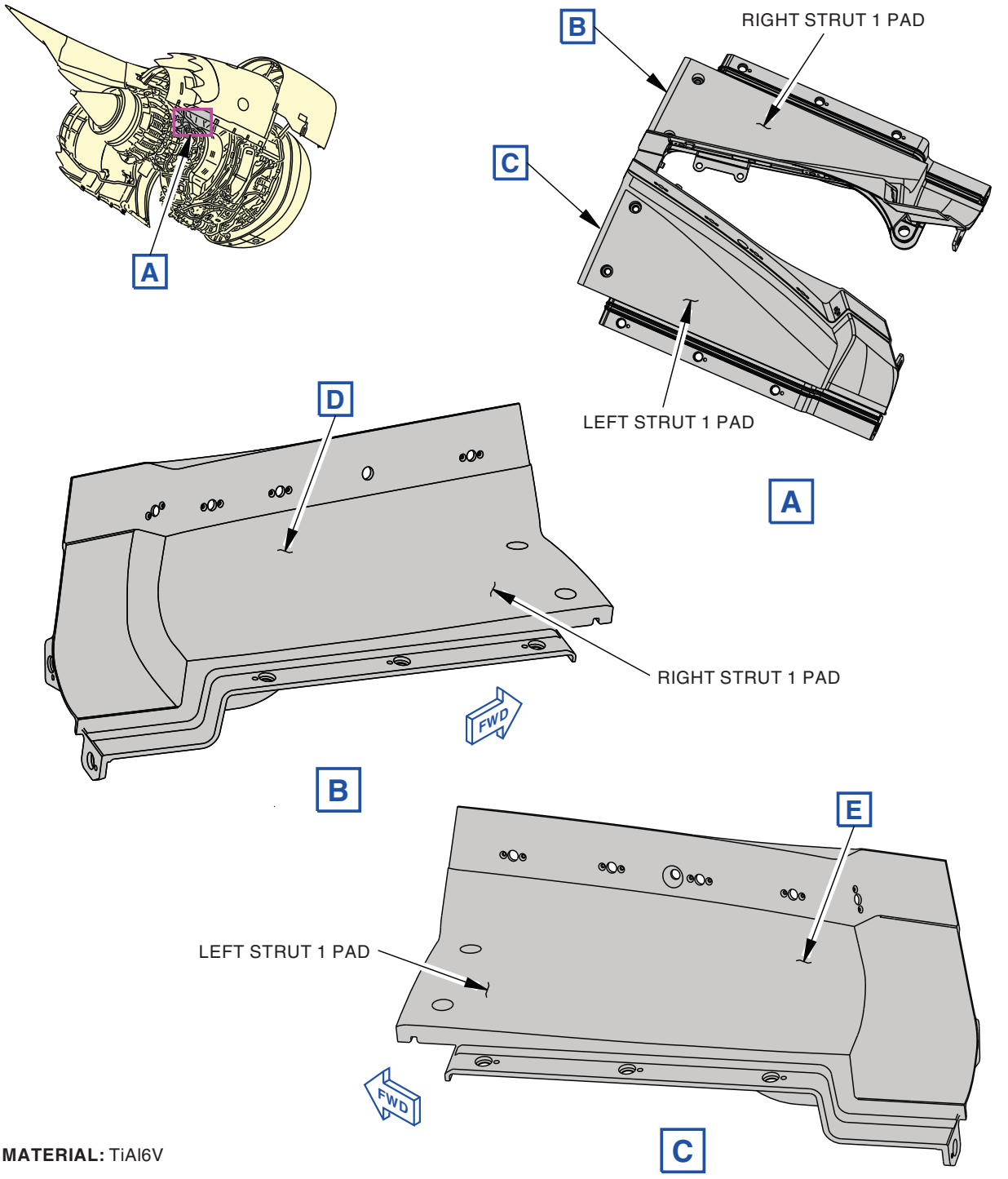
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

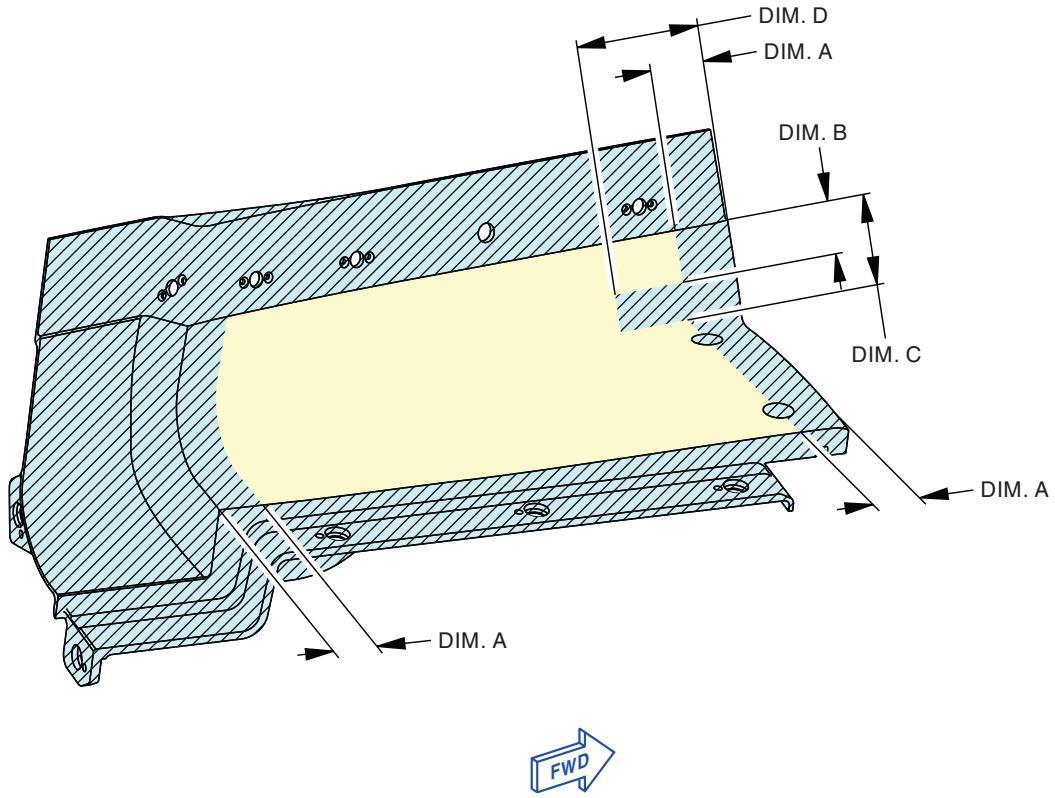
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut 1 Pad (Right and Left) Blending
Figure 811/72-23-06-990-831-G00 (Sheet 1 of 3)

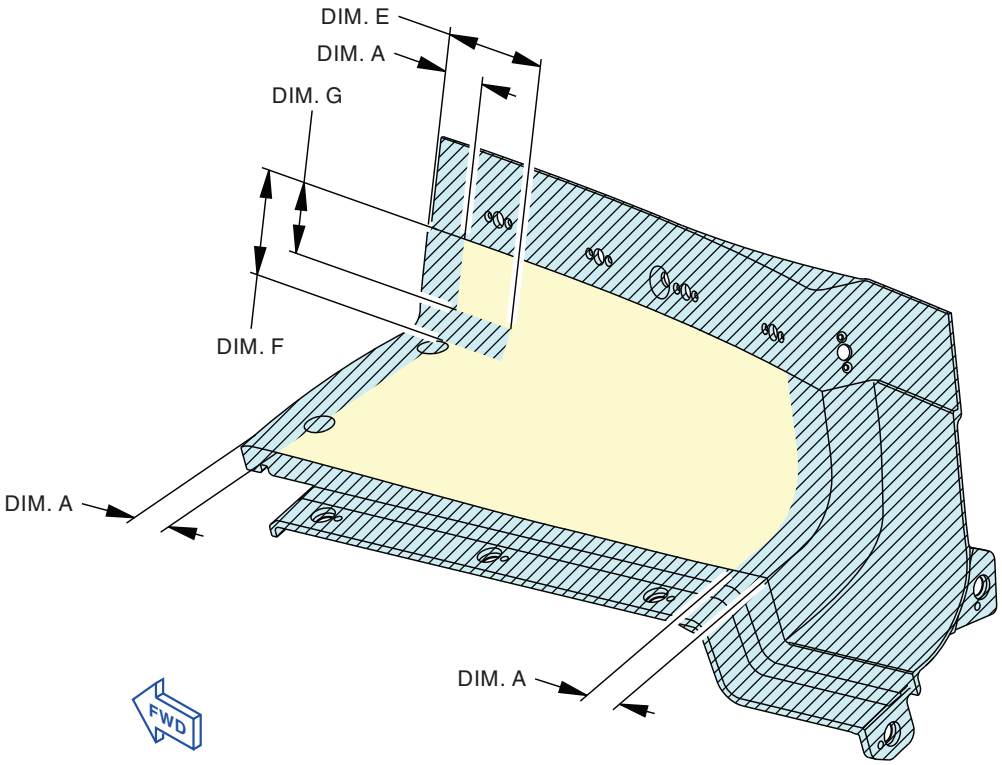
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2991506 S0000768396_V1

Fan Frame Strut 1 Pad (Right and Left) Blending
Figure 811/72-23-06-990-831-G00 (Sheet 2 of 3)

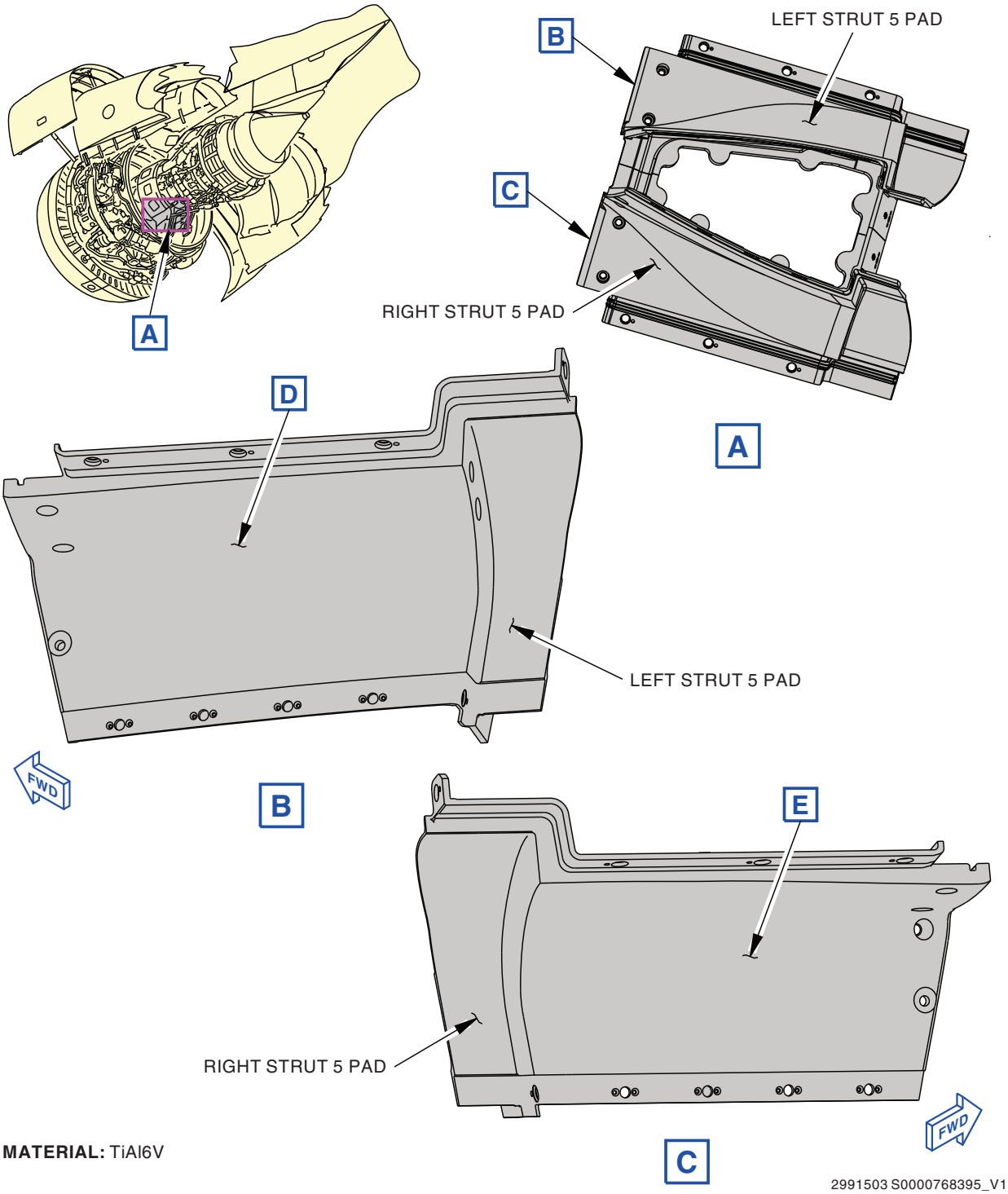
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2991510 S0000768397_V1

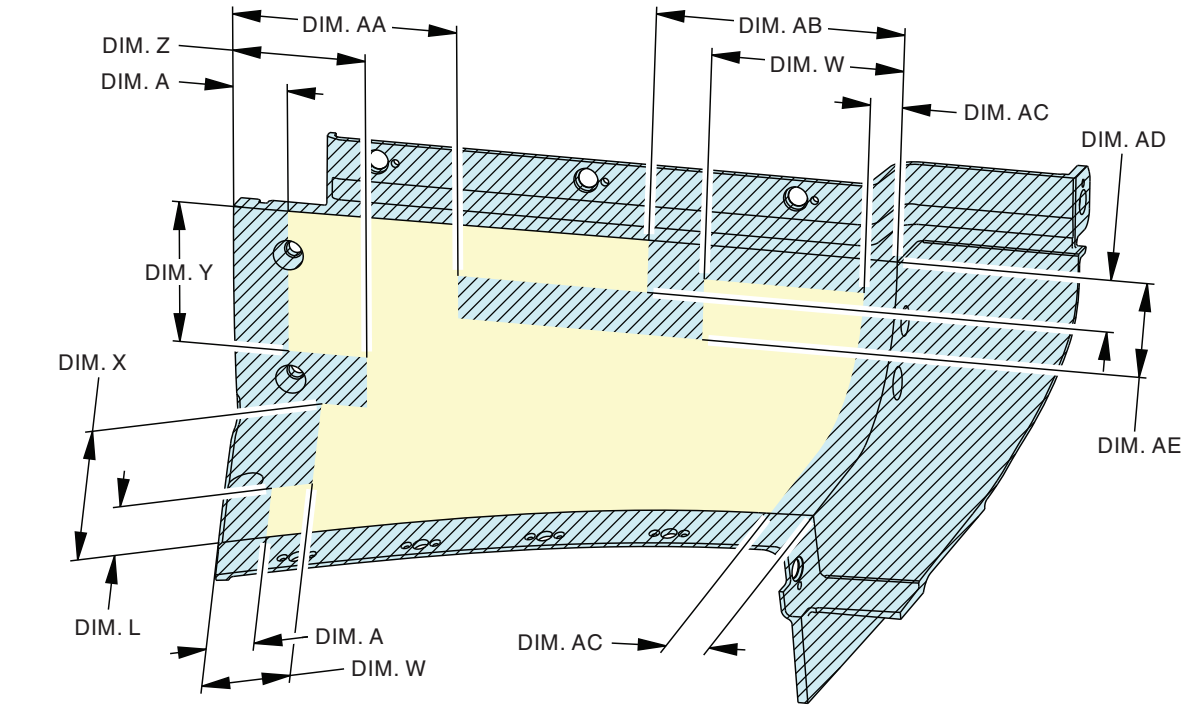
Fan Frame Strut 1 Pad (Right and Left) Blending
Figure 811/72-23-06-990-831-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut 5 Pad (Right and Left) Blending
Figure 812/72-23-06-990-832-G00 (Sheet 1 of 3)

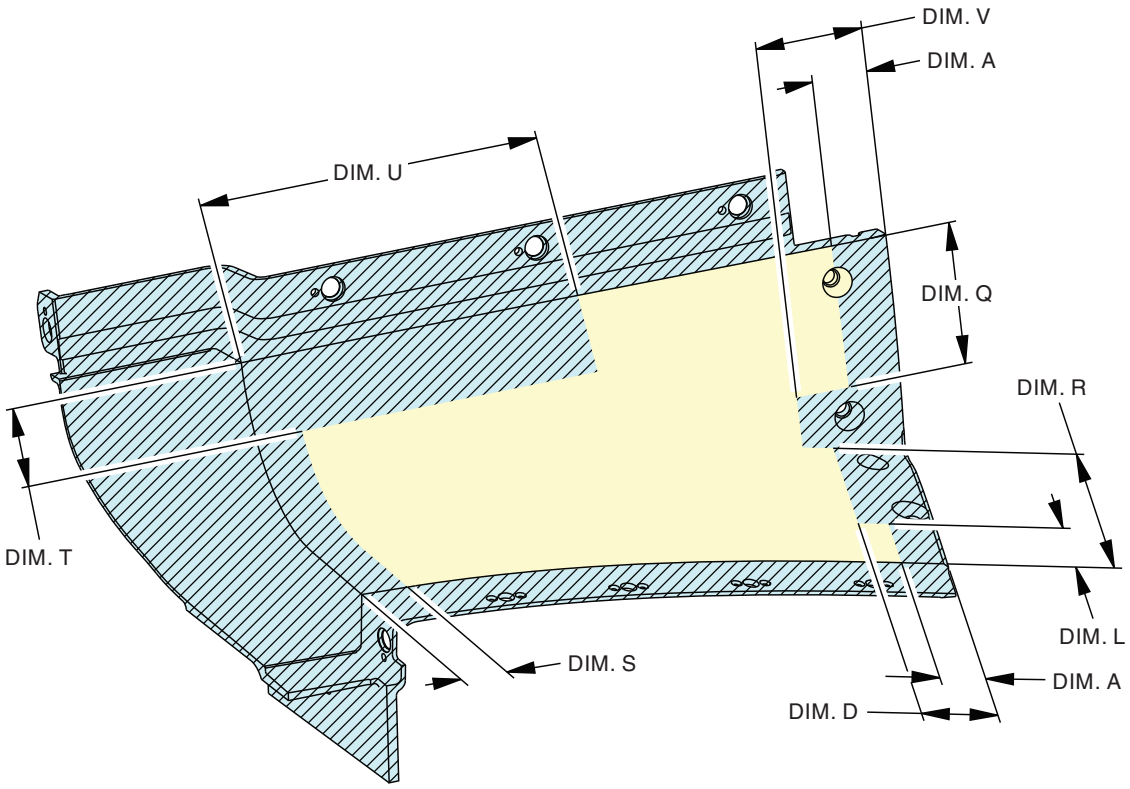
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2991515 S0000768399_V1

Fan Frame Strut 5 Pad (Right and Left) Blending
Figure 812/72-23-06-990-832-G00 (Sheet 2 of 3)

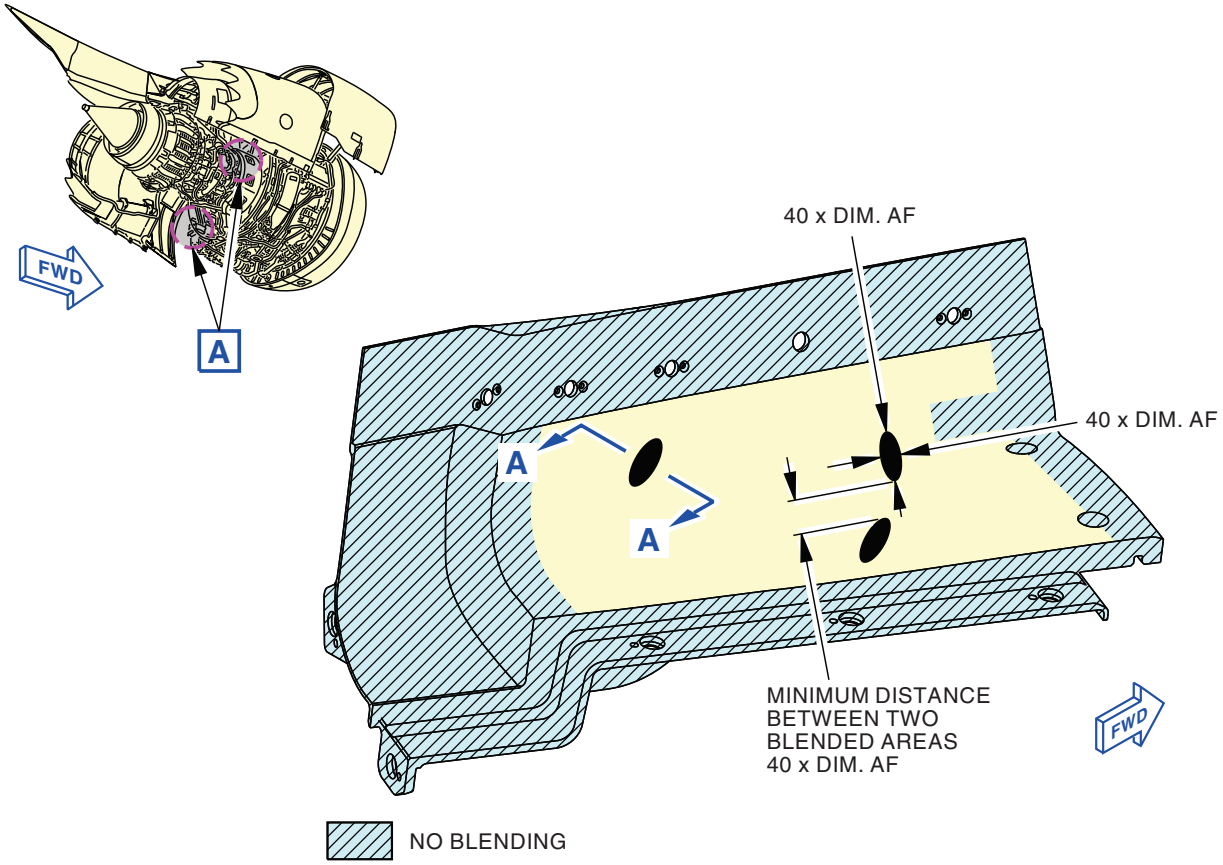
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2991513 S0000768398_V1

Fan Frame Strut 5 Pad (Right and Left) Blending
Figure 812/72-23-06-990-832-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ACCEPTABLE BLENDING PROFILE

A-A

2991518 S0000768400_V1

Dimensional Information
Figure 813/72-23-06-990-837-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-06-960-803-G00

6. Fan Frame Strut Pad Nut Replacement

(Figure 814)

A. General

(1) This task gives instructions to replace the nut of the SIMFAST floating bolt assembly or to replace the SIMFAST floating bolt assembly on the right Strut 1 Pad, on the left Strut 1 Pad, on the right Strut 5 Pad, and on the left Strut 5 Pad (Sheet 1 and 2).

NOTE: This repair can be apply only if the SIMFAST floating bolt assembly to be replaced is accessible.

(2) The table below gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 807/72-23-06-993-807-G00

Part number	Service Bulletin	CSN	Designation
364-080-510-0	-	(60-350, 72-00-00) (60A-350, 72-00-00)	RH Strut 1 Pad
364-082-810-0	-	(60-100, 72-00-00) (60A-100, 72-00-00)	LH Strut 1 Pad
364-082-910-0	-	(63-110, 72-00-00) 63A-110, 72-00-00)	RH Strut 5 Pad
364-083-010-0	-	(63-100, 72-00-00) (63A-100, 72-00-00)	LH Strut 5 Pad

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-34-300-801-G00	Simfast Floating Bolt Assemblies Rivet Repair (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G51479 [CP2696]	Tape - Masking - 3M 2525	

E. Location Zones

Zone	Area
411	Engine 1 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Replacement

SUBTASK 72-23-06-840-010-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-06-010-006-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-23-06-950-001-G00

(3) Make sure that you apply 3M 2525 tape, G51479 [CP2696], to protect the adjacent areas not to be repaired.

H. Fan Frame Strut Pad Nut Replacement

SUBTASK 72-23-06-960-001-G00

(1) If it is necessary to replace the nut of the SIMFAST floating bolt assembly, do these steps (Sheet 3):

(a) Remove and discard the nut (TASK 70-43-34-300-801-G00).

(b) Install the nut (TASK 70-43-34-300-801-G00).

SUBTASK 72-23-06-960-002-G00

(2) If it is necessary to replace the SIMFAST floating bolt assembly, do these steps (Sheet 3):

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

CAUTION DO NOT TOUCH THE PART WHEN YOU REMOVE THE SIMFAST FLOATING BOLT ASSEMBLY. DAMAGE TO THE PART CAN OCCUR.

(a) Remove and discard the SIMFAST floating bolt assembly (TASK 70-43-34-300-801-G00).

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(b) Clean the SIMFAST floating bolt assembly installation area with a cloth moistened with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(c) Do a visual inspection of the hole.

1) Deviation is not permitted.

(d) Install the SIMFAST floating bolt assembly (TASK 70-43-34-300-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-06-160-002-G00

(1) Remove the masking tape.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-23-06-410-005-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-06-840-011-G00

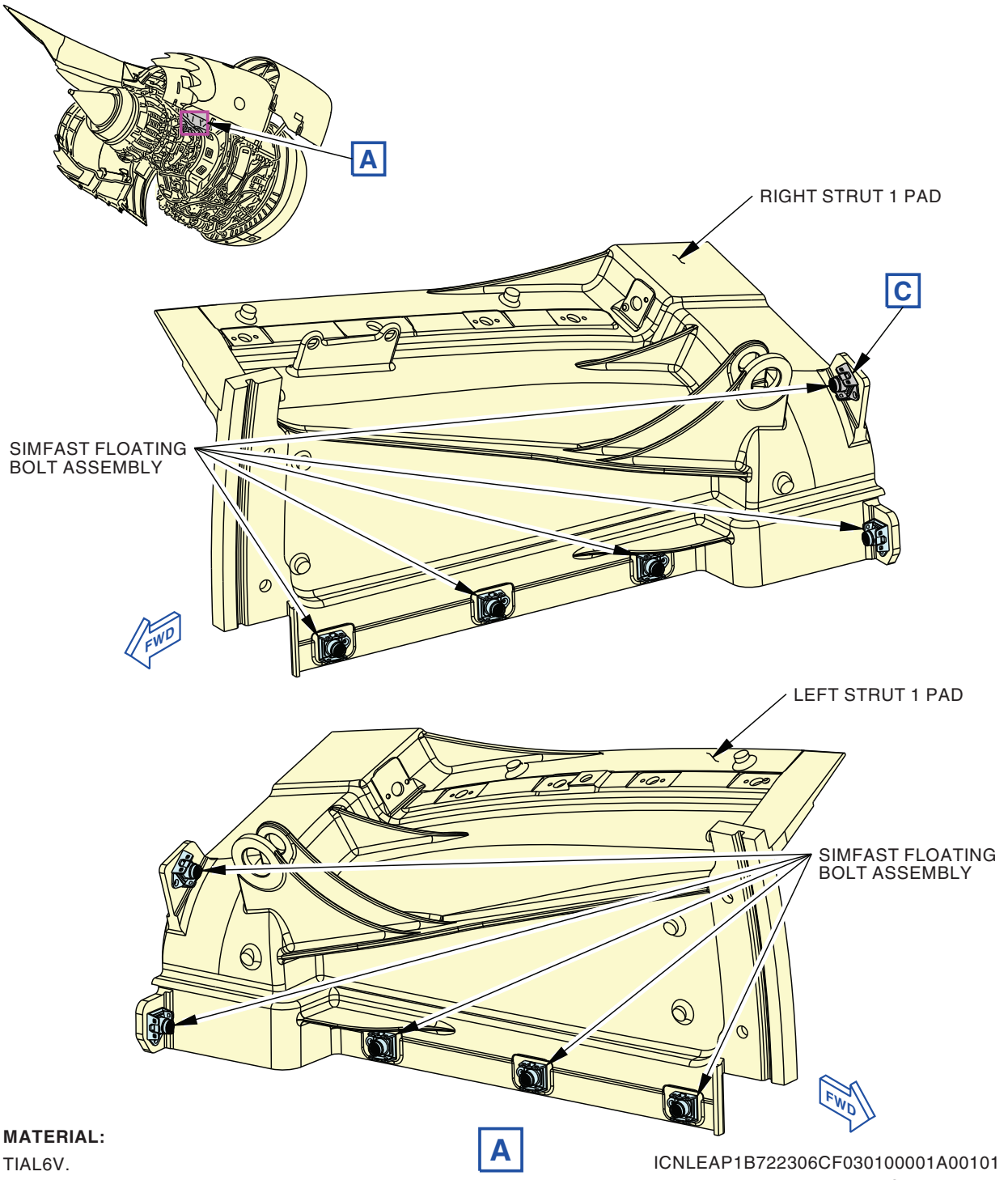
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

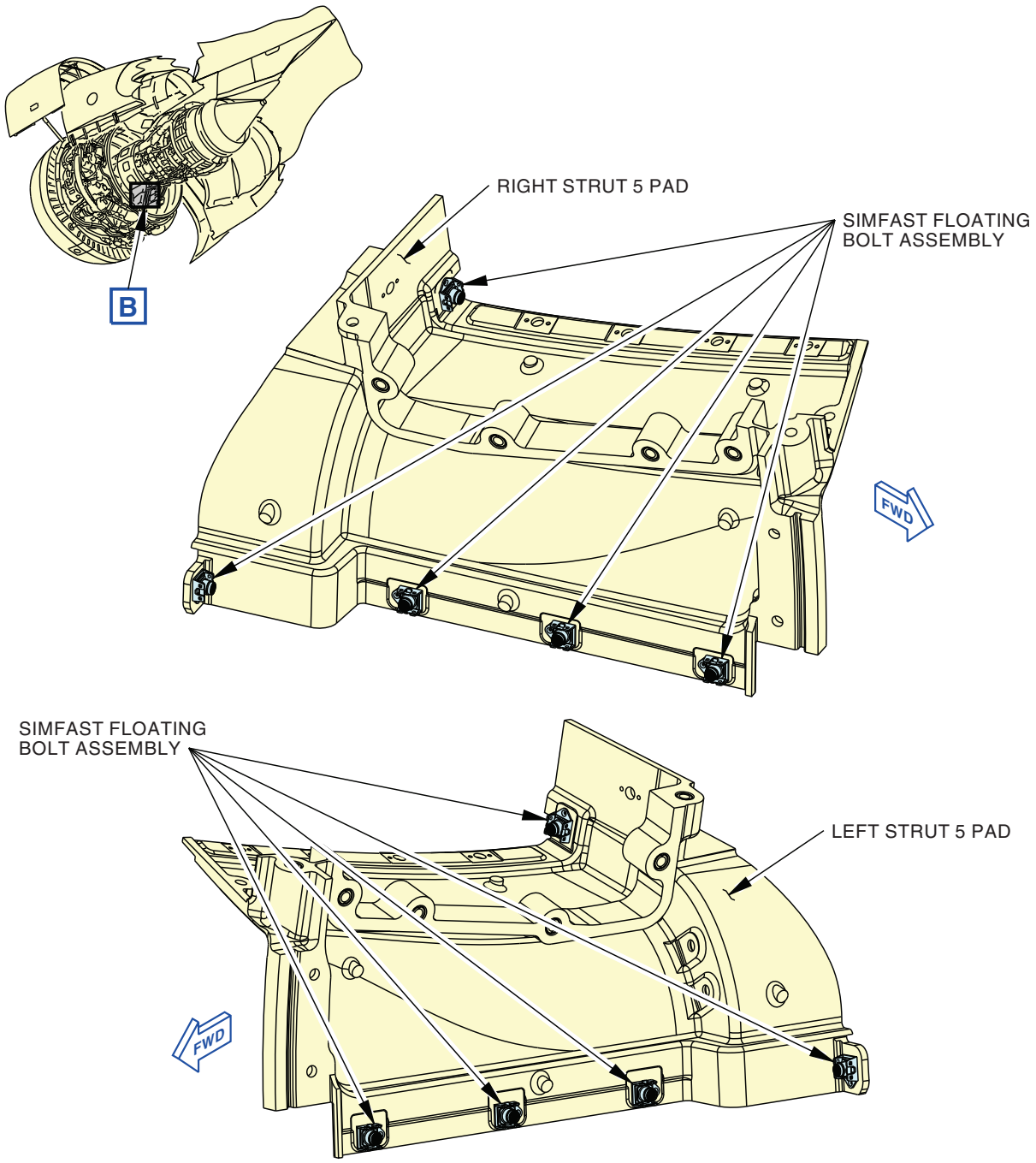


MATERIAL:
TIAL6V.

ICNLEAP1B722306CF030100001A00101
3058943 S0000822181_V1

Fan Frame Strut Pad Nut Replacement
Figure 814/72-23-06-990-846-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

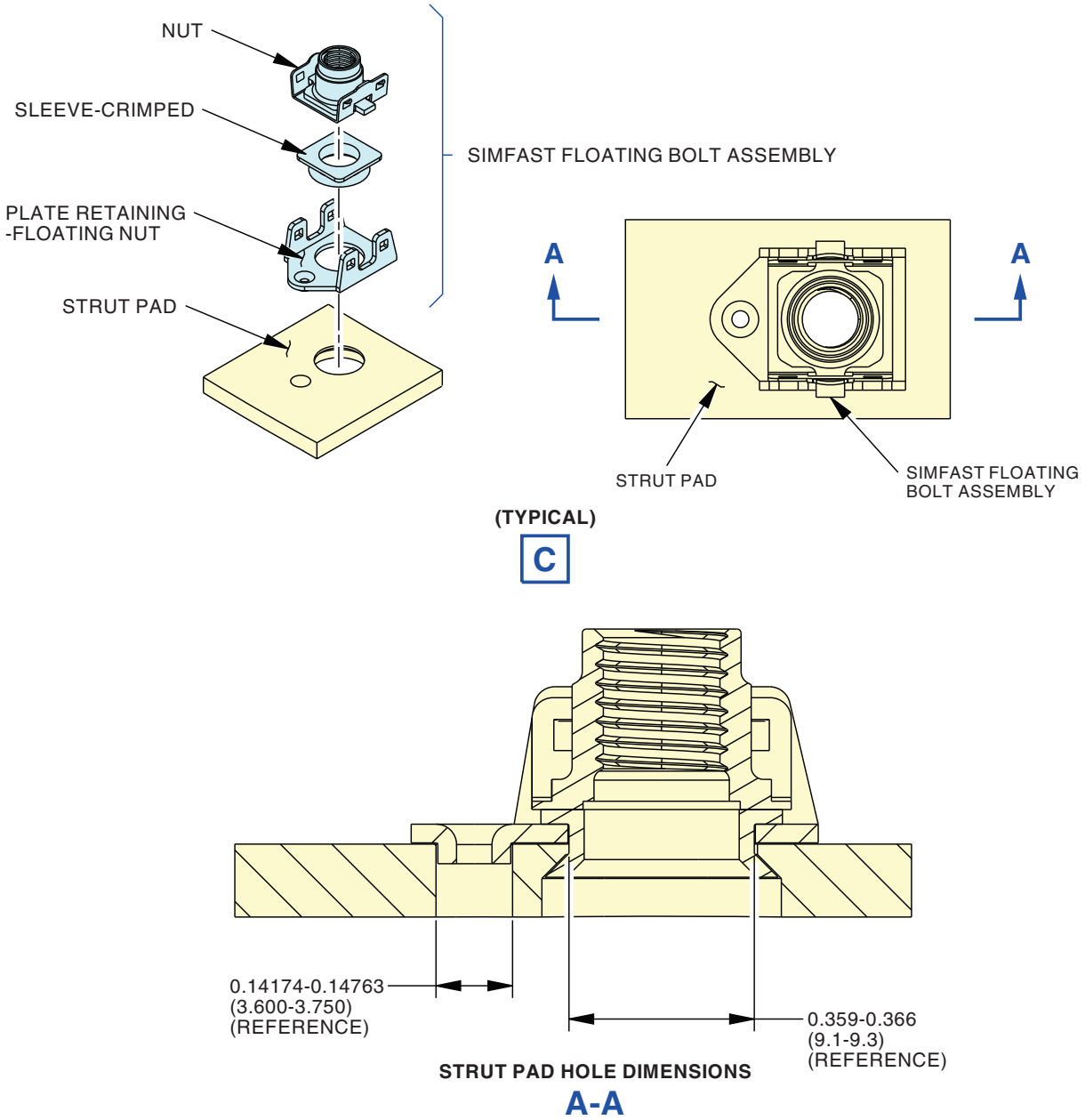


MATERIAL:
TIAL6V.

ICNLEAP1B722306CF030100001A00102
3058944 S0000822182_V1

Fan Frame Strut Pad Nut Replacement
Figure 814/72-23-06-990-846-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

ALL DIMENSIONS ARE IN INCHES
(MILLIMETERS ARE IN PARENTHESES).
REFERENCE DIMENSIONS DO NOT REQUIRE INSPECTION.

MATERIAL:

Z6NCT25 AND Z6CNT18.

ICNLEAP1B722306CF030100001A00103
3058942 S0000822183_V1

Fan Frame Strut Pad Nut Replacement
Figure 814/72-23-06-990-846-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN FRAME BIFI SUPPORT – INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) An inspection of the BIFI support.

TASK 72-23-07-200-801-G00

2. BIFI Support Inspection

(Figure 601 and Figure 602)

A. General

(1) This task provides the instructions to do a visual inspection of the BIFI support and harness block support.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-23-07-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-07-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. BIFI Support Visual Inspection

PTF ALL; AIRPLANES WITH PRE LEAP-1B SERVICE BULLETIN 72-0462

SUBTASK 72-23-07-210-003-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

(1) Do a visual inspection of the bracket assembly and support drain mast (Figure 601).

(a) Fretting on hole areas

1) Permitted if less than 0.0007 in. (0.02 mm) in depth.

(b) All indications on other areas.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH PRE LEAP-1B SERVICE BULLETIN 72-0462 (Continued)

1) Not serviceable.

PTF ALL; AIRPLANES WITH POST LEAP-1B SERVICE BULLETIN 72-0462

SUBTASK 72-23-07-210-001-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

(2) Do a visual inspection of the upper bracket assembly, front bracket assembly, central bracket assembly, drain tube support, and drain tube assembly (Figure 601).

(a) Fretting on hole areas

1) Permitted if less than 0.0007 in. (0.02 mm) in depth.

(b) All indications on other areas.

1) Not serviceable.

PTF ALL

H. Harness Block Supports Visual Inspection

SUBTASK 72-23-07-210-002-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

(1) Do a visual inspection of the harness block supports (Figure 602).

(a) Break, corrosion, crack, distortion, heating due to heat or discoloration, and missing parts

1) Not serviceable.

(b) Nick, dent, and scratch on the area A.

1) Permitted if equal to or smaller than 0.019 in. (0.5 mm).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-07-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-07-860-002-G00

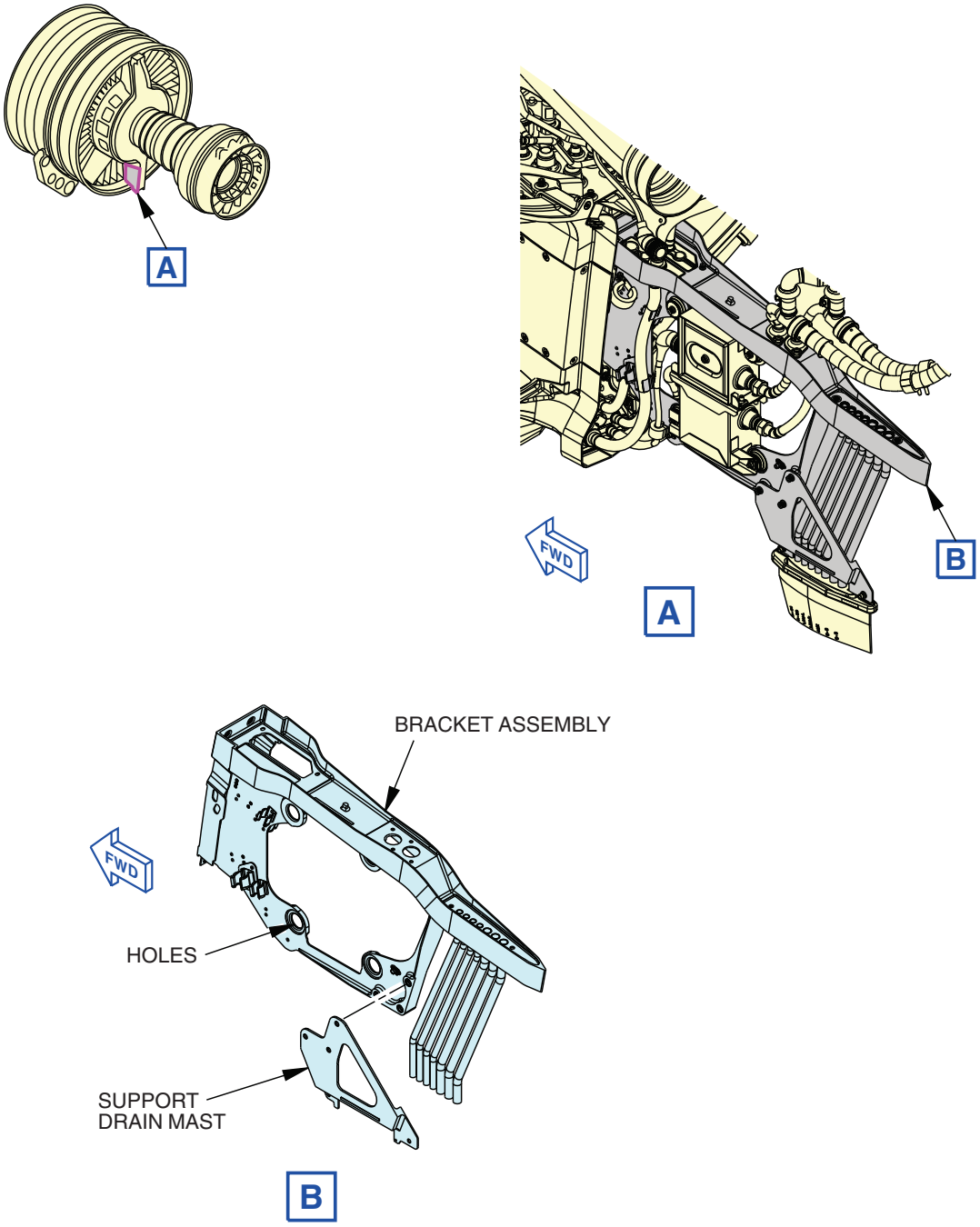
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

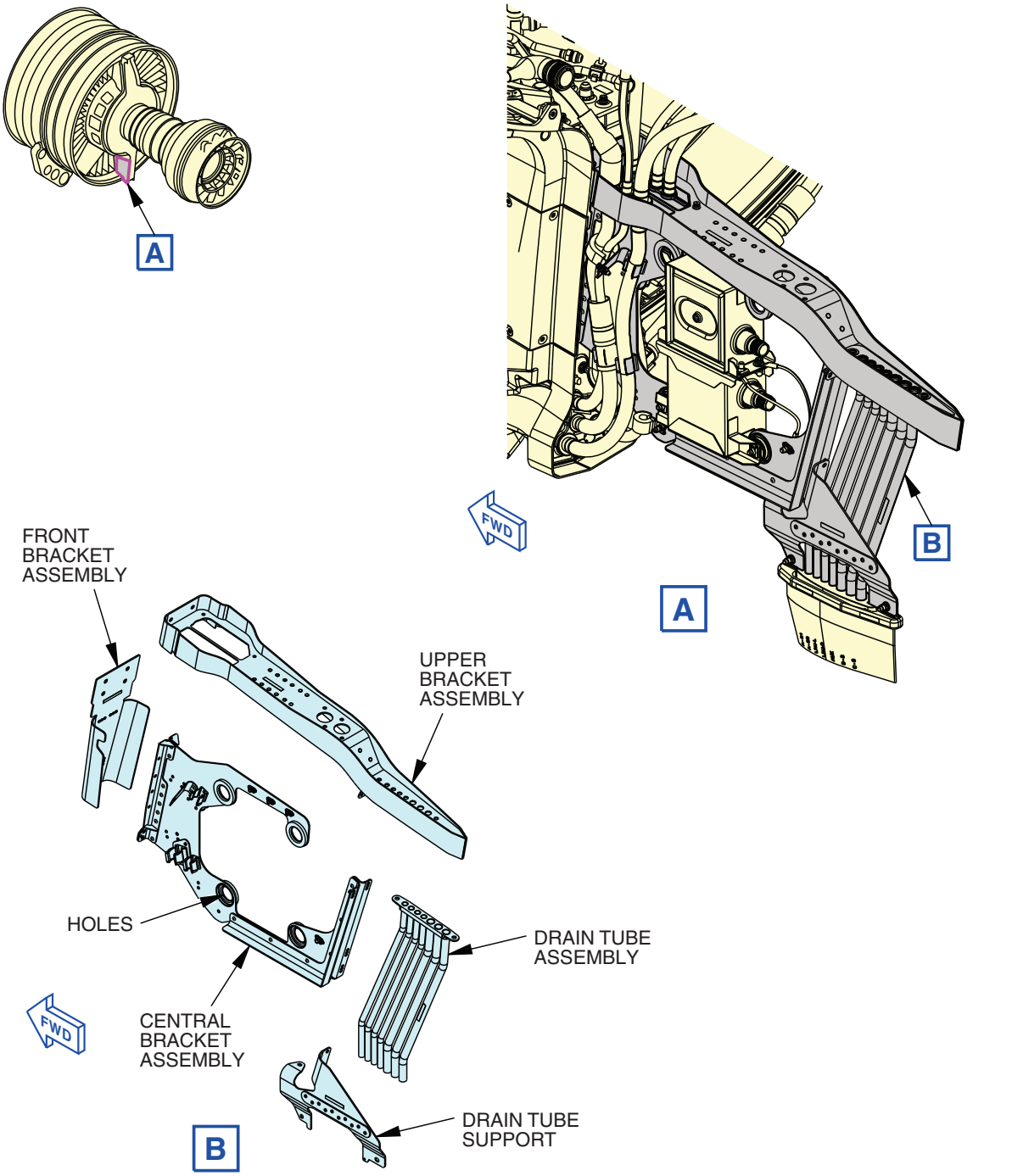
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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3120991 S0000855707_V1

BIFI Support Inspection
Figure 601/72-23-07-990-801-G00 (Sheet 1 of 2)

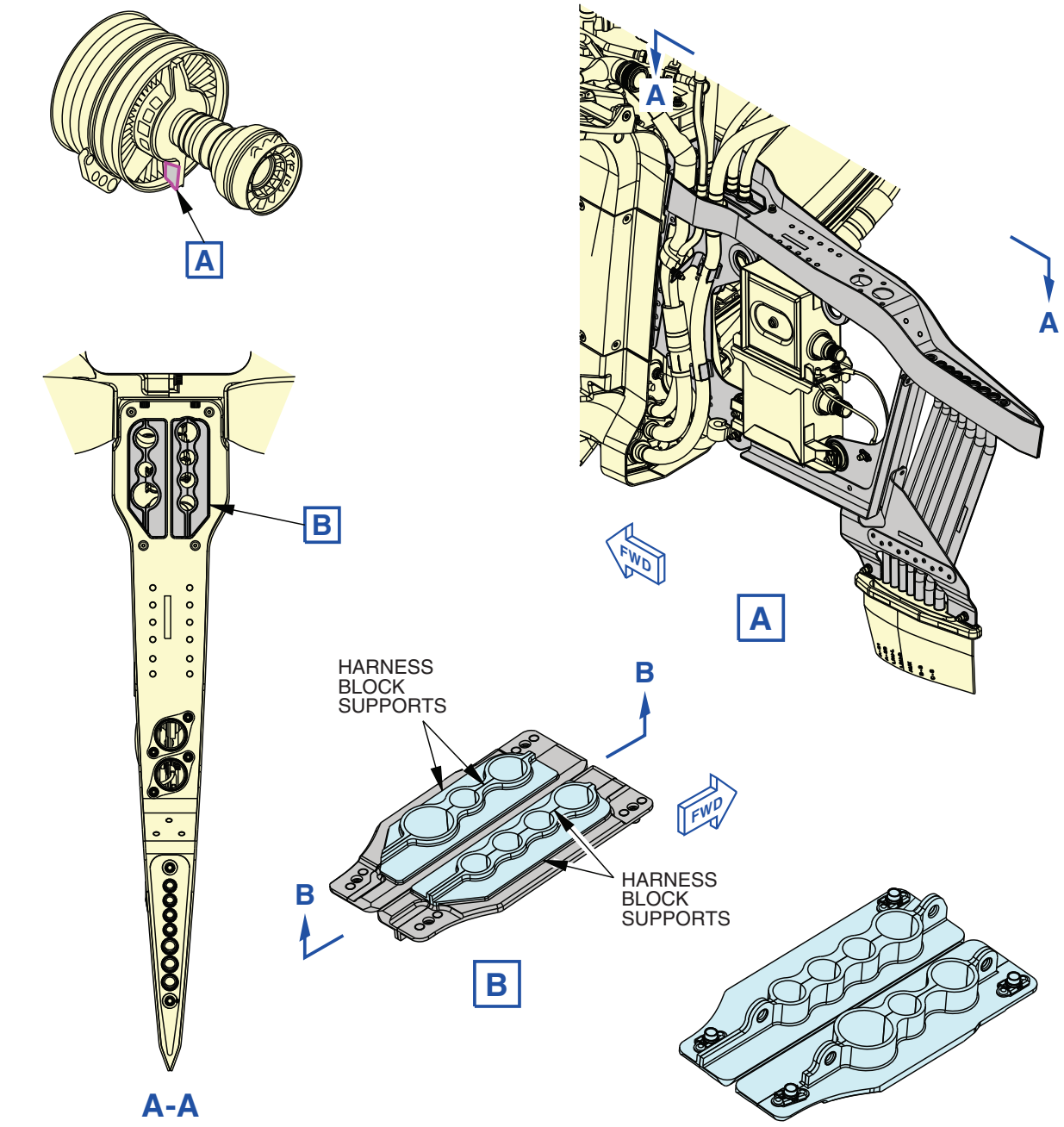
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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BIFI Support Inspection
Figure 601/72-23-07-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1BXXXXXXD5882800003A00101
3120996 S0000855709_V1

Harness Block Support Inspection
Figure 602/72-23-07-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HIGH PRESSURE COMPRESSOR SECTION - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) High Pressure Compressor (HPC) section inspection.

TASK 72-30-00-200-801-G00

2. High Pressure Compressor (HPC) Section Inspection

(Figure 601)

A. General

(1) This task gives the instructions to examine the High Pressure Compressor (HPC) section.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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F. Prepare for the Inspection

SUBTASK 72-30-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-30-00-860-002-G00

(2) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-30-00-040-001-G00

(3) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-30-00-010-001-G00

(4) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. High Pressure Compressor Section Inspection

SUBTASK 72-30-00-220-001-G00

(1) Examine the HPC case for cracks as follows:

(a) Cracks in the case skin

1) Not permitted.

(b) Cracks in the horizontal and circumferential flanges that go out from the bolt holes

1) Not permitted.

(c) Cracks in the horizontal and circumferential flanges that are between a bolt hole and the case skin

1) Not permitted.

(d) Cracks in the Environmental Control System (ECS) duct port or bracket pads and stage 4 bleed ports.

1) Not permitted.

SUBTASK 72-30-00-220-002-G00

(2) Examine the HPC case for nicks, gouges, scratches, pits, and scores as follows:

(a) Nicks, gouges, scratches, pits, and scores in the case skin and flanges

1) Blend the high metal smooth with no burrs or sharp edges (TASK 70-43-42-300-802-G00).

2) All amounts less than 0.005 in. (0.13 mm) deep are permitted.

3) Corrosion and rust discoloration on the HPC case and flange surfaces are permitted.

(b) Dents in the stage 4 bleed port of the HPC case

1) Blend the high metal smooth with no burrs or sharp edges (TASK 70-43-42-300-802-G00).

2) A maximum of two dents, 0.25 in. (6.35 mm) in diameter and 0.005 in. (0.13 mm) in depth, are permitted.

(c) Dents in the ECS port of the HPC case.

1) Blend the high metal smooth with no burrs or sharp edges (TASK 70-43-42-300-802-G00).

2) A maximum of one dent, 0.005 in. (0.13 mm) in depth, is permitted.

SUBTASK 72-30-00-220-003-G00

(3) Examine the HPC case bolts as follows:

(a) Loose, broken, or missing bolts and nuts

1) Not permitted.

SUBTASK 72-30-00-220-004-G00

(4) Examine the Inlet Guide Vane (IGV) and Variable Stator Vane (VSV) stage 1 thru stage 4 actuating rings as follows:

(a) Cracks in the actuating rings

1) Not permitted.

(b) Twisted or bent actuating rings

1) Not permitted.

(c) Actuating rings that show separation or are broken

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1) Not permitted.

(d) Nicks, dents, and scratches in all areas of the actuating rings

1) Blend the high metal smooth with no burrs or sharp edges (TASK 70-43-42-300-802-G00).

2) All amounts up to 0.01 in. (0.25 mm) deep are permitted after removal of the high metal.

(e) Damaged or broken actuating ring bushings

1) Bushing body shall not be damaged and give 360 degrees support to an actuating arm pin.

(f) Missing bushing on the IGV

1) One missing bushing is permitted.

2) Two missing bushing spaced more than 60 degrees apart are permitted with this Continue-In-Service (C-I-S) limit:

a) Five flight cycles or 30 hours, whichever occurs first.

3) Two missing bushing spaced less than 60 degrees apart, or three or more missing bushing are not permitted.

(g) Missing bushing on stage 1

1) Two missing bushings are permitted with no limit.

2) Three missing bushings are permitted with this Continue-In-Service (C-I-S) limit:

a) Five flight cycles or 30 hours, whichever occurs first.

3) Four or more missing bushings are not permitted.

(h) Missing bushings on stages 2 to 4

1) Three missing bushings are permitted with no limit.

2) Four missing bushings are permitted with this Continue-In-Service (C-I-S) limit:

a) Five flight cycles or 30 hours, whichever occurs first.

3) Five or more missing bushings are not permitted.

(i) Missing or broken bushing legs

1) Two missing or broken bushing legs are permitted.

(j) Cracked or distorted ring spacers

1) Not permitted.

(k) Wear on ring spacer buttons

1) 0.01 in. (0.25 mm) minimum at Dimension E is permitted.

(l) Looseness on ring spacer nuts

1) Not permitted, tighten the loose actuator ring spacer nut.

SUBTASK 72-30-00-220-005-G00

(5) Examine the actuating arms as follows:

(a) Bent, broken, or cracked actuating arms

1) Not permitted.

2) Remove the engine to replace the damaged actuating arm (TASK 71-00-02-000-801-G00).

(b) Nicks, dents, and scratches

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1) Nicks, dents, and scratches in area G are not permitted

2) Nicks, dents, and scratches 0.01 in. (0.25 mm) in Area H are permitted after removal of high metal.

(c) Nuts oxidation.

1) Any amount of surface oxidation permitted.

NOTE: Oxidation occurs due to copper-strike coating on VSV nuts that react with the environment.

SUBTASK 72-30-00-220-006-G00

(6) Examine the actuating ring bridges as follows:

(a) Actuating ring bridge that is broken or cracked on one or both sides of the engine

1) Not permitted.

2) Remove the engine to replace the broken actuating ring bridges (TASK 71-00-02-000-801-G00).

(b) Nicks, dents, and scratches in all areas of actuating ring bridges

1) Blend the high metal smooth with no burrs or sharp edges (TASK 70-43-42-300-802-G00).

2) All amounts up to 0.01 in. (0.25 mm) deep are permitted after removal of the high metal.

(c) Actuating ring bridge bushings with broken or missing legs

1) Two damaged or broken bushing legs are permitted.

(d) Missing or broken actuating ring bridge bushings

1) Not permitted.

SUBTASK 72-30-00-220-007-G00

(7) Examine the VSV actuating shaft as follows:

(a) Cracks

1) Not permitted.

(b) Nicks, dents, scratches, and scores

1) Not permitted.

SUBTASK 72-30-00-220-008-G00

(8) Examine the turnbuckles as follows:

(a) Cracks

1) Not permitted.

(b) Nicks, dents, scratches, and scores

1) Blend the high metal smooth with no burrs or sharp edges (TASK 70-43-42-300-802-G00).

2) All amounts up to 0.005 in. (0.13 mm) deep are permitted after removal of the high metal.

(c) Loose hex nuts

1) Not permitted, tighten the loose hex nut.

SUBTASK 72-30-00-220-011-G00

(9) Examine the spherical bearings as follows:

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(a) Looseness of the ball in the race

1) The maximum radial movement is permitted to 0.002 in. (0.05 mm).

SUBTASK 72-30-00-220-012-G00

(10) Examine the bellcrank assembly on the VSV actuating system as follows:

(a) Cracks

1) Not permitted.

(b) Nicks, dents, scratches, and scores on IGV to Stage 4 bellcranks or actuator link:

1) Permitted with these conditions:

a) Blend the high metal smooth with no burrs or sharp edges (TASK 70-43-42-300-802-G00).

b) All amounts and any length 0.015 in. (0.38 mm) in depth are permitted after removal of the high metal.

(c) Nicks, dents, scratches, and scores on bellcrank master plates and master rod:

1) Permitted with these conditions:

a) Blend the high metal smooth with no burrs or sharp edges (TASK 70-43-42-300-802-G00).

b) All amounts and any length 0.030 in. (0.76 mm) in depth are permitted after removal of the high metal.

(d) Lower plate, master rod, and Stage 2-4 bellcranks surface corrosion:

1) Any amount of surface corrosion is permitted as long as visual inspection shows the following conditions:

a) Continuous rust marks

b) No pitting or evident localized material corrosion (depth below surface) marks.

SUBTASK 72-30-00-220-014-G00

(11) Examine the vane spindle bushings as follows:

(a) Missing bushings

1) Not permitted.

SUBTASK 72-30-00-410-001-G00

(12) If it is necessary, do this task: Power Plant - Installation, TASK 71-00-02-400-801-G00.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-30-00-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-30-00-440-001-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-30-00-860-003-G00

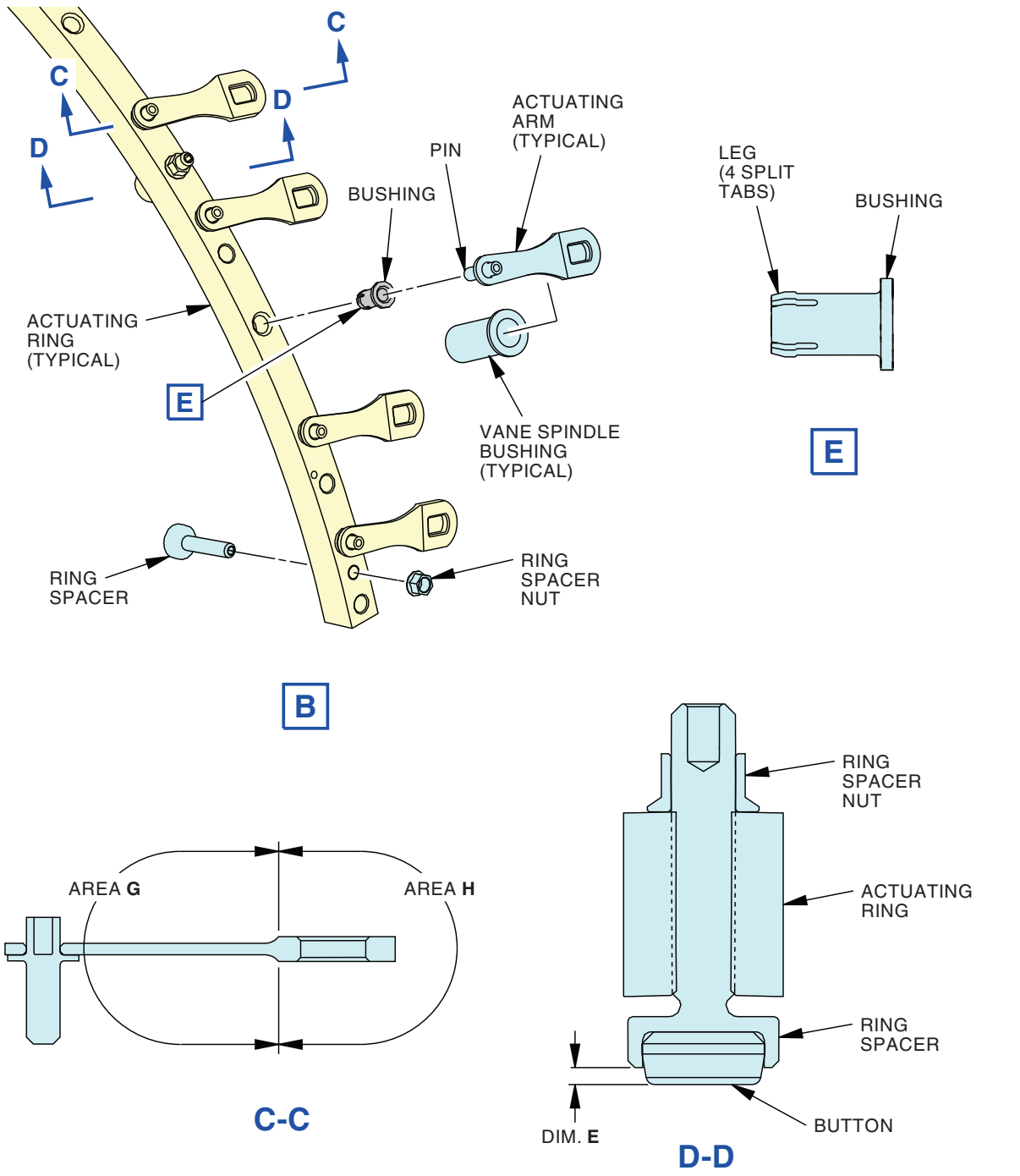
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

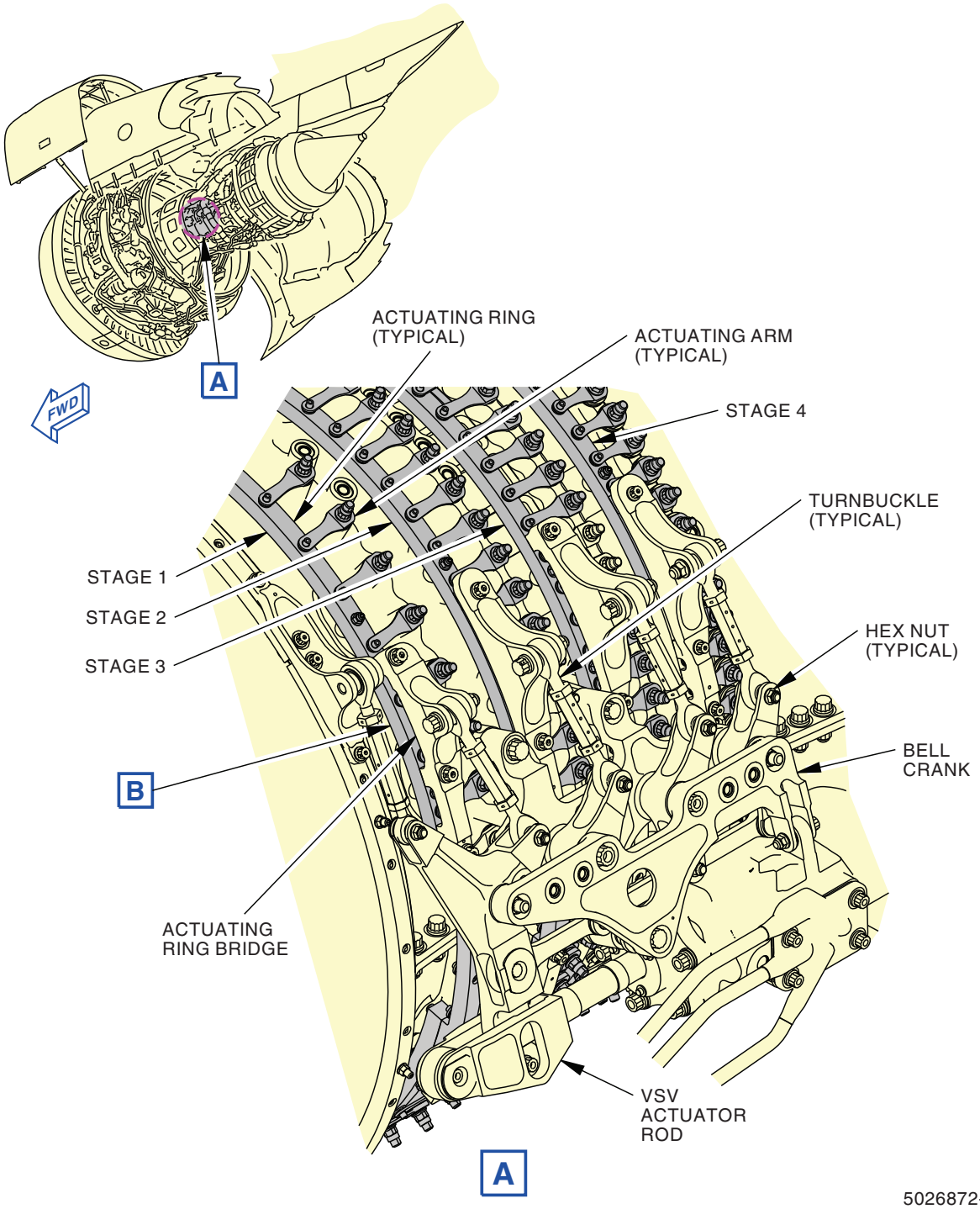
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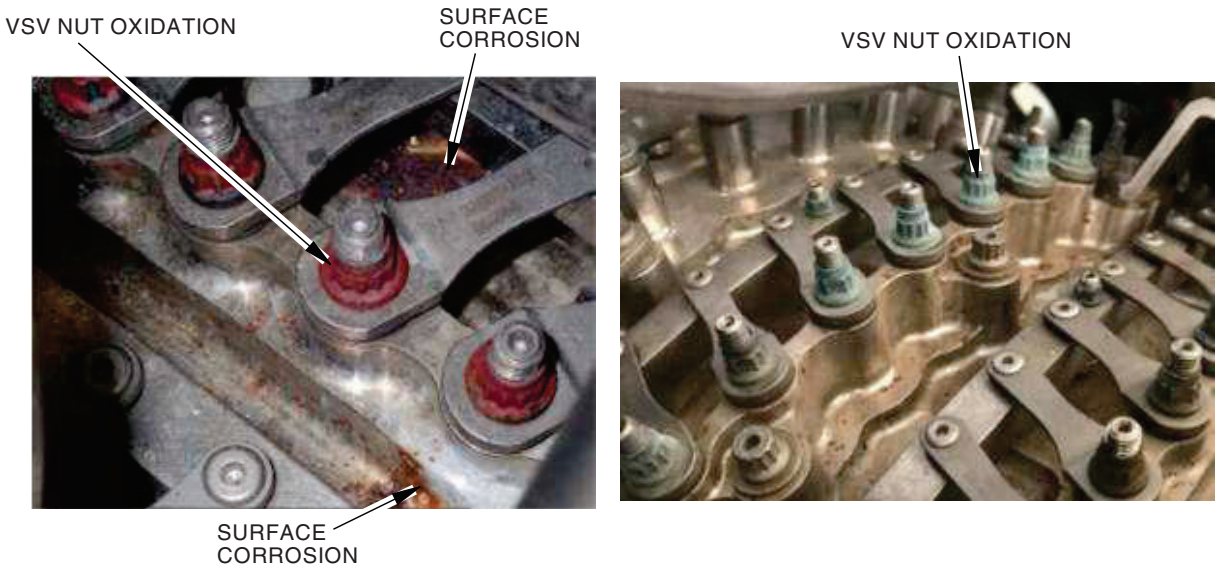
High Pressure Compressor Section Inspection
Figure 601/72-30-00-990-808-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

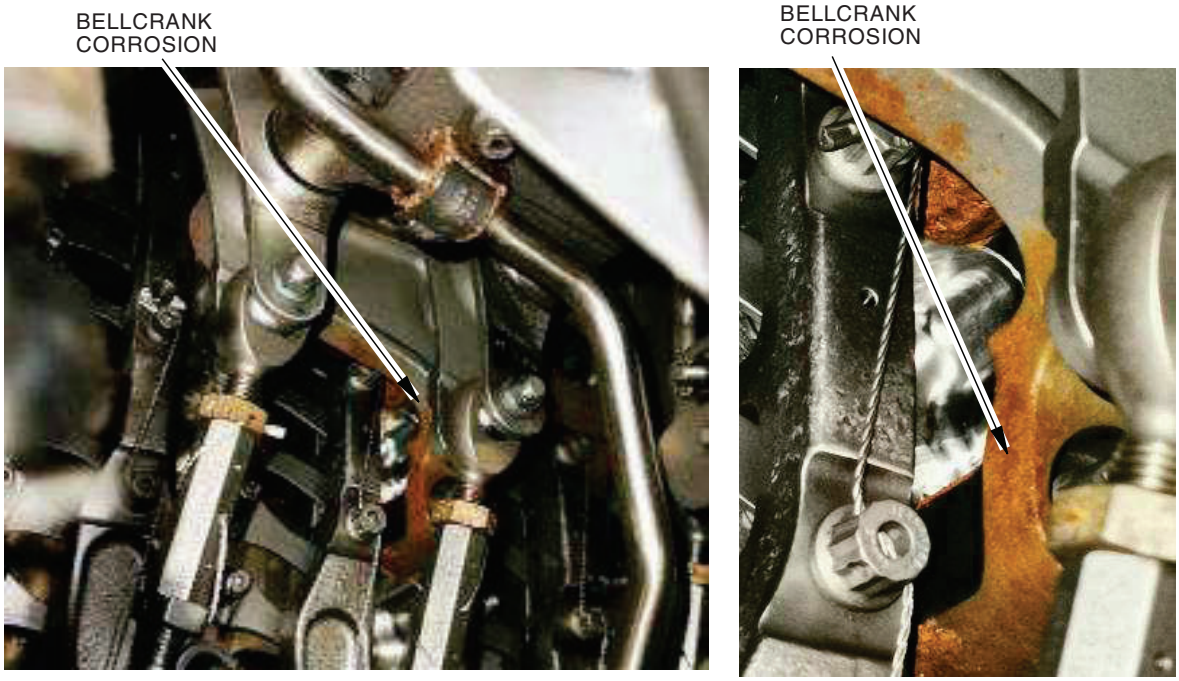


High Pressure Compressor Section Inspection
Figure 601/72-30-00-990-808-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**CONTINUOUS SURFACE CORROSION (RUST) MARKS ON HPC CASE.
RED, BLUE/GREEN OXIDATION PRODUCTS ON VSV COPPER-STRIKE COATED NUTS.**



CONTINUOUS SURFACE CORROSION (RUST) MARKS ON VSV BELLCRANK.

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**High Pressure Compressor Section Inspection
Figure 601/72-30-00-990-808-G00 (Sheet 3 of 3)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HIGH PRESSURE COMPRESSOR SECTION - REPAIR

1. General

A. This procedure has one task.

(1) High Pressure Compressor (HPC) Rotor Blades/Blisks Borescope Blending Repair.

TASK 72-30-00-350-801-G00

2. High Pressure Compressor (HPC) Rotor Blades/Blisks Borescope Blending Repair

(Figure 801, Figure 802, Figure 803, Figure 804, Figure 805, Figure 806, Figure 807, and Figure 808)

A. General

(1) This task gives the instructions to blend with a borescope blending tool to remove nicks, dents, and tears on the airfoil leading and trailing edges.

(2) Cracks are unserviceable and are not repairable with this task.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-290-801-G00	Preparation for Borescope Inspection (P/B 601)
72-00-00-290-804-G00	High Pressure Compressor Section Borescope Detailed Inspection (P/B 601)
72-00-00-822-801-G00	Zero-Index of the N2 Rotor Alignment (P/B 201)
72-00-00-910-801-G00	High Pressure Rotor (N2) Turning for Borescope (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
72-30-00-200-801-G00	High Pressure Compressor (HPC) Section Inspection (P/B 601)
72-63-04-000-801-G00	Accessory Gearbox Manual Control Cover Removal (P/B 401)
72-63-04-400-801-G00	Accessory Gearbox Manual Control Cover Installation (P/B 401)
75-31-00-910-801-G00	Variable Stator Vane Actuator Manual Operation (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-2332	Borescope Blend Tool Part #: 9471M44P01 Supplier: 58828
SPL-14908	Borescope Blend Tool Part #: 956A1604P01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Repair

SUBTASK 72-30-00-860-004-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-30-00-010-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-30-00-940-001-G00

(3) Prepare the borescope blend tool, SPL-14908, or the borescope blend tool, SPL-2332, and the engine assembly for the repair.

NOTE: Both borescope kits have vendor user manuals. The rigid borescope blend tool, SPL-2332 is recommended for all stages. However, the flexible borescope blend tool, SPL-14908 is recommended for stages 8-10 in the lower 50% span.

SUBTASK 72-30-00-010-003-G00

(4) Do this task: Accessory Gearbox Manual Control Cover Removal, TASK 72-63-04-000-801-G00.

SUBTASK 72-30-00-480-001-G00

(5) Do this task: High Pressure Rotor (N2) Turning for Borescope, TASK 72-00-00-910-801-G00.

SUBTASK 72-30-00-010-004-G00

(6) Remove the borescope plugs C through M as necessary from the HPC stator case assembly (TASK 72-00-00-290-801-G00).

SUBTASK 72-30-00-910-001-G00

(7) Put a tag on the borescope plugs that you will remove.

NOTE: This will identify and match the plug to its access port.

SUBTASK 72-30-00-290-005-G00

(8) Do an inspection of the blisks and blades (TASK 72-00-00-290-804-G00).

(a) If it is necessary, manually shift the Variable Stator Vane (VSV) system to do the inspection of the stages 1 through 4 blisks (TASK 75-31-00-910-801-G00).

SUBTASK 72-30-00-010-005-G00

(9) If you will repair stages 1 through 3, remove the shroud segment 3 (TASK 72-23-04-000-801-G00).

SUBTASK 72-30-00-820-001-G00

(10) Do this task: Zero-Index of the N2 Rotor Alignment, TASK 72-00-00-822-801-G00.

G. High Pressure Compressor (HPC) Rotor Blades/Blisks Borescope Blending Repair

SUBTASK 72-30-00-970-001-G00

(1) Find the dimensions of the necessary blend repair as follows:

(a) Put the borescope blend tool, SPL-2332, drive shaft head in the 90 degree position (Figure 805).

(b) Lock the drive shaft head in place with the supplied lock pin.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Install the measuring grid tip by hand tightening in the clockwise direction.

1) Make sure that the measuring grid tip with drive shaft head is in the 90 degree position.

NOTE: If the grid tip with drive shaft head is in a different position, the drive belt will disengage.

NOTE: All measurements of damage must be made with the borescope blend tool, SPL-2332. The borescope blend tool, SPL-14908, does not use measurement tips. After you measure the damage, you can use the tool if necessary for better access to stages 8 -10.

(d) Put the borescope blend tool, SPL-2332, drive shaft head in the 0 degree position.

(e) Put the borescope tool into the compressor through the applicable borescope access port.

(f) Turn ON the borescope light source.

(g) Monitor the borescope blend tool, SPL-2332, image of the measuring grid tip while you move it from 0 to the 90 degree position as follows:

1) Put the measuring grid tip near each blisk and blade airfoil damaged area.

2) Use the grid tip to find and record the dimensions of the damage of these areas:

NOTE: Each square of the measuring grid tip is 0.039 in. (1.0 mm) X 0.039 in. (1.0 mm).

a) The maximum depth of the damaged area.

b) The maximum length of the damaged area.

c) The location of the edges of the damaged area from the nearest edge of the airfoil.

SUBTASK 72-30-00-970-002-G00

(2) Find if a blend repair is permitted for each damaged area in the blisk and blade airfoil as follows:

(a) Examine the blend repair limits for the blisk and blade airfoils (Figure 802).

(b) Compare the data for the location of each damaged area to the blend repair limits.

1) Check of the entire stage to account for blades that were previously blended.

NOTE: This counts toward the overall stage limit in Figure 802.

(c) For each damaged area that you are permitted to repair in a blisk or blade airfoil, do these steps:

1) Calculate the minimum blend repair length as six times dimension A (the maximum depth of the damage).

2) Adjust the damaged area to the center in the blend repair that you calculate.

a) Make sure that the minimum blend area does not extend into area A (Figure 802 and Figure 803).

NOTE: Area A extends above the airfoil root fillet for all stages. All cracks are not permitted and cannot be blended.

(d) Put the borescope blend tool, SPL-2332, head in the 0 degree position and remove the borescope shaft from the access port.

(e) Put the borescope blend tool, SPL-2332, drive shaft head in the 90 degree position (Figure 805).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Lock the tool spindle in place with the supplied lock pin.

(g) Turn the measuring grid tip counter-clockwise to remove it from drive shaft head.

SUBTASK 72-30-00-320-003-G00

(3) Blend each blisk and blade airfoil damaged area that is permitted for repair as follows:

NOTE: Use Figure 802 as reference for the limits on the number of airfoils and adjacent airfoils that can be blended for each stage. Blends cannot extend into area A or DIM F, and cannot exceed depth limits. All cracks are not permitted and cannot be blended.

(a) Get the correct blending tip for the size and/or the shape of the blisk and blade damaged area (Figure 807 or Figure 808).

(b) If you will use the borescope blend tool, SPL-2332, do these steps (Figure 807):

1) Put the drive shaft head in the 90 degree position.

a) Make sure that the blending tip with the drive shaft head is in the 90 degree position.

NOTE: If the tip with the drive shaft head is in a different position, the drive belt will disengage.

NOTE: Do not use tips that have a red colored base. These tips are not for use on the borescope blend tool, SPL-2332.

2) Lock the tool spindle in place with the supplied lock pin.

3) Install the grind tip by hand tightening in the clockwise direction.

4) With the blending tip in the 90 degree position, move the tool until it points away from you.

5) Briefly operate the tool motor as follows:

a) Turn on the drive motor using the ON/OFF switch.

b) Make sure that the blending tip turns in the counter-clockwise direction when looking at the tool tip (Figure 805).

NOTE: The counter-clockwise direction will tighten the tip during operation.

c) Turn off the drive motor.

d) If the tool tip turns in the clockwise direction, use the tool manufacturer's instruction to reverse the direction to counter-clockwise.

6) Put the drive shaft head in the 0 degree position.

(c) If you will use the borescope blend tool, SPL-14908, do these steps:

1) Make sure the tool is in the 0 degree (straight) position.

2) Use the blending tip tool (BKT-C) to loosen the collar from the collet end to insert the tip with threads facing out (Figure 806).

CAUTION

MAKE SURE THAT YOU DO NOT BEND THE SHAFT WHEN YOU SCREW ON THE TIP OR HOLD THE THREADED SHAFT. DAMAGE TO THE SHAFT WILL CAUSE TOO MUCH TIP VIBRATION.

3) Tighten the collar to let the collet end hold the tip.

NOTE: Use only tips with a red band at its bottom. The tips with the red band are for this tool only.

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4) Use the blending shaft grip tool (BKT-T) to hold the first thread and secure the threaded section of the shaft.

NOTE: This will prevent rotation of the threaded section.

5) Use the BKT-C with the tip on it to hand tighten the blending tip to the shaft.

NOTE: Hand tighten the blending tip until you can longer screw the tip with the tool.

6) Loosen the collar to release the grinding tip from the tool.

(d) Put the borescope drive shaft head into the compressor through the borescope access port.

(e) Turn on the borescope light source.

(f) If using the borescope blend tool, SPL-2332, put the head in the 90 degree position.

(g) Monitor the borescope image of the blending tip while you put the blending tip near the blisk or blade damaged area.

(h) Put the blending tip in the location of the damaged area with the borescope kit motor off to find the minimum blend area as follows:

1) Use the blending tip to put two marks at three times dimension A (the maximum depth of the damage) on either side of the damage center (Figure 803).

NOTE: Only blend to a depth that is necessary to completely remove the damage.

2) Make sure that the minimum blend length does not extend into area A or DIM F which extend above the root fillet.

3) The distance between the two marks will be the minimum blend length of the repair.

CAUTION

DO NOT TURN THE MOTOR TO ON WHEN YOU SIMULATE THE MOVEMENT. DAMAGE TO THE BLISK AND BLADE CAN OCCUR.

(i) Move the blending tip to simulate the movement necessary to start the blend repair across the minimum blend repair length.

1) Make sure that the necessary movements for the blending tip are known.

2) Make sure that the borescope view is in the center of the blend area and it is clear.

(j) After you do the necessary practice with the grind tool on the damage surface, do these steps:

1) Turn on the drive motor using the ON/OFF switch.

2) Use the speed control as necessary.

3) Start the initial blend movements slowly and apply only light pressure to the blisk and blade damaged area.

a) If the grind tool does not turn freely, apply less pressure to the blade edge.

4) When the blend movements become correct, increase the speed and pressure of the blending tip as necessary.

5) Make sure that you put a contour on the edge of the compressor blade and remove the sharp edges (Figure 804).

NOTE: There are no polishing tips for the borescope blend tool, SPL-14908.

a) Make sure that there are no sharp edges or high metal after the grinding operation.

6) Turn off the drive motor.

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(k) Put the borescope tool drive shaft head in the 0 degree position.

(l) Remove the blending tool from the borescope access port.

SUBTASK 72-30-00-080-001-G00

(4) If the borescope blend tool, SPL-2332, was used, remove the grinding tip from the drive shaft head as follows:

(a) Put the borescope drive shaft head in the 90 degree position.

1) Make sure that you remove the blending tip with the drive shaft head in the 90 degree position.

NOTE: If you remove the blending tip with the drive shaft head in a different position, the drive belt will disengage.

(b) Lock the tool spindle in place with the supplied lock pin.

(c) Turn the grind tip counter-clockwise to remove it from the drive shaft.

SUBTASK 72-30-00-080-002-G00

(5) If the borescope blend tool, SPL-14908, was used, remove the grinding tip from the drive shaft head as follows:

(a) Make sure that the drive shaft head is in the 0 degree (straight) position (Figure 806).

(b) Insert the blending tip into the blending tip tool.

(c) Tighten the collar at the collet end to hold the blend tip.

(d) Use the blending shaft grip tool (BKT-T) to hold the first thread and secure the threaded section of the shaft.

NOTE: This will prevent rotation of the threaded section.

(e) Use the blending tip tool to loosen and remove the blending tip.

(f) Loosen the collar to release the grinding tip from the tool.

SUBTASK 72-30-00-320-004-G00

(6) If the borescope blend tool, SPL-2332, was used, polish the blended compressor airfoil as follows (Figure 805):

NOTE: There are no polishing steps for the borescope blend tool, SPL-14908. Due to the lower rotational speeds of the tool, there is no requirement for the polishing steps.

(a) Put the borescope blend tool, SPL-2332, drive shaft head in the 90 degree position.

(b) Lock the tool spindle in place with the supplied lock pin.

(c) Install the polishing tip by hand tightening in the clockwise direction

1) Make sure that the polishing tip with the drive shaft head is in the 90 degree position.

NOTE: If the polishing tip with the drive shaft head is in a different position, the drive belt will disengage.

(d) With the polishing tip in the 90 degree position, move the tool until it points away from you.

(e) Briefly operate the tool motor as follows:

1) Turn on the drive motor using the ON/OFF switch.

2) Make sure that the blending tip turns in the counter-clockwise direction when looking at the tool tip (Figure 805).

NOTE: The counter-clockwise direction will tighten the tip during operation.

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3) Turn off the drive motor.

4) If the tool tip turns in the clockwise direction, use the tool manufacturer's instruction to reverse the direction to counter-clockwise.

(f) Put the drive shaft head in the 0 degree position.

(g) Put the borescope drive shaft head into the compressor through the correct borescope access port.

(h) Put the borescope blend tool, SPL-2332, drive shaft head in the 90 degree position.

(i) Turn the motor ON and polish the surface of the blisks and blades repaired area with the polishing tip (Figure 804).

(j) Turn off the motor.

(k) Put the borescope drive shaft head in the 0 degree position and remove the polishing tip from the borescope access port.

(l) Do a borescope inspection of the blisk and blade airfoil blended areas to make sure that you remove all damages and no sharp edges remain (TASK 72-30-00-200-801-G00).

(m) Turn OFF the borescope light source after you complete all of the repairs.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-30-00-410-003-G00

(1) Install the borescope plugs C through M in the HPC stator case assembly (TASK 72-00-00-290-801-G00).

SUBTASK 72-30-00-080-003-G00

(2) Remove the N2 rotor turning equipment (TASK 72-00-00-910-801-G00).

SUBTASK 72-30-00-410-004-G00

(3) Do this task: Accessory Gearbox Manual Control Cover Installation, TASK 72-63-04-400-801-G00.

SUBTASK 72-30-00-210-001-G00

(4) Make sure that there are no borescope electrical leads or video cables that remained on the engine assembly.

SUBTASK 72-30-00-410-005-G00

(5) Install the shroud segment 3 (TASK 72-23-04-400-801-G00).

SUBTASK 72-30-00-410-006-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(6) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-30-00-860-005-G00

(7) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

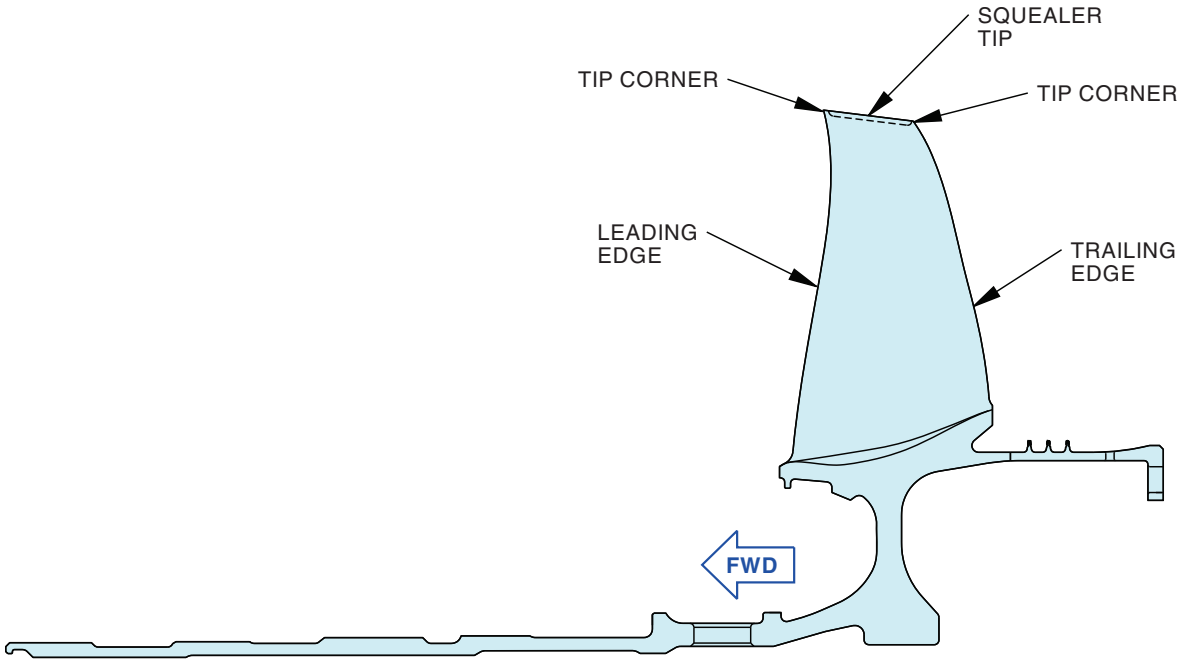
I. HPC Rotor Blades Blisks with Borescope Blending Repair Test

SUBTASK 72-30-00-710-002-G00

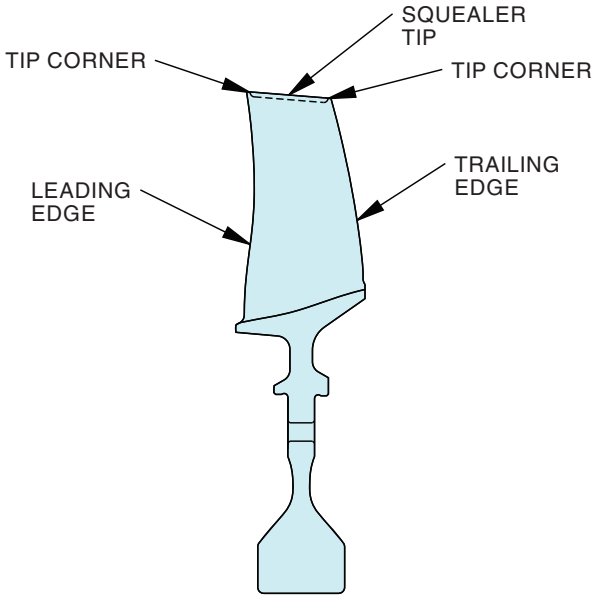
(1) Do the applicable tests given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STAGE 1 BLISK



STAGE 2 BLISK

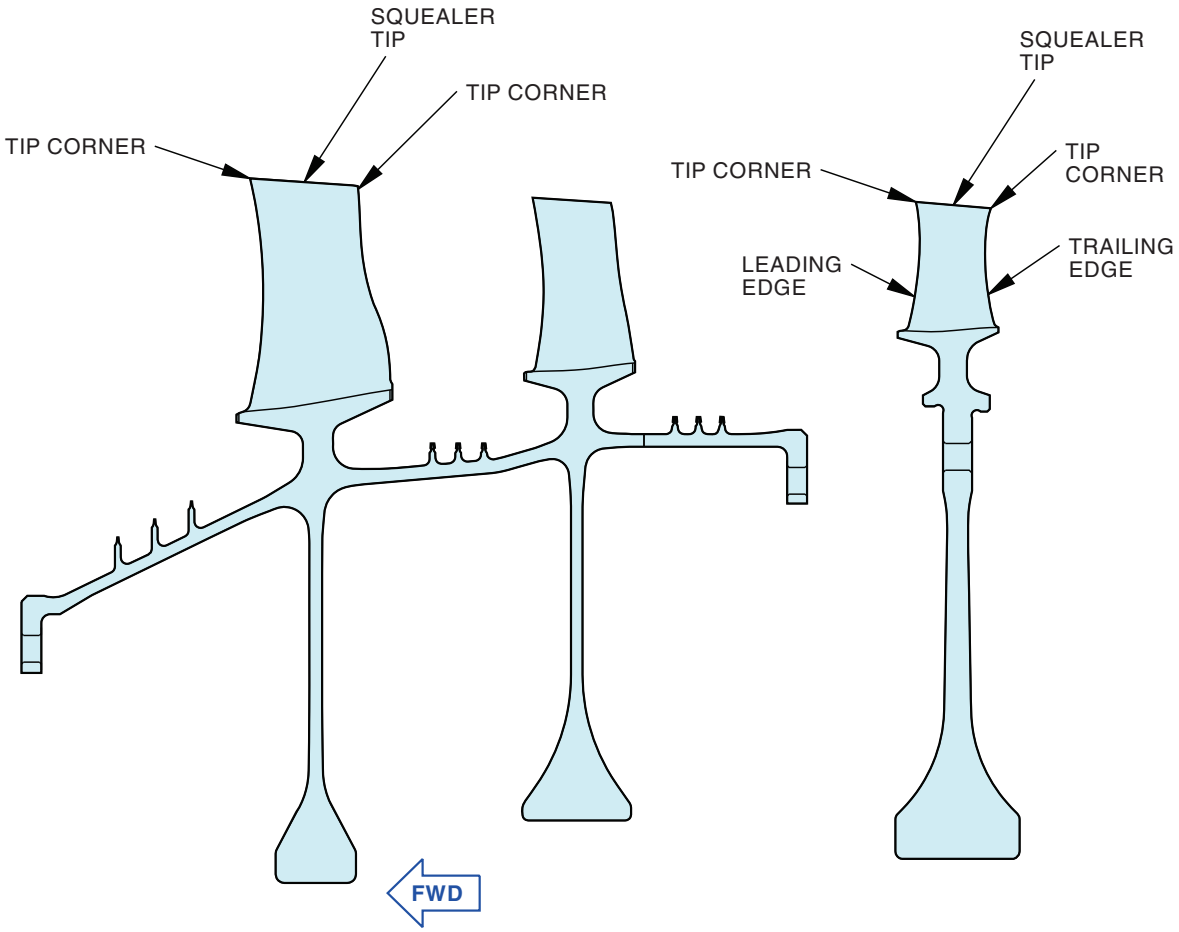
MATERIAL:
TI 17 (AIRFOIL ONLY)

5035931-00

2569038 S0000615672_V1

Nomenclature of Blisks and Blades (Stages 1-10)
Figure 801/72-30-00-990-809-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



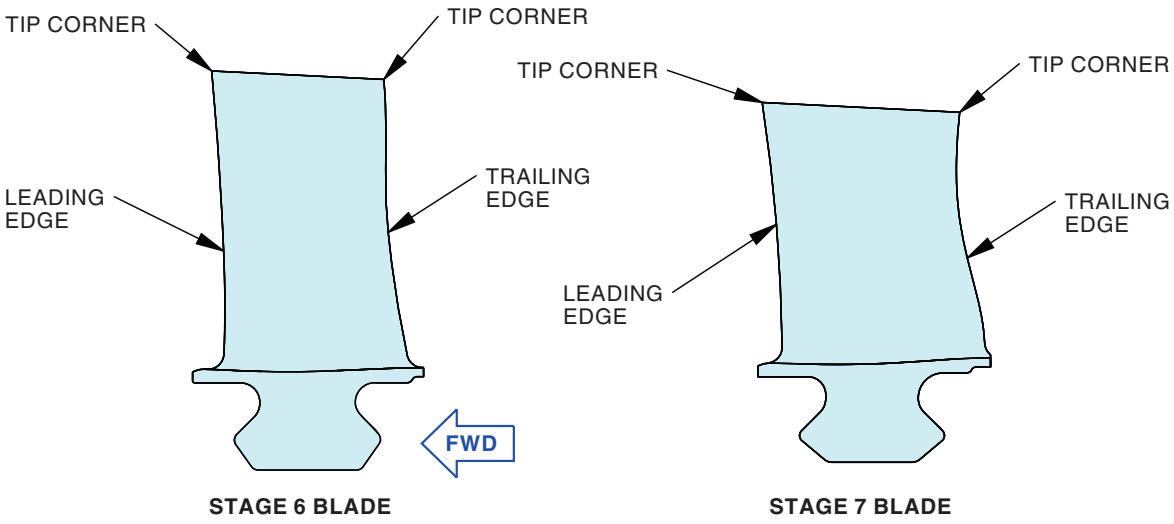
MATERIAL:
STAGE 3-4: TI 17 (AIRFOIL ONLY)
STAGE 5: DA 718

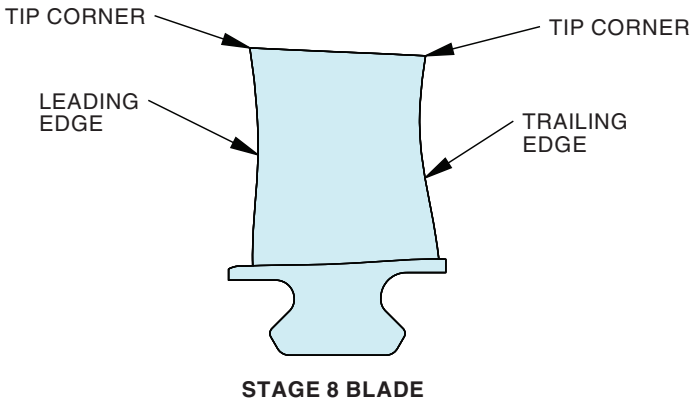
5035932-00

2569046 S0000615673_V1

Nomenclature of Blisks and Blades (Stages 1-10)
Figure 801/72-30-00-990-809-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





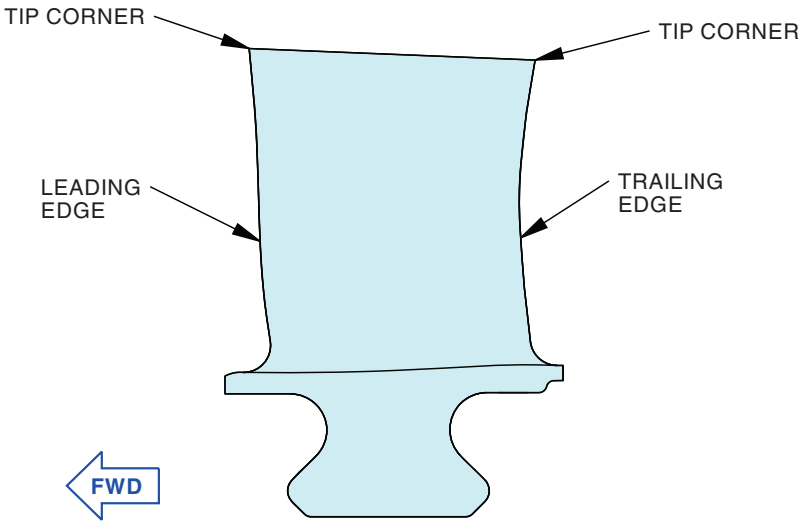
MATERIAL:
STAGE 6, 7: IN 718
STAGE 8: 718+

5035933-00

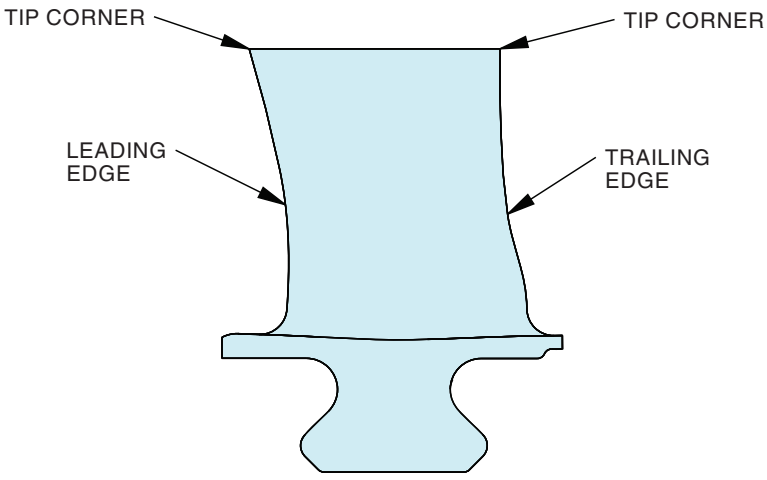
2569069 S0000615674_V1

Nomenclature of Blisks and Blades (Stages 1-10)
Figure 801/72-30-00-990-809-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STAGE 9 BLADE



STAGE 10 BLADE

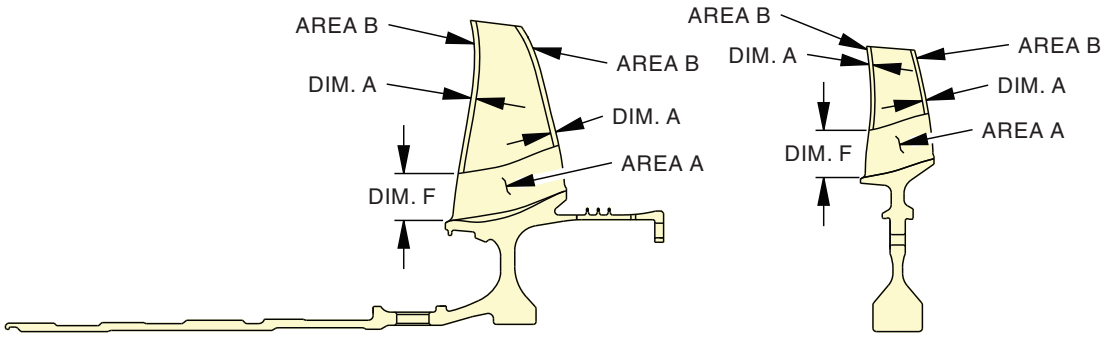
MATERIAL:
STAGE 9: 718+
STAGE 10: U720

5035934-00

2569080 S0000615675_V1

Nomenclature of Blisks and Blades (Stages 1-10)
Figure 801/72-30-00-990-809-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STAGE 1 BLISK

STAGE 2 BLISK

STAGES	DIM. A MAX BLEND REPAIR PERMITTED	DIM. F (AREA A) BLEND REPAIR NOT PERMITTED	MAXIMUM NUMBER OF BLENDS PERMITTED PER AIRFOIL	MAXIMUM NUMBER OF BLENDED AIRFOILS PERMITTED PER BLISK	MAXIMUM NUMBER OF ADJACENT BLENDED AIRFOILS
1 1	0.05 INCH (1.3 mm)	0.850 INCH (21.59 mm)	2	3	2
2 1	0.05 INCH (1.3 mm)	0.850 INCH (21.59 mm)	2	3	2
2	0.05 INCH (1.3 mm)	0.545 INCH (13.84 mm)	2	3	2

1 PRE LEAP-1B SERVICE BULLETIN 72-0182, 25 AIRFOIL COUNT

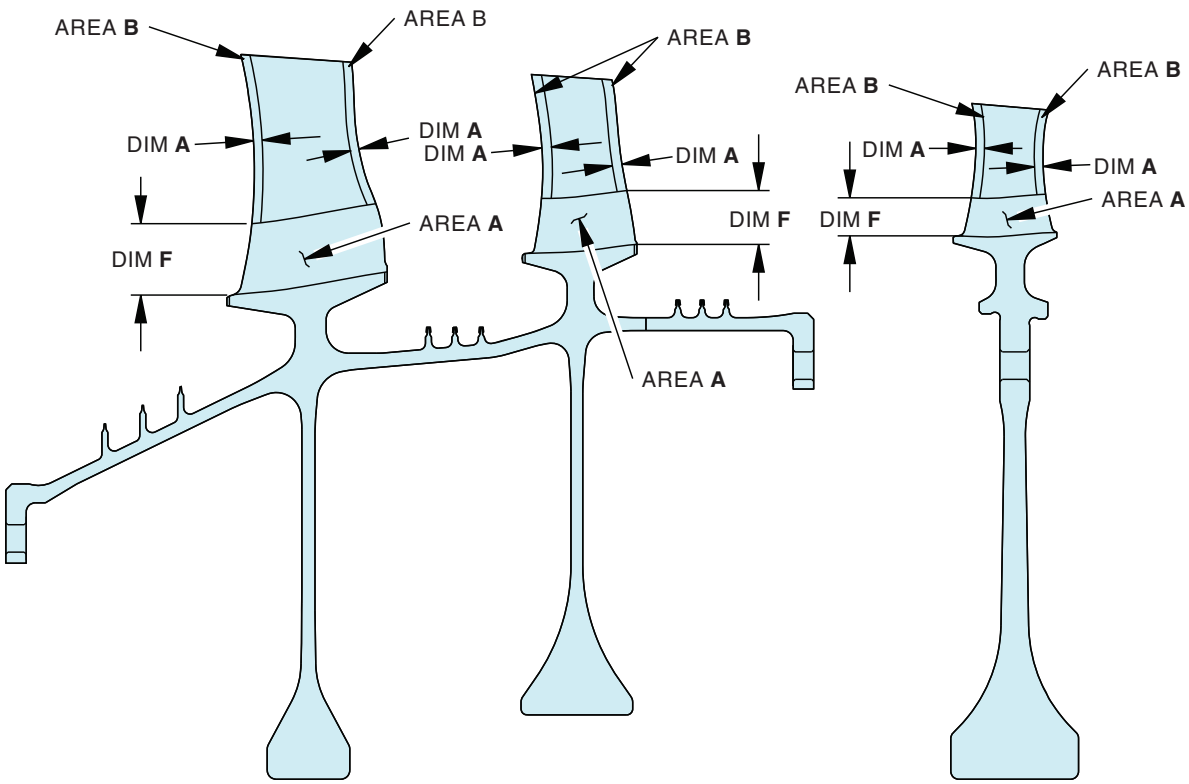
2 POST LEAP-1B SERVICE BULLETIN 72-0182, 21 AIRFOIL COUNT

5035915-01

2569423 S0000615676_V3

Blend Repair Areas (Stages 1-10)
Figure 802/72-30-00-990-810-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STAGES 3 AND 4 BLISKS

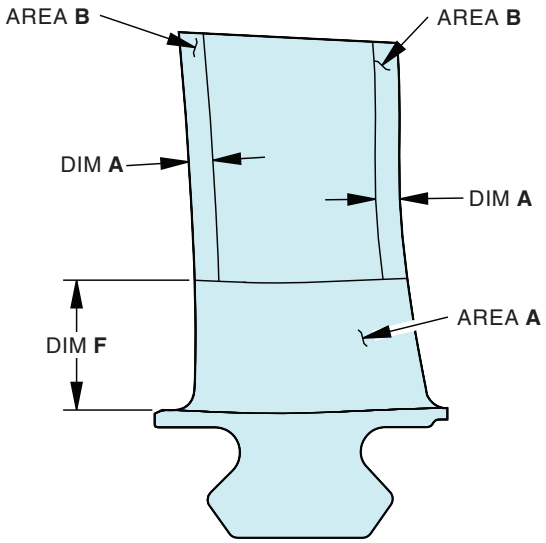
STAGE 5 BLISK

STAGES	DIM. **A** MAX BLEND REPAIR PERMITTED	DIM. **F** (AREA **A**) BLEND REPAIR NOT PERMITTED	MAXIMUM NUMBER OF BLENDS PERMITTED PER AIRFOIL	MAXIMUM NUMBER OF BLENDED AIRFOILS PERMITTED PER BLISK	MAXIMUM NUMBER OF ADJACENT BLENDED AIRFOILS
3	0.05 IN. (1.3 mm)	0.375 IN. (9.53 mm)	2	3	2
4	0.05 IN. (1.3 mm)	0.295 IN. (7.49 mm)	2	4	2
5	0.05 IN. (1.3 mm)	0.220 IN. (5.59 mm)	2	4	2

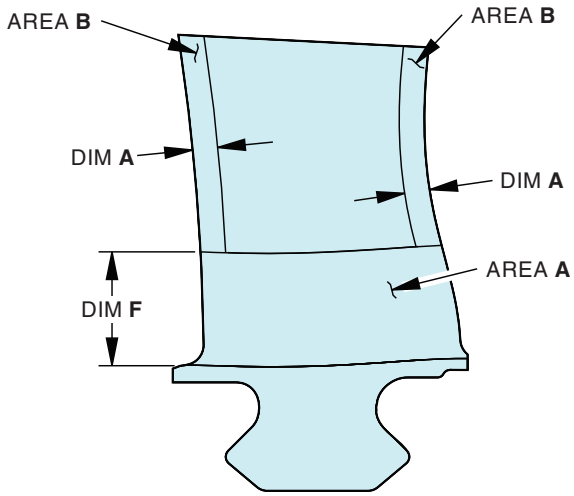
5035916-00
2569506 S0000615677_V1

Blend Repair Areas (Stages 1-10)
Figure 802/72-30-00-990-810-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STAGES 6 BLADE



STAGE 7 BLADE

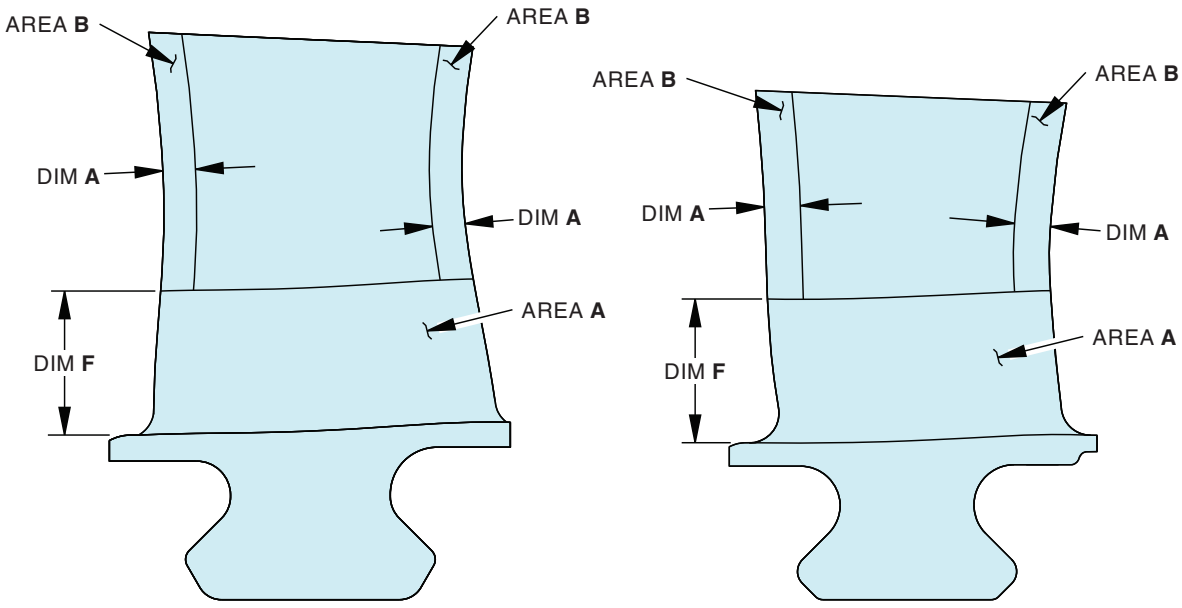
STAGES	DIM. **A** MAX BLEND REPAIR PERMITTED	DIM. **F** (AREA **A**) BLEND REPAIR NOT PERMITTED	MAXIMUM NUMBER OF BLENDS PERMITTED PER AIRFOIL	MAXIMUM NUMBER OF BLENDED AIRFOILS PERMITTED PER STAGE	MAXIMUM NUMBER OF ADJACENT BLENDED AIRFOILS
6	0.05 IN. (1.3 mm)	0.090 IN. (2.29 mm)	1	4	4
7	0.05 IN. (1.3 mm)	0.090 IN. (2.29 mm)	1	4	4

5035917-00

2569638 S0000615678_V3

Blend Repair Areas (Stages 1-10)
Figure 802/72-30-00-990-810-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

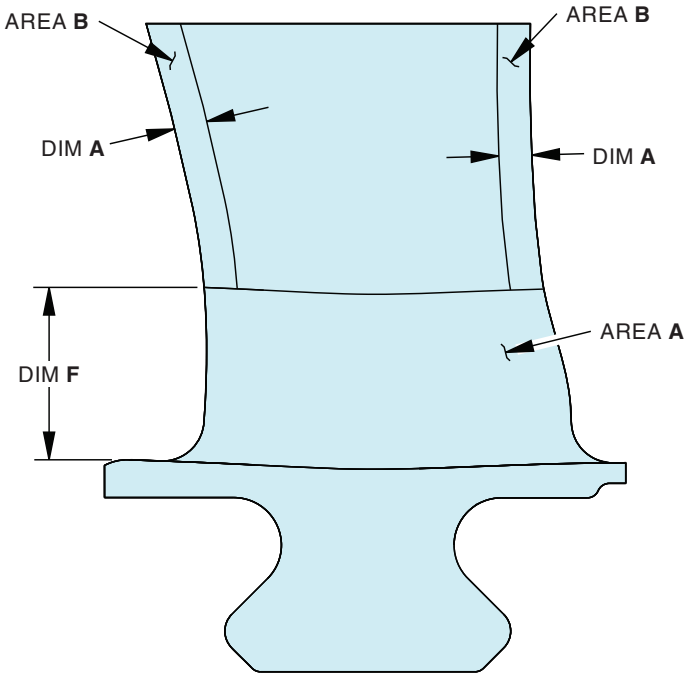


STAGES	DIM. **A** MAX BLEND REPAIR PERMITTED	DIM. **F** (AREA **A**) BLEND REPAIR NOT PERMITTED	MAXIMUM NUMBER OF BLENDS PERMITTED PER AIRFOIL	MAXIMUM NUMBER OF BLENDED AIRFOILS PERMITTED PER STAGE	MAXIMUM NUMBER OF ADJACENT BLENDED AIRFOILS
8	0.05 IN. (1.3 mm)	0.090 IN. (2.29 mm)	1	4	4
9	0.05 IN. (1.3 mm)	0.090 IN. (2.29 mm)	1	4	4

5035918-00
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Blend Repair Areas (Stages 1-10)
Figure 802/72-30-00-990-810-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STAGE 10 BLADE

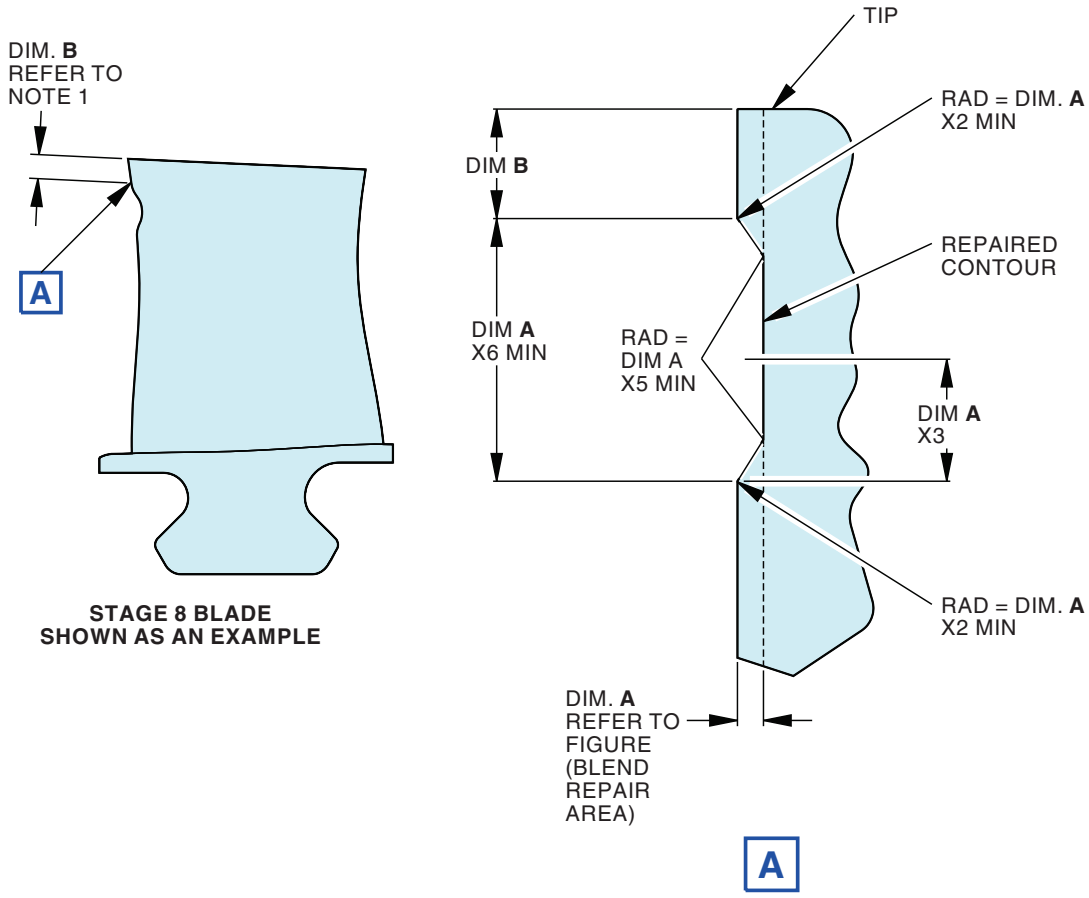
STAGES	DIM. **A** MAX BLEND REPAIR PERMITTED	DIM. **F** (AREA **A**) BLEND REPAIR NOT PERMITTED	MAXIMUM NUMBER OF BLENDS PERMITTED PER AIRFOIL	MAXIMUM NUMBER OF BLENDED AIRFOILS PERMITTED PER STAGE	MAXIMUM NUMBER OF ADJACENT BLENDED AIRFOILS
10	0.05 IN. (1.3 mm)	0.090 IN. (2.29 mm)	1	4	4

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Blend Repair Areas (Stages 1-10)
Figure 802/72-30-00-990-810-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



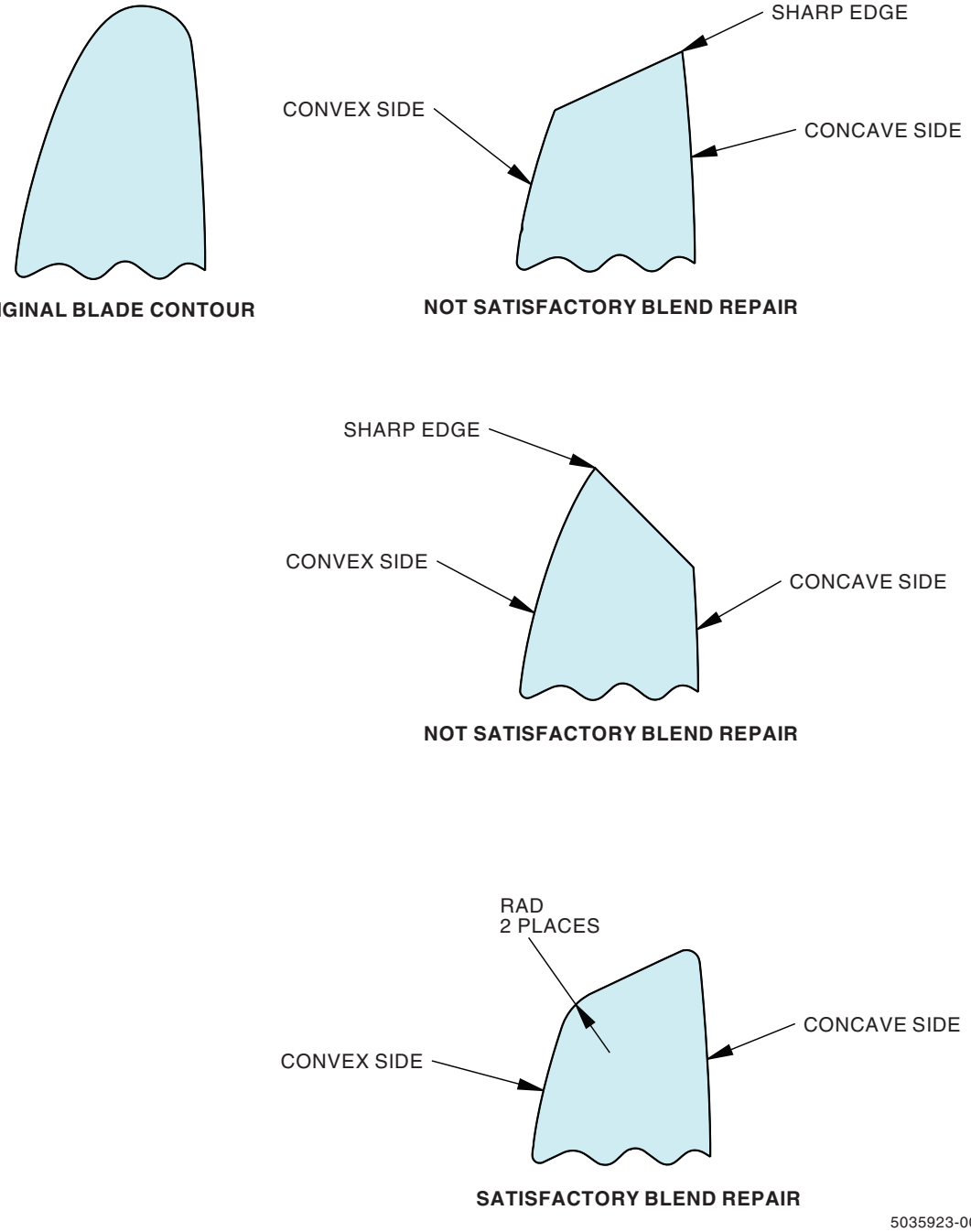
NOTE:

1. WHEN DIMENSION **B** (THE DISTANCE BETWEEN THE END OF THE BLEND REPAIR AREA AND THE AIRFOIL TIP) IS LESS THAN 0.100 IN. (2.54 mm). USE THE BLENDING TIP TO REMOVE THE DIMENSION **B** AREA.
2. RADIUS DIM. **A** x2 AND DIM. **A** x5 MUST BE TANGENT AND/OR LINKED TOGETHER WITHOUT STEP.
3. RADIUS DIM. **A** x2 MUST BE TANGENT WITH LEADING EDGE.

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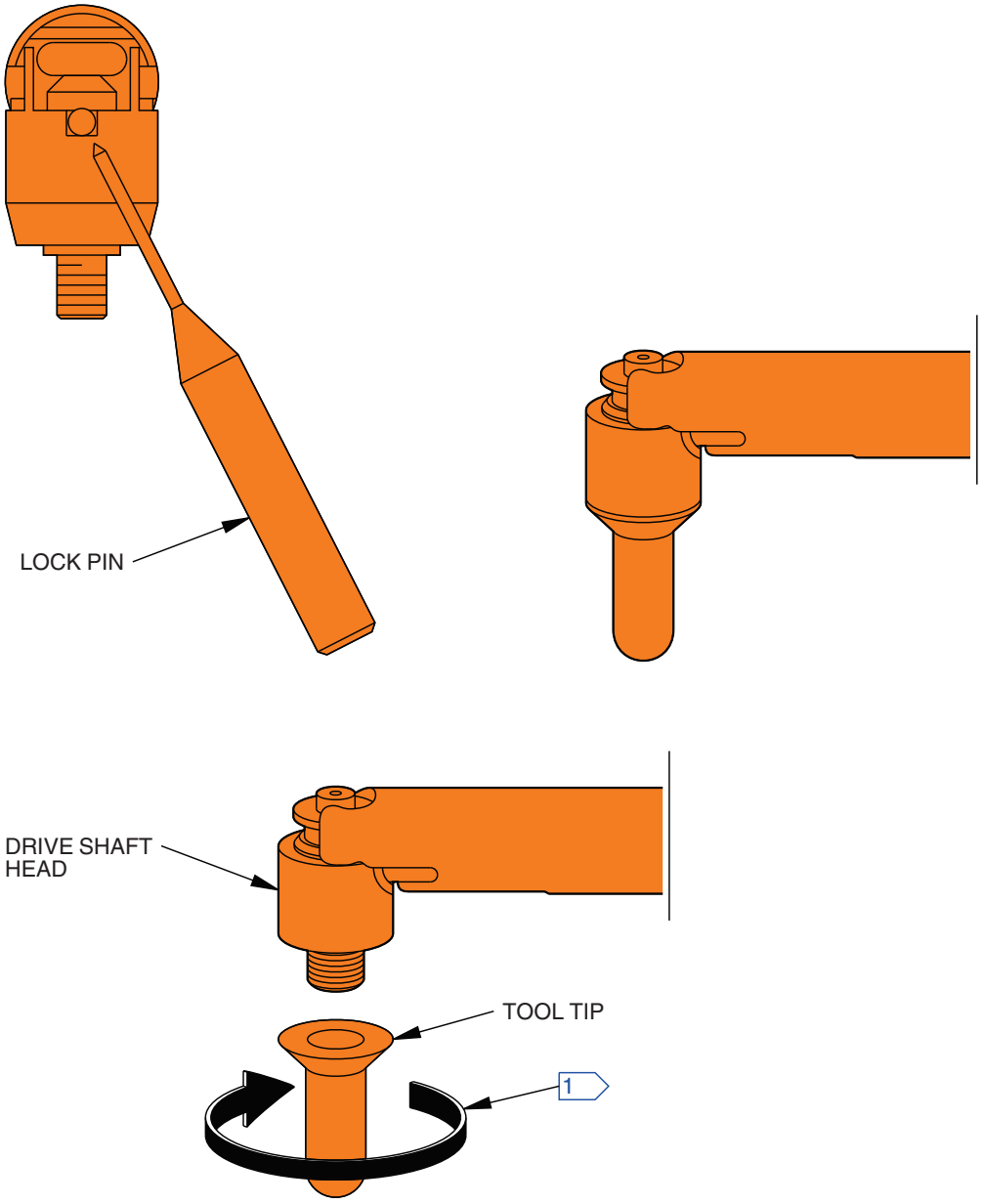
Minimum Blend Length of the Repaired Area
Figure 803/72-30-00-990-811-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Leading Edge Blend Repair Contours
Figure 804/72-30-00-990-812-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

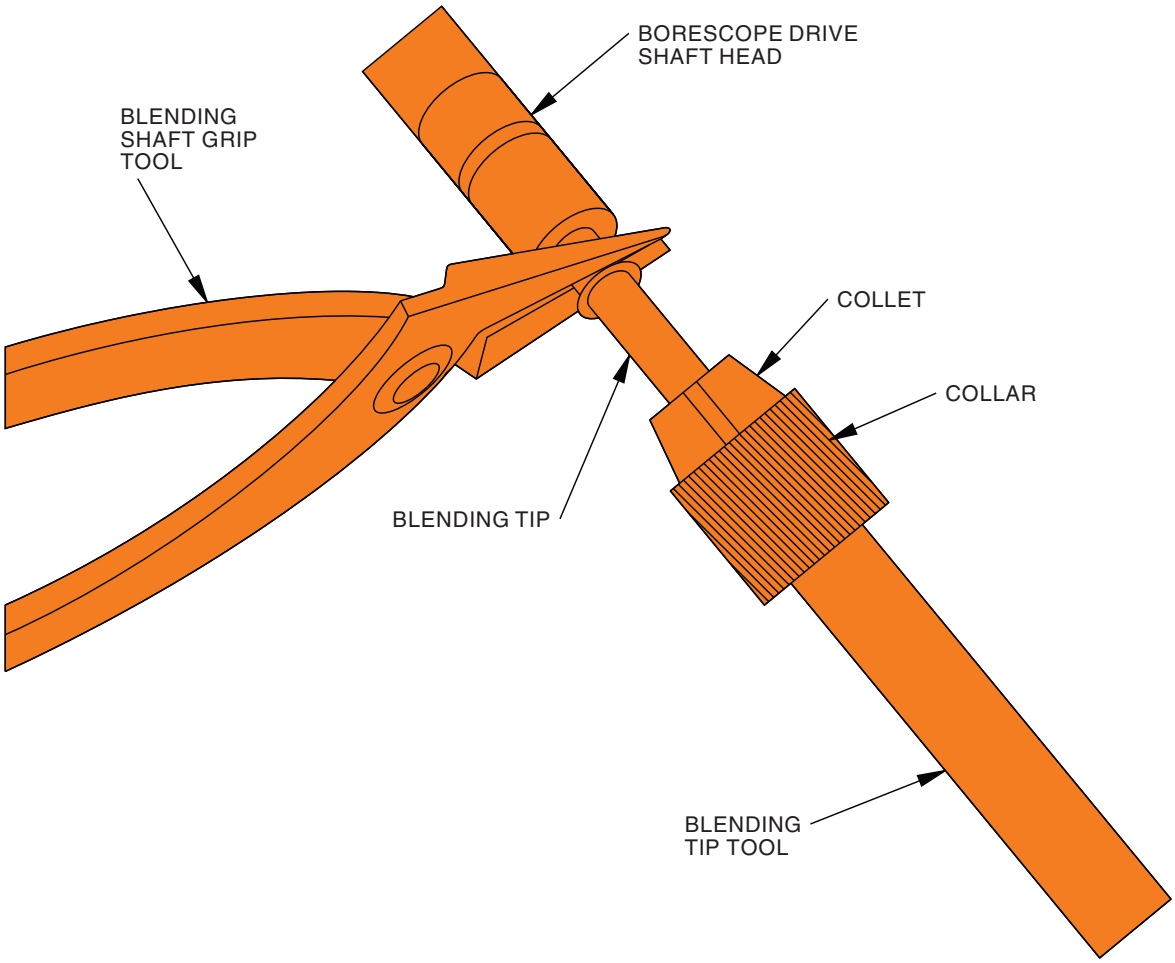
A TOOL TIP ALWAYS NEEDS TO BE INSTALLED (EVEN DURING STORAGE) TO PREVENT THE DRIVE BELT FROM DISENGAGING.

1 ROTATION OF BORESCOPE BLENDING TOOL TIP DURING OPERATION (AFTER INSTALLATION)

5035927-00
2569884 S0000615683_V2

9471M44 Borescope Blending Tool Tip
Figure 805/72-30-00-990-813-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

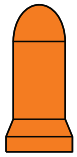
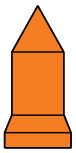
MAKE SURE THE TIP OF THE SCOPE IS STRAIGHT. HOLD THE DRIVE SHAFT WITH THE BLENDING SHAFT GRIP TOOL AND USE THE BLENDING TIP TOOL TO HAND TIGHTEN THE BLENDING TIP ON THE THREADED SHAFT.

5035928-00

2569900 S0000615686_V1

956A1604 Borescope Blending Tool Tip Installation
Figure 806/72-30-00-990-814-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

			
LENGTH **L**	DIAMOND ROUNDED CUTTER (FORM NO. 6)	DIAMOND CONE CUTTER (FORM NO. 7)	ABRASIVE POLISHER (FORM NO. 19)
4 mm (0.16 IN)	X		
5 mm (0.2 IN)	X		
6 mm (0.24 IN.)	X	X	
7 mm (0.28 IN.)		X	
8 mm (0.32 IN.)	X	X	
10 mm (0.4 IN.)	X	X	
12 mm (0.48 IN.)	X	X	
13 mm (0.52 IN.)			X
15 mm (0.6 IN.)	X	X	X
17 mm (0.68 IN.)			X
20 mm (0.8 IN.)	X	X	X
23 mm (0.92 IN.)			X
25 mm (1 IN.)	X	X	
26 mm (1.04 IN.)			X
29 mm (1.16 IN.)			X
30 mm (1.2 IN.)	X	X	
32 mm (1.28 IN.)			X
35 mm (1.4 IN.)	X	X	X
40 mm (1.6 IN.)			

NOTE:

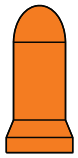
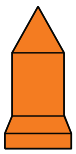
1. "X" DENOTES AN ACCEPTABLE USE TIP.
2. MAKE SURE YOU DO NOT USE A TOOL TIP WITH A RED COLOR BAND AT THE BASE WHICH IS FOR THE FLEXIBLE TOOL (956A1604).

5035929-00

2569976 S0000615684_V2

Tool Tips for the 9471M44 Rigid Borescope Blending Tool
Figure 807/72-30-00-990-815-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

			
LENGTH **L**	DIAMOND ROUNDED CUTTER (FORM NO. 6)	DIAMOND CONE CUTTER (FORM NO. 7)	ABRASIVE POLISHER (FORM NO. 19)
4 mm (0.16 IN)	X		
5 mm (0.2 IN)	X		
6 mm (0.24 IN.)	X	X	
7 mm (0.28 IN.)		X	
8 mm (0.32 IN.)	X	X	
10 mm (0.4 IN.)		X	

NOTE:

1. "X" DENOTES AN ACCEPTABLE USE TIP.
2. THERE ARE NO POLISHING TIPS AVAILABLE FOR THIS TOOL.
3. MAKE SURE THE TOOL TIP HAS A RED COLOR BAND AT THE BASE.

5035930-00

2569981 S0000615685_V2

Tool Tips for the 956A1604 Flexible Borescope Blending Tool
Figure 808/72-30-00-990-816-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

COMBUSTION CASE - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Combustion Case Inspection.

TASK 72-40-00-200-801-G00

2. Combustion Case Inspection

(Figure 601)

A. General

(1) This task gives the instructions to examine the combustion case.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-48-02-020-801-G00	Ring Locked Self Locking Insert Removal (P/B 201)
70-48-13-960-801-G00	Key-Locked Insert Replacement (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-290-805-G00	Combustion Section and High Pressure Turbine Section Stage One Nozzle Leading Edge Detailed Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-40-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-40-00-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-40-00-860-005-G00

(3) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-40-00-040-001-G00

(4) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-40-00-010-001-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Combustion Case Inspection

SUBTASK 72-40-00-220-001-G00

(1) Examine the combustion case forward and aft flanges for these conditions:

(a) Bolt hole radial cracks

1) Not permitted.

(b) Nicks, scores, and scratches

1) Remove the high metal (TASK 70-43-42-300-802-G00).

2) Damage that continues to stay after you remove the high metal is permitted if the depth is less than 0.005 in. (0.127 mm).

SUBTASK 72-40-00-220-002-G00

(2) Examine the combustion case for these conditions:

(a) Cracks in the parent metal

1) Not permitted.

(b) Discoloration

1) Discoloration is permitted only if the metal is not crazed.

2) Do a borescope inspection of the combustor case inner surface and the combustor chamber when there are hot spots, and bulges in the outer case (TASK 72-00-00-290-805-G00).

SUBTASK 72-40-00-020-001-G00

(3) Examine the combustor case pads, bores, and ports when the line replaceable units are removed for other maintenance for these conditions:

(a) Igniter ports, ECS (Environmental Control System) ports, and TBV (Transient Bleed Valve) port (includes ports, bores, and pads) for:

1) Cracks:

a) Not permitted.

2) Nicks, scores, and scratches (does not include insert holes):

a) Remove the high metal (TASK 70-43-42-300-802-G00).

b) Any amount up to 0.005 in. (0.127 mm) in depth, if damage does not go across more than 25% of the sealing surface.

3) Galling or pickup on the threads of the threaded insert at the igniter ports (.250-28 threads):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Five threads (cumulative or continuous) must remain of the first seven entry threads.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0082 (COMBUSTOR CASE WITH RING LOCK INSERTS)

4) Loose or missing threaded inserts:

a) Loose

<1> Turning until locking ring engages is permitted.

b) Missing

<1> Not permitted.

c) Replace the damaged inserts only (TASK 70-48-02-020-801-G00).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0082 (COMBUSTOR CASE WITH KEY LOCK INSERTS)

5) Loose or missing threaded inserts:

a) Loose

<1> Turning insert and pushing protruding locking keys is permitted only when insert and all keys are not damaged.

b) Missing

<1> Not permitted.

c) Replace the damaged inserts only (TASK 70-48-13-960-801-G00).

PTF ALL

6) Galling or pickup on the threads of the threaded insert at the ECS ports and TBV ports (.3125-24 threads):

a) Six threads (cumulative or continuous) must remain along the full insert length.

7) Galling or pickup on the threads of the threaded insert at the TBV ports (.250-28 threads):

a) Five threads (cumulative or continuous) must remain along the full insert length.

(b) Examine the PS3/T3 sensor ports for:

1) Cracks:

a) Not permitted.

2) Galling or pickup on the threads:

a) Permitted on two threads all along the thread section.

3) Nicks, scores, and scratches:

a) Remove the high metal (TASK 70-43-42-300-802-G00).

b) Any amount up to 0.005 in. (0.127 mm) in depth, if damage does not go across more than 25% of the sealing surface.

(c) Examine the fuel nozzle ring, fuel nozzle ring pads, fuel nozzle ring ports, and Modulated Turbine Cooling (MTC) pads and ports for:

1) Cracks, nicks, scores, and scratches:

a) Not permitted.

2) Galling of pickup on the threads on the threaded insert:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Five threads (cumulative or continuous) must remain along the full insert length.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0082 (COMBUSTOR CASE WITH RING LOCK INSERTS)

3) Loose or missing threaded inserts:

a) Loose

<1> Turning until locking ring engages is permitted.

b) Missing

<1> Not permitted.

c) Replace the damaged inserts only (TASK 70-48-02-020-801-G00).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0082 (COMBUSTOR CASE WITH KEY LOCK INSERTS)

4) Loose or missing threaded inserts:

a) Loose

<1> Turning insert and pushing protruding locking keys is permitted only when insert and all keys are not damaged.

b) Missing

<1> Not permitted.

c) Replace the damaged inserts only (TASK 70-48-13-960-801-G00).

PTF ALL

SUBTASK 72-40-00-220-003-G00

(4) If you remove the plugs for other maintenance, examine the combustor case borescope ports (TASK 72-00-00-290-805-G00).

SUBTASK 72-40-00-220-004-G00

(5) Examine areas that you can see of the combustor thermal insulation blankets (insulation) for these conditions:

(a) Discoloration

1) Any amount.

(b) Varnish film (soot or oil buildup) (external)

1) Any amount.

(c) Burning (charring)

1) Not permitted.

(d) Missing and loose capstan

1) Not permitted.

(e) Cuts, tears, and holes

1) For forward insulation blankets (SIN: 12090), (SIN: 12091), and (SIN: 12092) and aft thermal insulation blankets (SIN: 12093), (SIN: 12094), and (SIN: 12095):

a) In Area A of the forward thermal insulation blankets, the axial length of missing material must not be more than0.60 in. (15.2 mm) from the edge.

b) Missing material is not permitted in area B and area C.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) In the other areas (excluding Area A, Area B, and Area C), the total area of missing material over the forward insulation blankets and aft thermal insulation blankets must not be more than 15 in^2 (96.8 cm^2).

d) The area of a single location of missing material must not be more than 5 in^2 (32.3 cm^2).

e) When there are multiple locations of missing material, the spacing between them must be at least three times the maximum dimension of a missing material location.

f) Missing material across the full axial length of thermal insulation blankets is not permitted.

2) Continue-In-Service (C-I-S) Limits

NOTE: This over serviceable limit extension is for a one time Continue-In-Service (C-I-S) application.

a) Total missing material up to 71 in^2 (458.1 cm^2) over the forward and aft thermal insulation blankets is acceptable to C-I-S for 1000 flight cycles.

b) Missing material is permitted in Area A and Area B.

c) Missing material is not permitted in Area C.

(f) Looseness

1) Any amount, if the insulation can be correctly installed.

(g) Dents in the outer skin

1) Any amount, if dents are less than 50 percent of the thermal blanket thickness.

(h) Distortion caused by trapped air

1) Not permitted.

SUBTASK 72-40-00-220-005-G00

(6) If exposed, examine the MTC manifold assembly for these conditions:

(a) Broken support straps or cracks in support straps located at 5 O'clock and 11 O'clock aft looking forward

1) Permitted.

(b) Broken straps or cracks in support straps at other locations

1) Not permitted.

2) Continue-In-Service limit: 18000 flight cycles is permitted.

NOTE: This over serviceable limit extension is for a one-time Continue-In-Service (C-I-S) application.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-40-00-440-001-G00

(1) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-40-00-410-001-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-40-00-860-003-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-40-00-860-004-G00

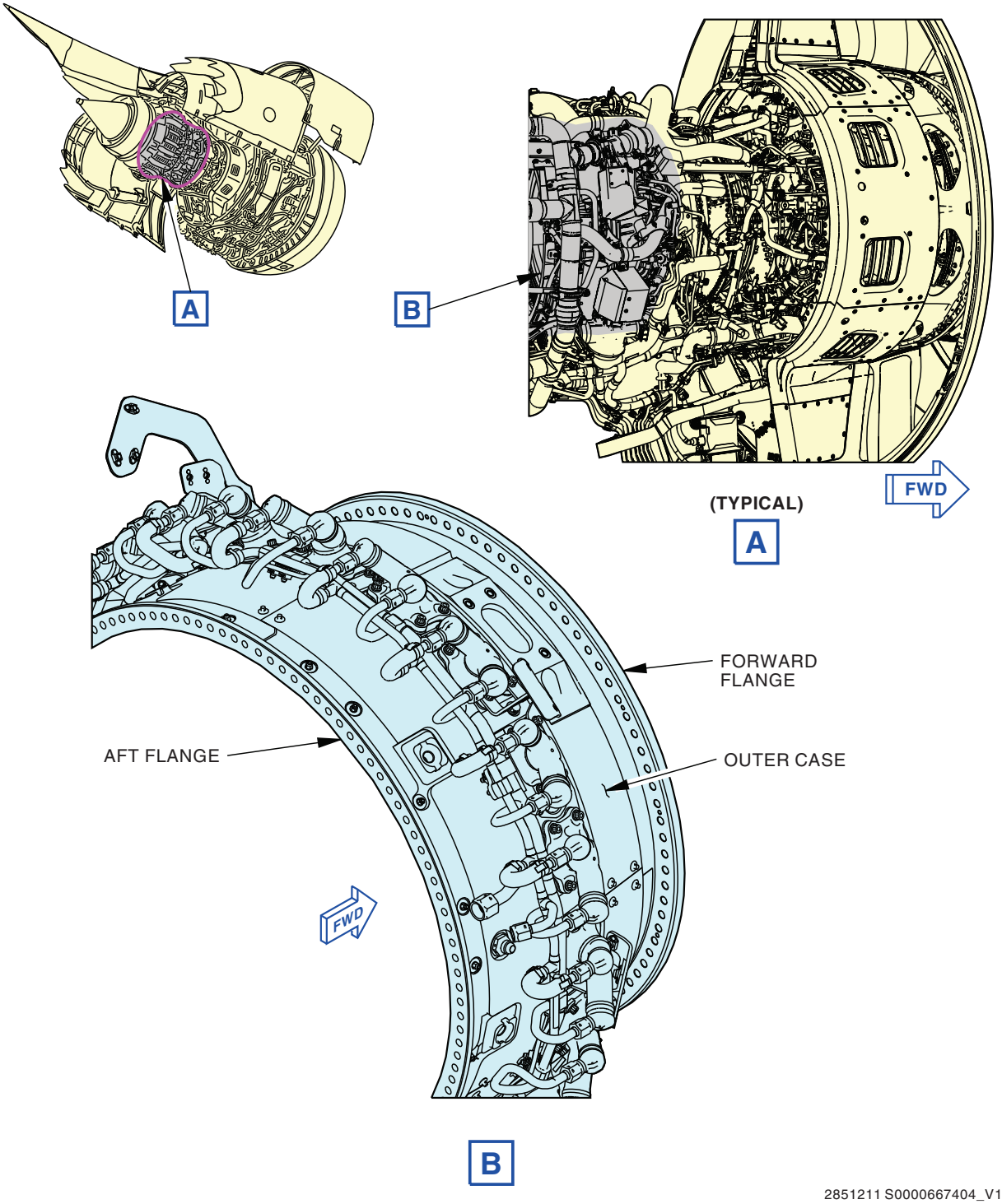
(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

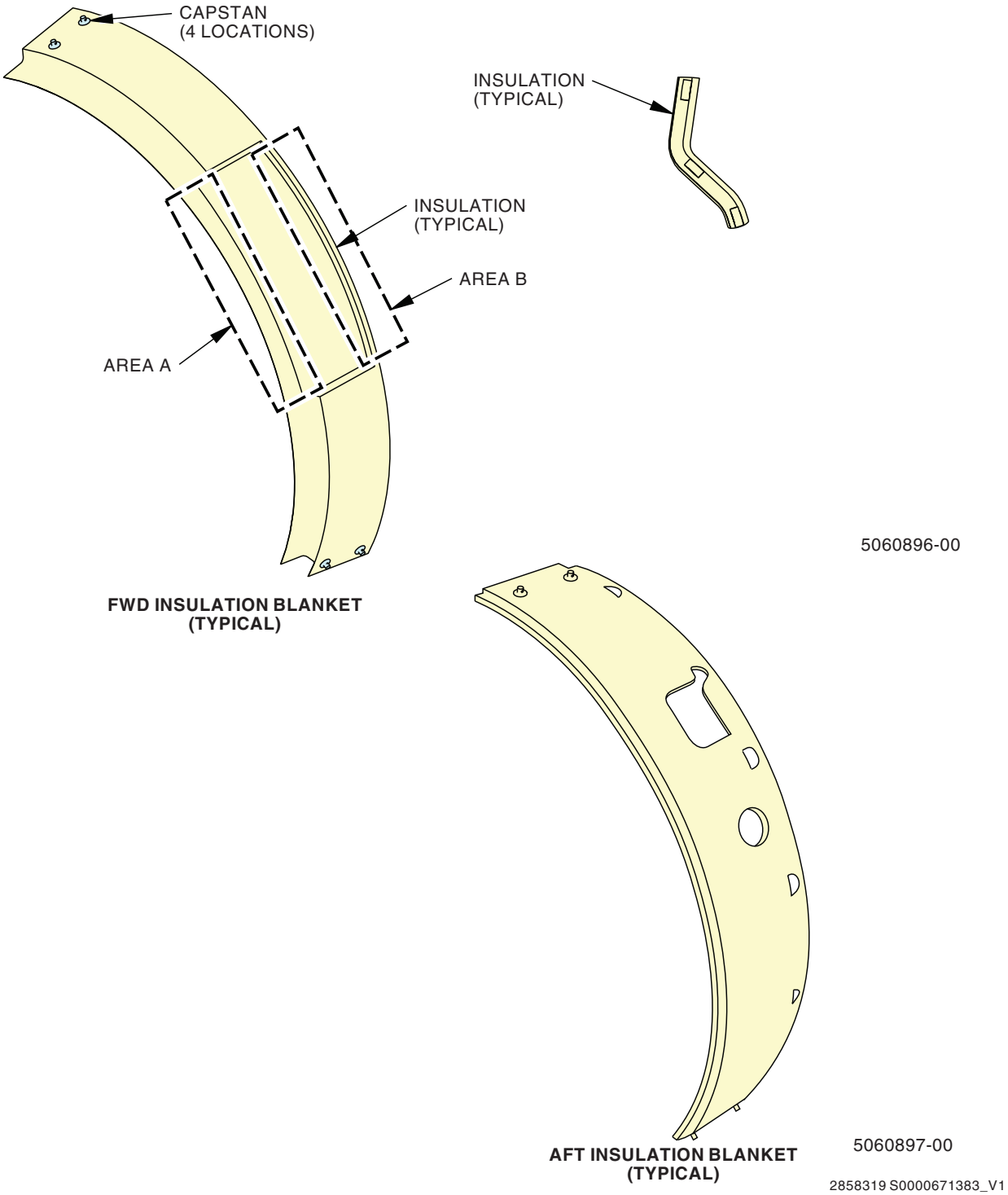
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



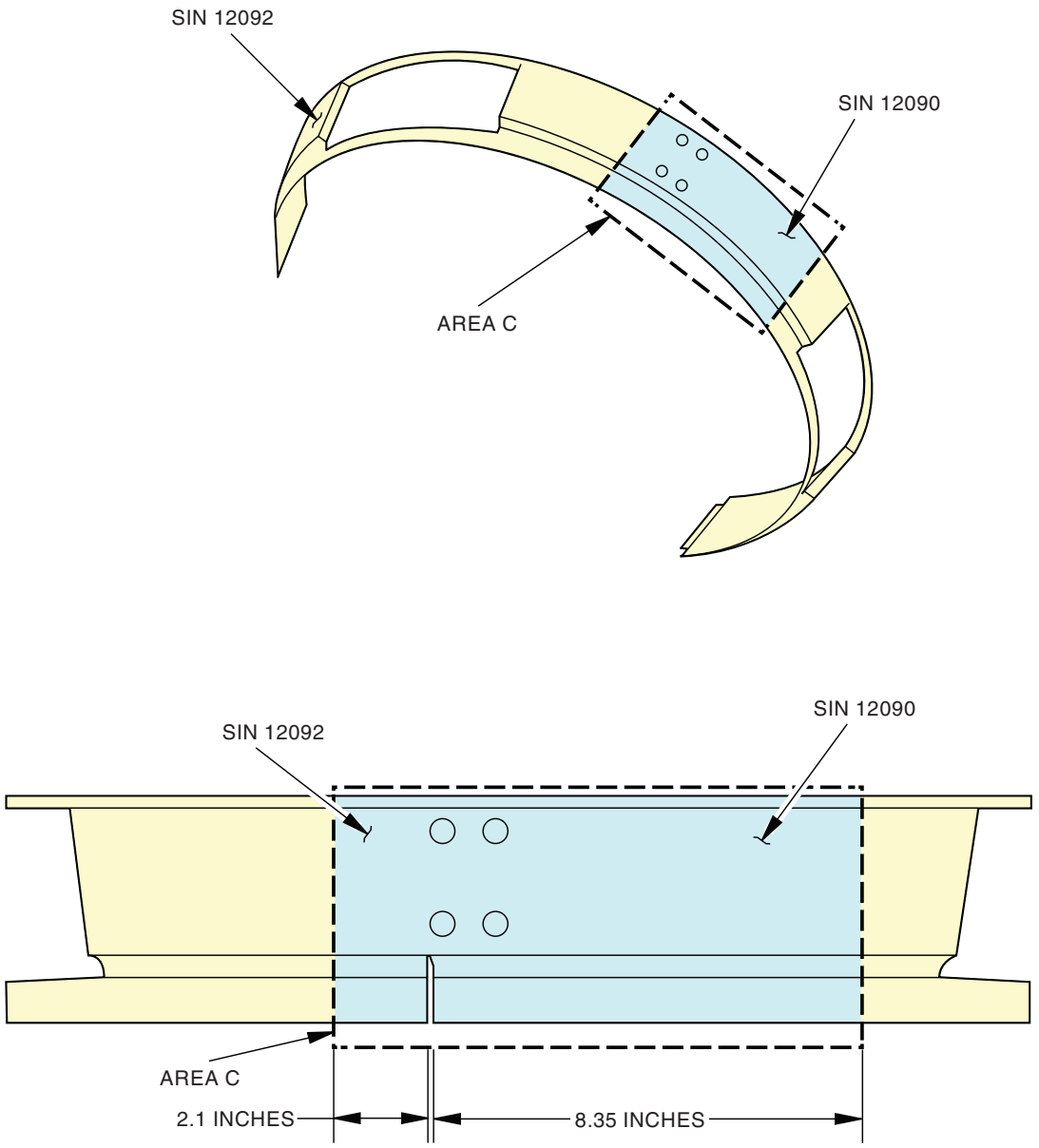
Combustion Case Inspection
Figure 601/72-40-00-990-802-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Combustion Case Inspection
Figure 601/72-40-00-990-802-G00 (Sheet 2 of 5)

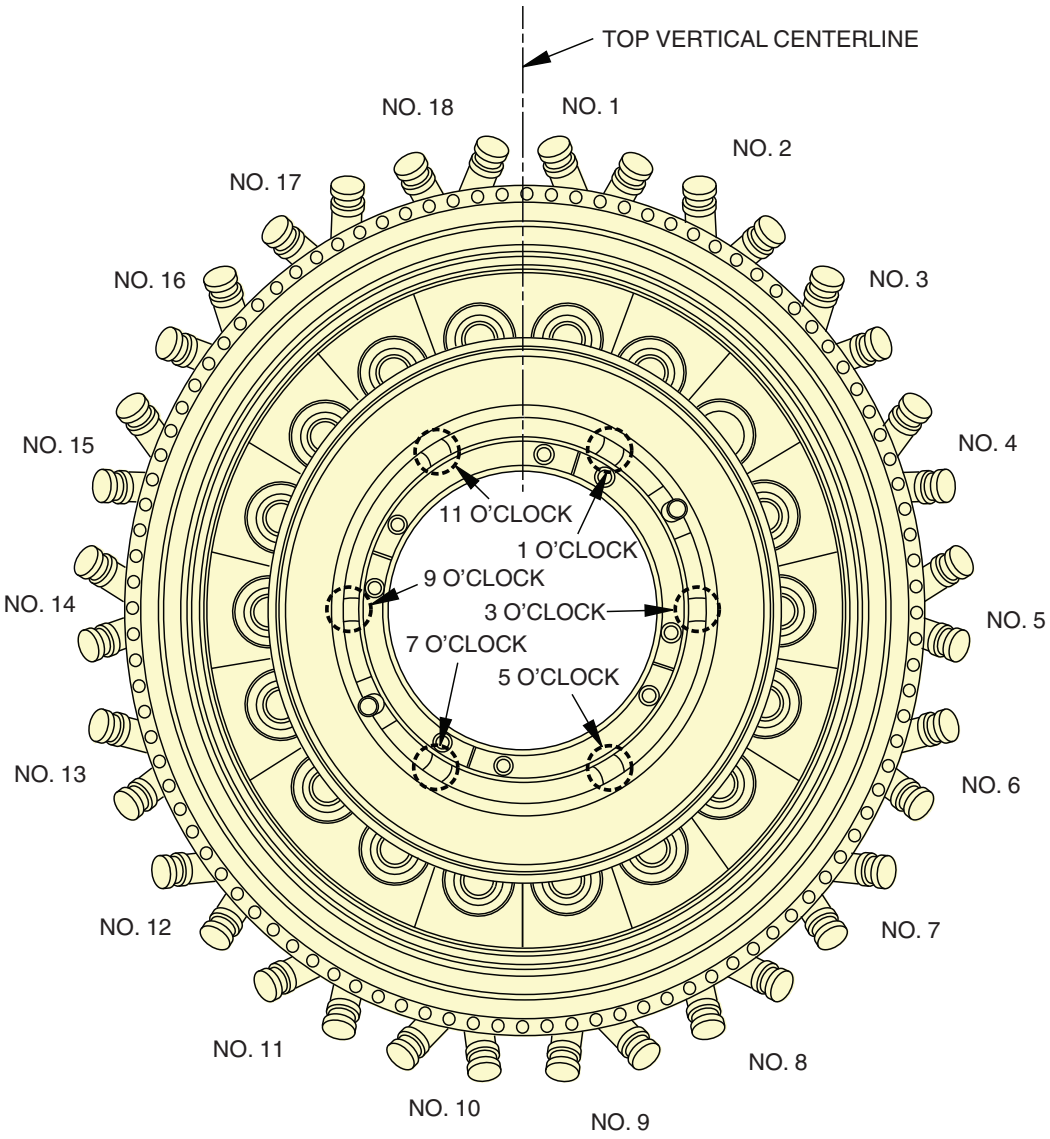
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Combustion Case Inspection
Figure 601/72-40-00-990-802-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

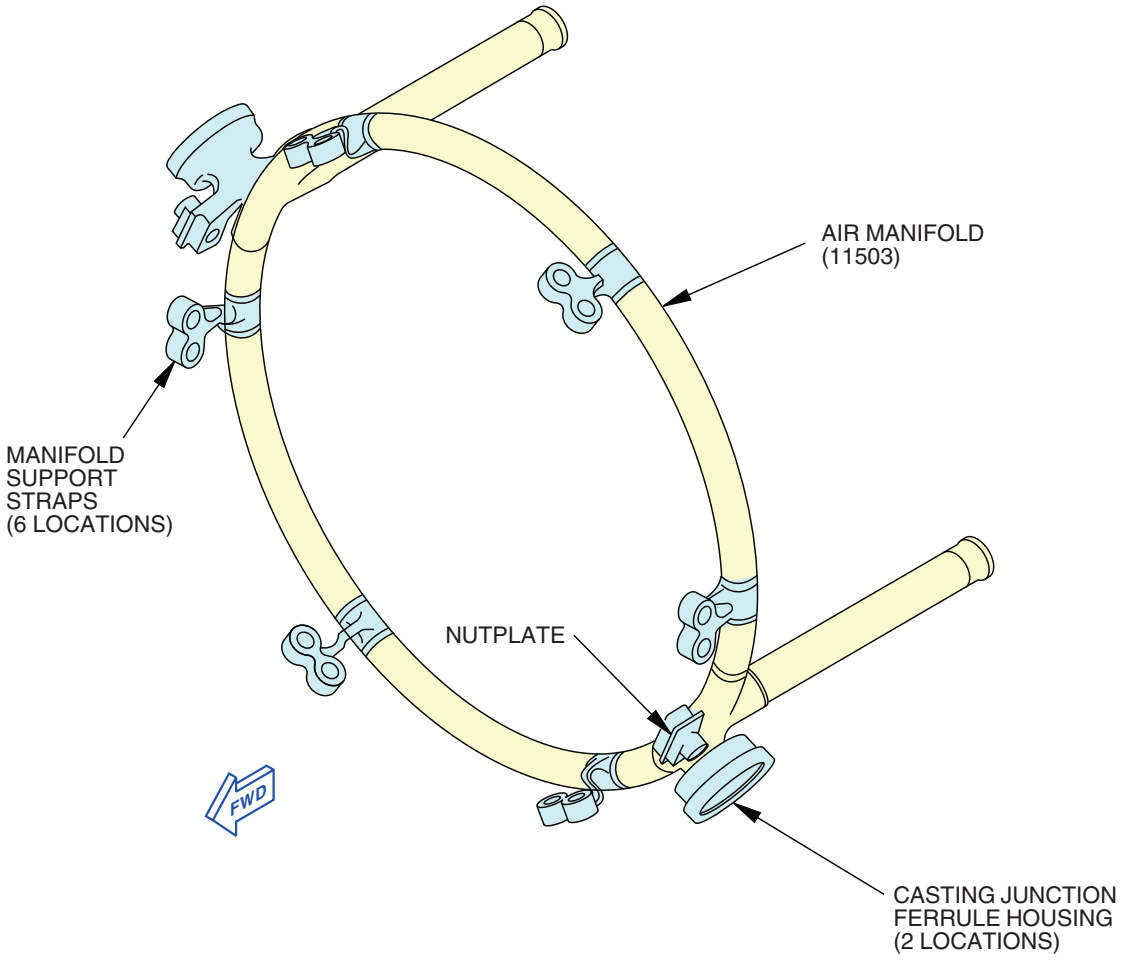


(VIEW IN THE FORWARD DIRECTION)

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Combustion Case Inspection
Figure 601/72-40-00-990-802-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



3071135 S0000831869_V1

Combustion Case Inspection
Figure 601/72-40-00-990-802-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TURBINE SECTION - INSPECTION/CHECK

1. General

A. This procedure has three tasks:

(1) High Pressure Turbine (HPT) section inspection

(2) Turbine Center Frame (TCF) aft engine mount lugs (scheduled maintenance requirement) inspection

(3) Turbine Center Frame (TCF) section inspection.

TASK 72-50-00-200-801-G00

2. High Pressure Turbine (HPT) Section Inspection

A. General

(1) This task gives the instructions to examine the High Pressure Turbine (HPT) section.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
71-00-02-000-801-G00	Power Plant - Removal (P/B 401)
71-00-02-400-801-G00	Power Plant - Installation (P/B 401)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-50-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-50-00-865-001-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-50-00-860-002-G00

(3) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-50-00-040-001-G00

(4) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-50-00-010-001-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. High Pressure Turbine (HPT) Inspection

SUBTASK 72-50-00-900-001-G00

(1) Replace the engine for all damage that is more than the limits (TASK 71-00-02-000-801-G00 and TASK 71-00-02-400-801-G00).

SUBTASK 72-50-00-220-001-G00

(2) Examine the HPT case as follows:

(a) If you find a possible crack in the turbine section, do a spot fluorescent penetrant inspection of the possible crack (TASK 70-11-06-230-802-G00).

NOTE: This step is necessary only if you are not sure of the crack in the steps that follows.

(b) Cracks that go from the bolt hole to the edge of the forward flange

1) Not permitted.

(c) Cracks that go from the bolt hole to the edge of the aft flange

1) Not permitted.

(d) Nicks, scores, and scratches on the flanges

1) Damage is permitted after you remove the high metal and the depth is less than 0.005 in. (0.127 mm) (TASK 70-43-42-300-802-G00).

2) The damage must not go into or below the bolt head.

(e) Bulges and dents on the forward or aft flanges

1) Not permitted.

(f) Cracks on the HPT case

1) Not permitted.

(g) Nicks or scratches on the HPT case

1) Damage is permitted after you remove the high metal and the depth is less than 0.005 in. (0.127 mm) (TASK 70-43-42-300-802-G00).

2) The damage must not go into or below the bolt head.

(h) Dents on the case

1) Dents with a smooth contour to 0.005 in. (0.127 mm) in depth are permitted.

(i) Cracks in the cooling air ports

1) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-50-00-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-50-00-865-002-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-50-00-440-001-G00

(3) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-50-00-860-003-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-50-00-200-802-G00

3. Turbine Center Frame (TCF) Aft Engine Mount Lugs (Scheduled Maintenance Requirement) Inspection

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task provides the instructions on how to examine the aft engine mount lugs on the Turbine Center Frame (TCF).

(2) The turbine center frame is at the aft end of the engine in front of the Low Pressure Turbine (LPT).

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-572	Light - White
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Inspection

SUBTASK 72-50-00-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-50-00-865-003-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-50-00-860-006-G00

(3) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-50-00-040-002-G00

(4) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 72-50-00-010-003-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Turbine Center Frame (TCF) Aft Engine Mount Lugs Inspection

SUBTASK 72-50-00-211-009-G00

(1) Do a visual check of the TCF aft engine mounting ring and lugs for damage:

NOTE: The TCF aft engine mounting ring is a 360 degree ring around the entire engine.

(a) Cracks in the visible portions of the TCF engine mounting ring and lugs

1) Not permitted.

2) If you find a possible crack in the mounting ring or lugs, do these steps:

a) Examine the area with a white light, STD-572 to make sure that it is a crack.

b) If it is necessary, do a spot fluorescent penetrant inspection to make sure it is or it is not a crack (Fluorescent Penetrant Inspection, TASK 70-11-06-230-802-G00).

3) Indications interpreted as through are not permitted.

(b) Nicks, scores, scratches, and dents in the visible portions of the TCF engine mounting ring and lugs

1) All amounts that are not more than 0.005 in. (0.13 mm) in depth after removal of high metal are permitted.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-50-00-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-50-00-440-002-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-50-00-865-004-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-50-00-860-019-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 72-50-00-200-804-G00

4. Turbine Center Frame (TCF) Section Inspection

A. General

(1) This task gives the instructions to do the inspection of the Turbine Center Frame (TCF) section.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-50-00-860-008-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-50-00-860-009-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-50-00-860-010-G00

(3) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 72-50-00-040-003-G00

(4) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-50-00-860-012-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Turbine Center Frame (TCF) Section Inspection

SUBTASK 72-50-00-211-010-G00

(1) Examine the visible portions of the turbine center frame as follows:

(a) Cracks in the case flanges

1) Not permitted.

(b) Cracks in the outer case

1) Not permitted.

(c) Punctures or tears in the outer case

1) Not permitted.

(d) Nicks, gouges, scores and scratches

1) All amounts that are not more than 0.005 in. (0.127 mm) in depth are permitted.

(e) Dents

1) All amounts are permitted with these conditions:

a) The depth of the dent is not more than 0.005 in. (0.127 mm) and width is not more than 10 times the depth.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-50-00-211-011-G00

(2) Examine the visible portions of the Turbine Center Frame (TCF) strut caps and service tubes as follows:

(a) Cracks

1) Not permitted.

(b) Nicks, scores, scratches and chafing

1) Permitted if no more than 0.005 in. (0.127 mm) in depth.

2) All amounts permitted on the B-nut fitting wrenching flats and corners.

(c) Adapter cap jumper tube dents

1) All amounts are permitted if not more than 0.005 in. (0.127 mm) in depth and 0.05 in. (1.27 mm) in length.

(d) Loose adapter cap jumper tubes

1) Permitted if a light hand force applied to the jumper tubes does not cause it to slide within the adapter cap sockets.

SUBTASK 72-50-00-211-012-G00

(3) Examine the visible TCF case flange bolts as follows:

(a) Missing or broken bolts

1) Not permitted.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-50-00-860-013-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-50-00-440-003-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

SUBTASK 72-50-00-860-015-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 72-50-00-860-020-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LOW PRESSURE TURBINE (LPT) ROTOR AND STATOR MODULE - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Low Pressure Turbine (LPT) rotor and stator module inspection.

TASK 72-50-02-200-801-G00

2. Low Pressure Turbine (LPT) Rotor and Stator Module Inspection

(Figure 601 and Figure 605)

A. General

(1) This task provides the instructions on how to examine the Low Pressure Turbine (LPT) rotor and stator module.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-290-801-G00	Preparation for Borescope Inspection (P/B 601)
72-00-00-290-811-G00	Low Pressure Turbine (LPT) Section Borescope Inspection (P/B 601)
72-50-02-960-801-G00	LPT Cooling Manifold Assembly Loop Clamp and Bracket - Replacement (P/B 801)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Inspection

SUBTASK 72-50-02-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-50-02-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-50-02-840-001-G00

(3) Do this task: Preparation for Borescope Inspection, TASK 72-00-00-290-801-G00.

G. Low Pressure Turbine (LPT) Rotor and Stator Module Inspection

SUBTASK 72-50-02-211-015-G00

(1) Do a visual examination of the LPT borescope plugs (TASK 72-00-00-290-811-G00 and Figure 605).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-50-02-220-001-G00

(2) Do a visual examination of the LPT case forward flange for these conditions (Figure 601):

(a) Cracks in all areas

1) Not permitted.

(b) Nicks

1) Permitted up 0.0047 in. (0.12 mm) in depth.

(c) Scores and scratches

1) Permitted up 0.0027 in. (0.07 mm) in depth.

SUBTASK 72-50-02-211-016-G00

(3) Do a visual examination of the LPT case aft flange for these conditions:

(a) Cracks that connect the bolt holes

1) Not permitted.

(b) Cracks that extend from the bolt holes to the outer edge of the flange

1) Not permitted.

(c) Cracks that extend from the bolt holes to the casing skin

1) Not permitted.

(d) Cracks in the skin radius of the flange and case

1) Not permitted.

(e) Cracks in other areas

1) Not permitted.

(f) Nicks

1) Permitted up 0.0047 in. (0.12 mm) in depth.

(g) Scores and scratches

1) Permitted up 0.0027 in. (0.07 mm) in depth.

(h) Dents

1) Permitted up to 0.0047 in. (0.12 mm) in depth and dent diameter is not greater than 10 times the depth.

SUBTASK 72-50-02-220-002-G00

(4) Do a visual examination of the LPT case skin for these conditions:

(a) Cracks in the parent metal

1) Not permitted.

(b) Nicks and scratches

1) Permitted up 0.0023 in. (0.06 mm) in depth.

2) Not permitted if nicks and scratches overlay on "wear in areas opposite to the cooling manifold clamps".

(c) Dents

1) Permitted up 0.0023 in. (0.06 mm) in depth and dent diameter is not greater than 10 times the depth.

2) Not permitted if dents overlay on "wear in areas opposite to the cooling manifold clamps".

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(d) Heat or other discoloration

1) Permitted if there are no distortion bulges.

(e) Bulges

1) Not permitted.

(f) Wear in areas opposite to the cooling manifold clamps for the tubes (#2, #3, and #4)

1) Permitted a maximum of 0.0015 in. (0.04 mm) in depth, not included area A.

2) The Continue-In-Service limit is 10000 cycles with these conditions: any amount, up to 0.0043 in. (0.11 mm) in depth (with area A).

(g) Wear in areas opposite to the cooling manifold clamps for the tubes (#1, #5, #6, #7, and #8)

1) Permitted a maximum of 0.0015 in. (0.04 mm) in depth, not included area A.

2) The Continue-in-Service limit is:

a) 5000 cycles with these conditions: any amount, up to 0.0027 in. (0.07 mm) in depth, with area A.

b) 4000 cycles with these conditions: any amount, up to 0.0043 in. (0.11 mm) in depth, with area A.

(h) Wear in other areas

1) Not permitted.

SUBTASK 72-50-02-220-003-G00

(5) Do a visual examination of the LPT case brackets for these conditions:

(a) Cracks

1) Not permitted.

(b) Broken brackets

1) Not permitted.

(c) Nicks, scores, scratches

1) Permitted up 0.0027 in. (0.07 mm) in depth.

(d) Dents

1) Permitted up 0.0027 in. (0.07 mm) in depth and the dent diameter is not greater than 10 times the depth.

SUBTASK 72-50-02-211-017-G00

(6) Do a visual examination of all areas of the upper cooling manifold and lower cooling manifold for cracks.

(a) Cracks are not permitted.

SUBTASK 72-50-02-211-018-G00

(7) Do a visual examination of the upper cooling manifold and the lower cooling manifold tubes for these conditions (Figure 602):

(a) Nicks and scratches

1) Permitted if there is no tube wall perforation.

(b) Dents and collapse

1) Permitted up to 0.062 in. (1.6 mm) in depth and these conditions as follows:

- There are no holes in the tube wall

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- The dent is no more than 0.24 in. (6 mm) in length and width
- There are no dents or collapses in area C
- There is no dent or collapse on the same section of tube.

(c) Rubbing

1) Permitted up to 0.0059 in. (0.15 mm) in depth.

SUBTASK 72-50-02-211-019-G00

(8) Do a visual examination of the upper cooling manifold and the lower cooling manifold forward lugs for breakage (Figure 602).

(a) Breakage is not permitted.

SUBTASK 72-50-02-211-020-G00

(9) Do a visual examination of the upper cooling manifold and the lower cooling manifold rear lugs for breakage.

(a) Breakage is not permitted.

SUBTASK 72-50-02-220-004-G00

(10) Do a visual examination of the upper cooling manifold and the lower cooling manifold box for these conditions:

(a) Scratches

1) Permitted up to 0.0031 in. (0.08 mm) in depth.

(b) Rub

1) Permitted up to 0.0031 in. (0.08 mm) in depth.

(c) Dent

1) Permitted up to 0.094 in. (2.4 mm) in depth without a sharp edge.

SUBTASK 72-50-02-211-021-G00

(11) Do a visual examination of the upper cooling manifold, lower cooling manifold clamps, and cushions for these conditions:

(a) Wear on cushions

1) Permitted.

2) Maximum repairable limit: any amount.

3) Repair method: LPT Cooling Manifold Assembly Loop Clamp and Bracket - Replacement, TASK 72-50-02-960-801-G00.

(b) Contact on clamps

1) Permitted if there is no wear.

2) Maximum repairable limit: any amount.

3) Repair method: LPT Cooling Manifold Assembly Loop Clamp and Bracket - Replacement, TASK 72-50-02-960-801-G00.

SUBTASK 72-50-02-211-022-G00

(12) Do a visual examination of all areas of the air duct for breaks and cracks (Figure 603).

(a) Breaks and cracks are not permitted.

SUBTASK 72-50-02-211-025-G00

(13) Do a visual examination of the air duct areas A and B brackets for nicks and scratches.

(a) Any number permitted up to 0.0051 in. (0.13 mm) in depth.

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SUBTASK 72-50-02-220-005-G00

(14) Do a visual examination of the air duct areas A and B but not the brackets for dents.

(a) Dents up to 0.0299 in. (0.76 mm) in depth and with radius greater than 0.056 in. (1.42 mm) are permitted.

SUBTASK 72-50-02-211-023-G00

(15) Do a visual examination of the air duct area A brackets for these conditions:

(a) Wear

1) Not permitted.

(b) Distortion

1) Permitted if assembly can be installed without stresses.

SUBTASK 72-50-02-211-024-G00

(16) Do a visual examination of the bracket for missing rivets (Figure 604).

(a) Missing rivets are not permitted.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-50-02-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-50-02-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

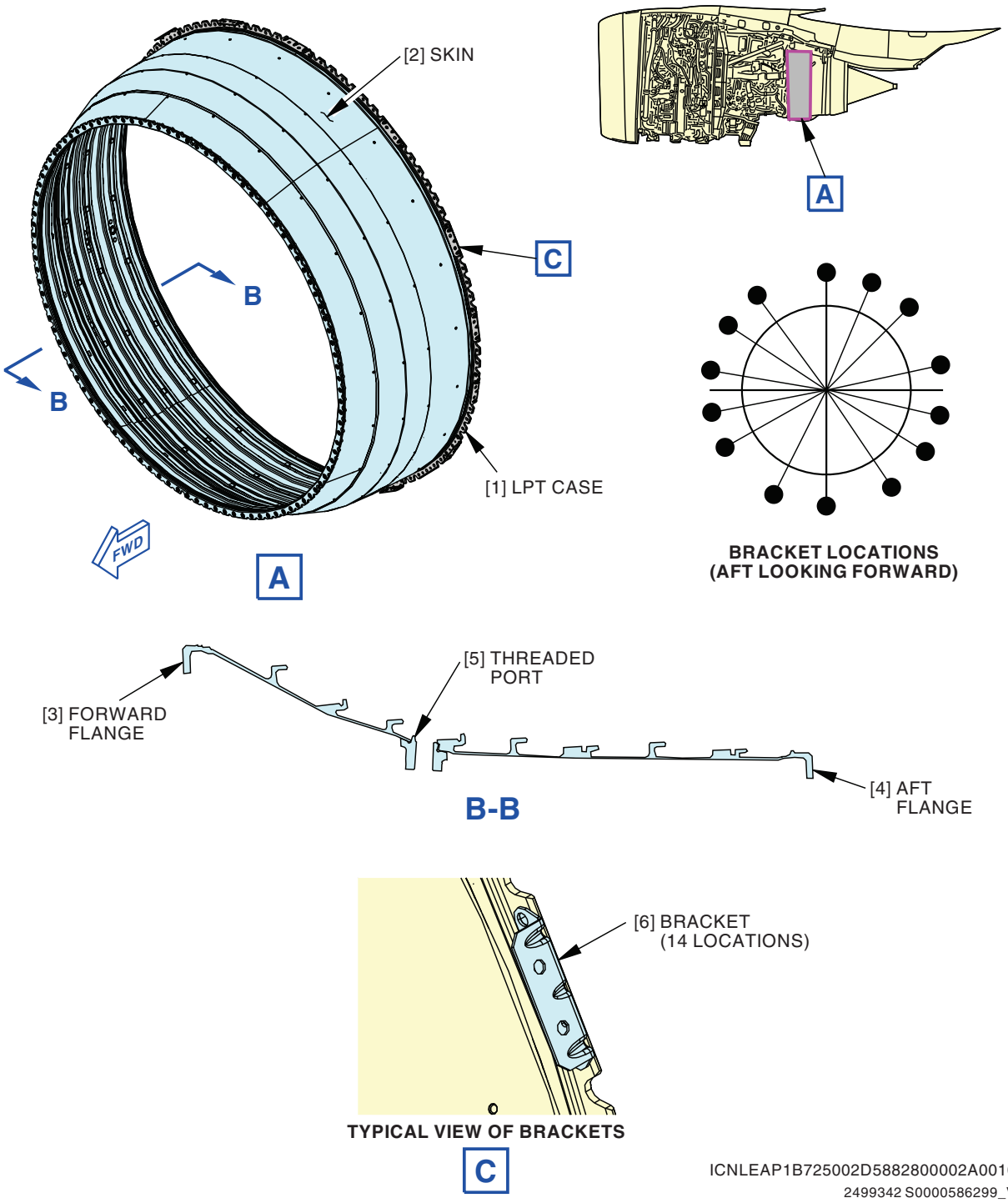
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

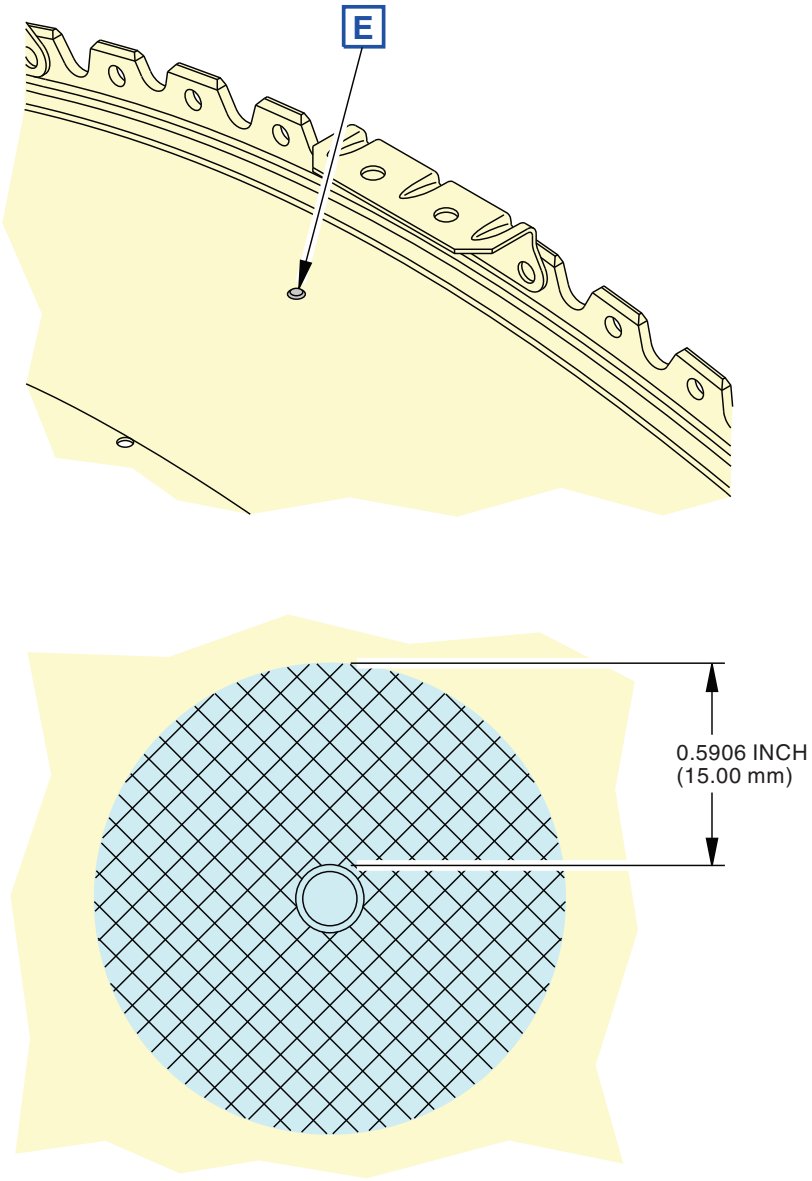
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LPT Rotor and Stator Module Inspection
Figure 601/72-50-02-990-801-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

AREA A

NOTE:

ALL DIMENSIONS ARE IN INCHES (mm).

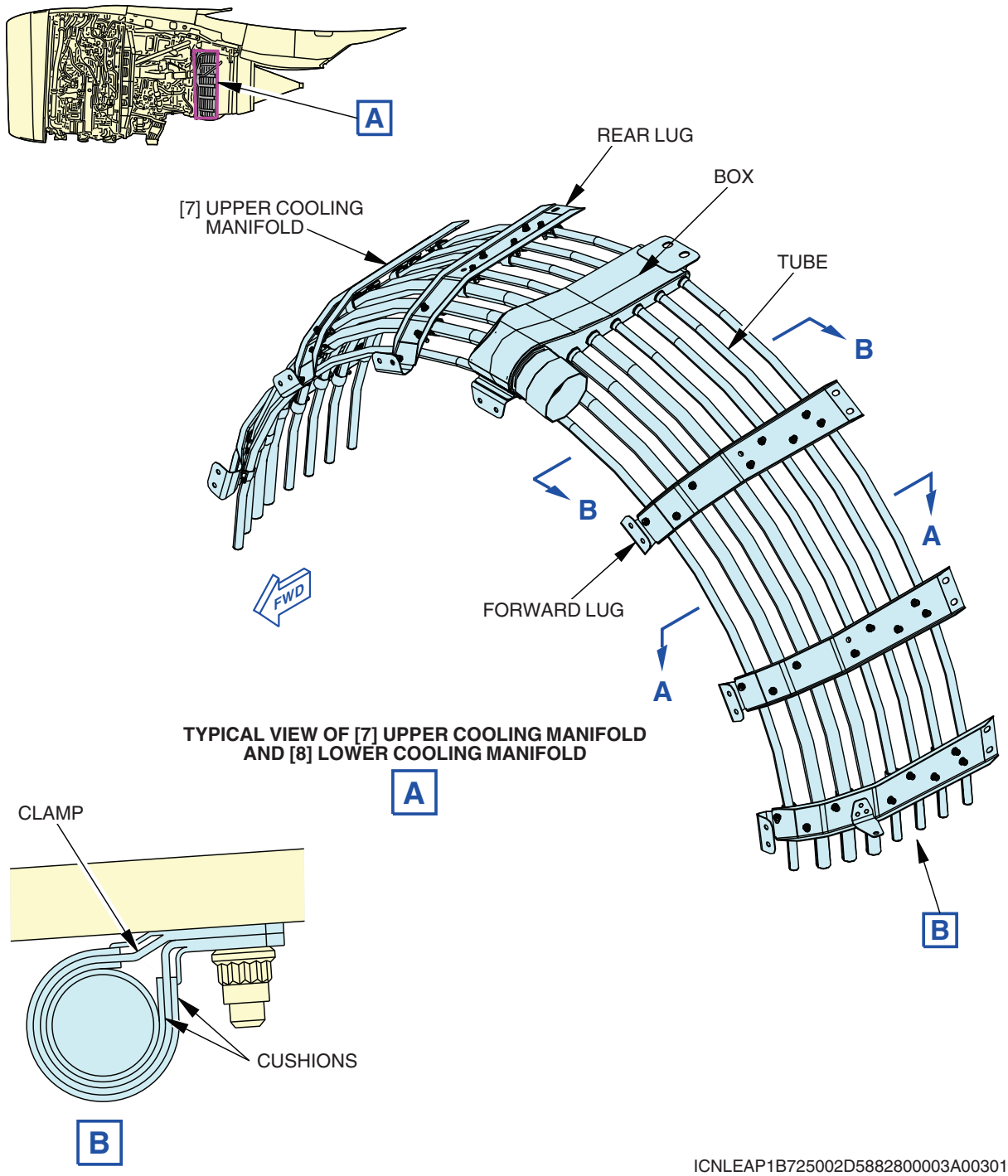
THE AREA A IS APPLICABLE TO ALL PIN-HOLES ACROSS THE LPT CASE.

ICNLEAP1B725002D5882800003A00101

2566830 S0000613926_V1

LPT Rotor and Stator Module Inspection
Figure 601/72-50-02-990-801-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

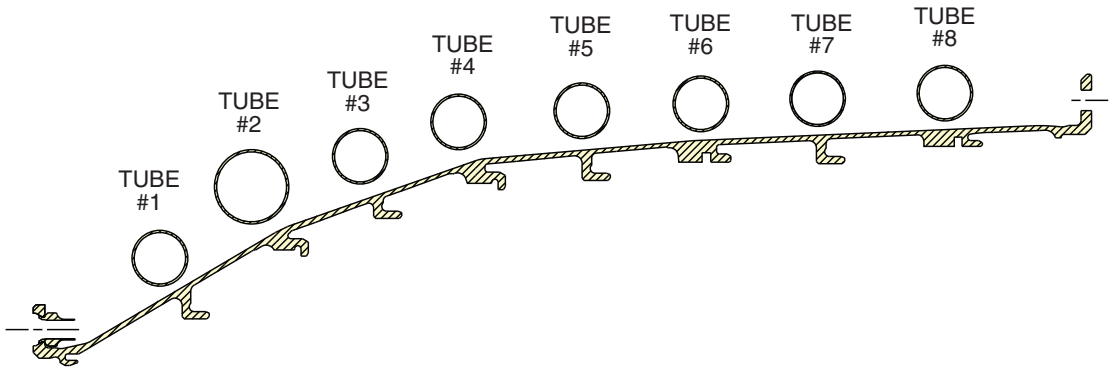


TYPICAL VIEW OF [7] UPPER COOLING MANIFOLD AND [8] LOWER COOLING MANIFOLD

ICNLEAP1B725002D5882800003A00301
2499345 S0000586302_V5

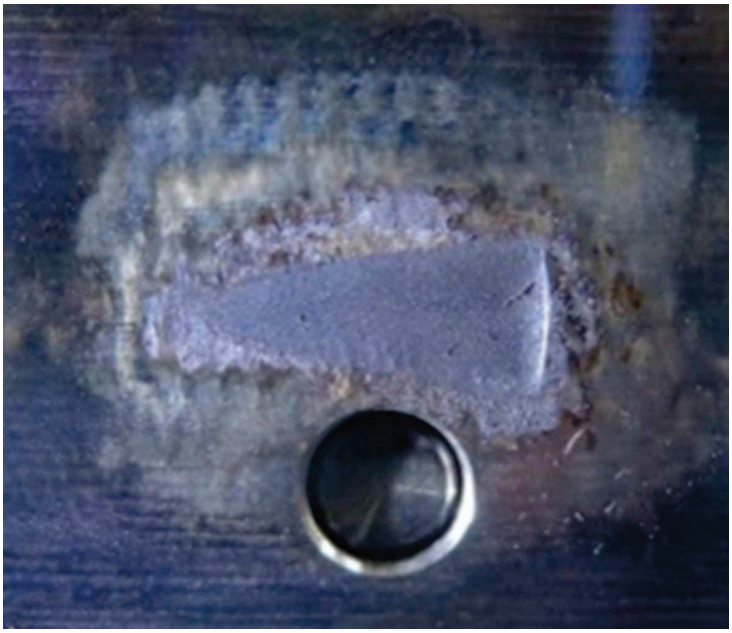
Cooling Manifold Inspection
Figure 602/72-50-02-990-804-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



POSITION OF THE COOLING MANIFOLD TUBES

A-A

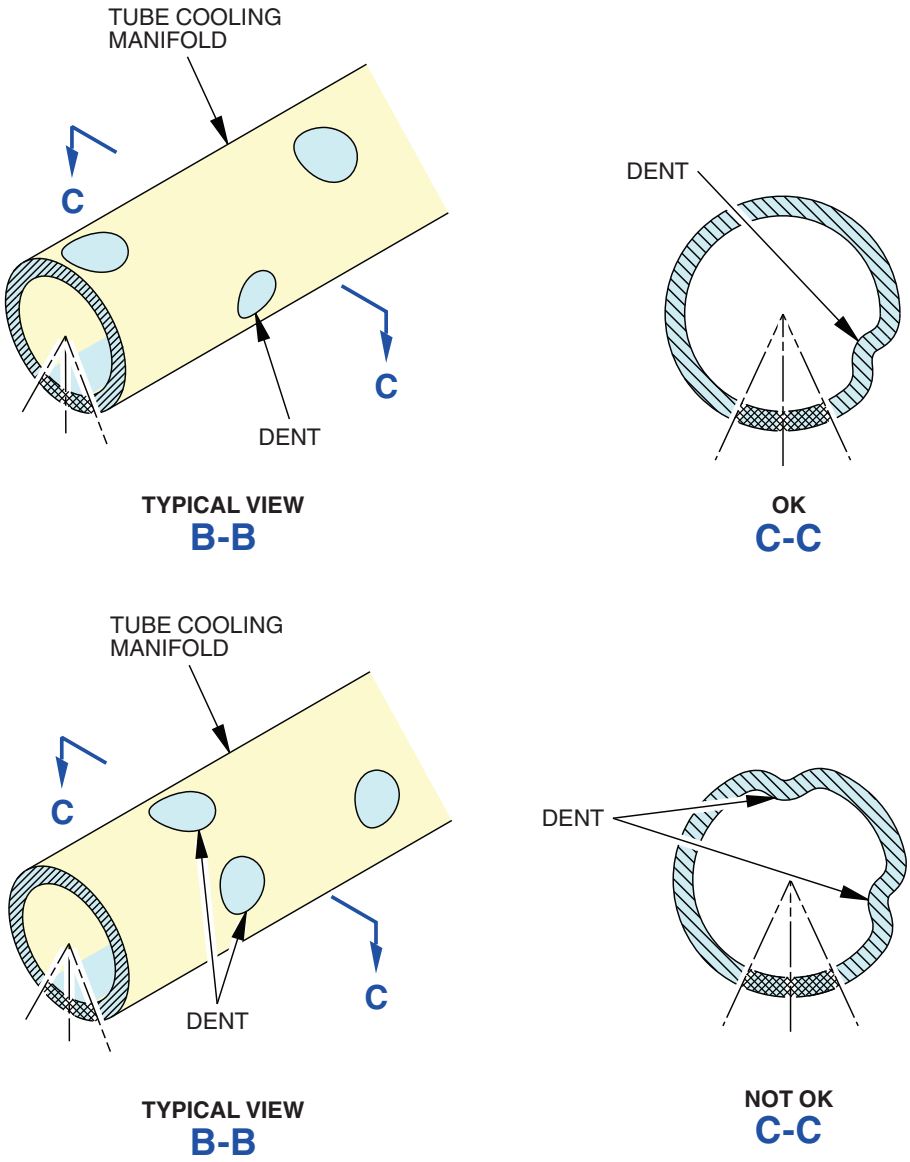


EXAMPLE OF WEAR IN AREAS OPPOSITE TO THE CLAMPS LOOPS

ICNLEAP1B725002D5882800017A00101
3085137 S0000839014_V2

Cooling Manifold Inspection
Figure 602/72-50-02-990-804-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

AREA C

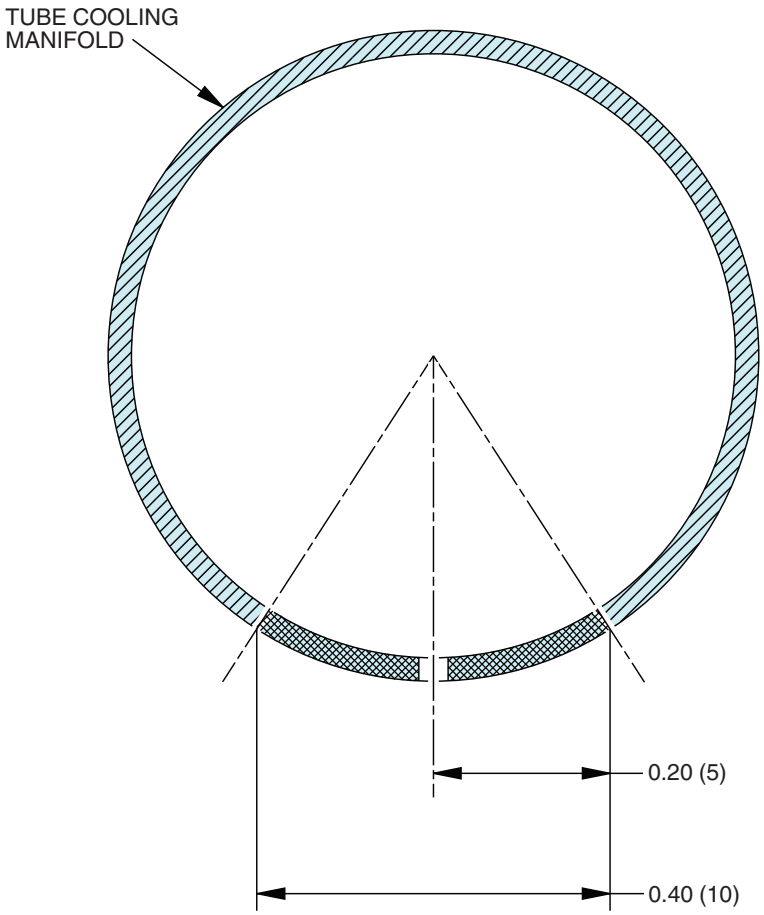
NOTE:

THIS VIEW IS APPLICABLE FOR ALL THE TUBES.

ICNLEAP1B725002D5882800018A00101
3123094 S0000857042_V1

Cooling Manifold Inspection
Figure 602/72-50-02-990-804-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW
B-B

LEGEND:

AREA C

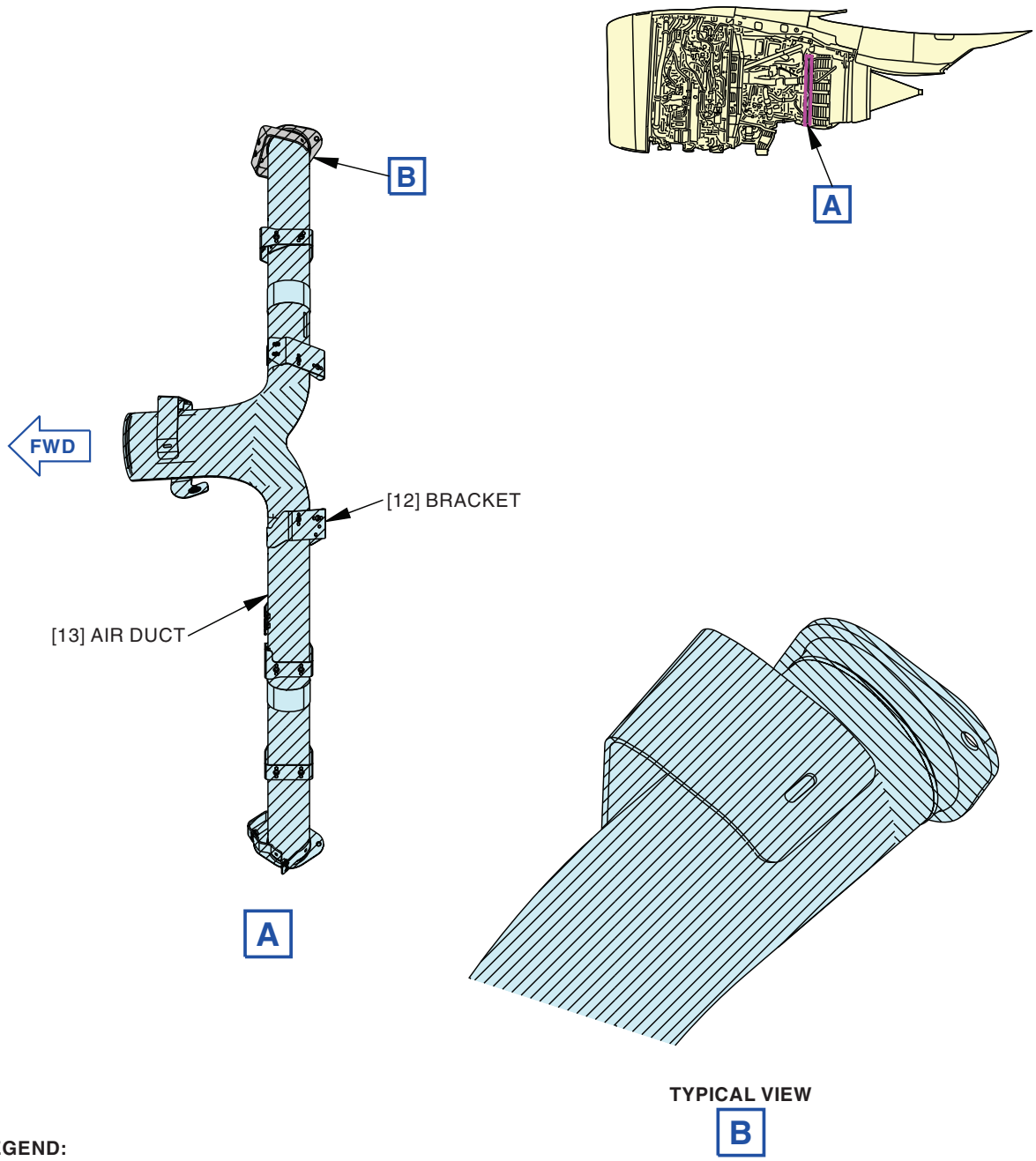
NOTE:

ALL DIMENSIONS ARE IN INCHES (mm).

ICNLEAP1B725002D5882800019A00101
3123068 S0000857043_V1

Cooling Manifold Inspection
Figure 602/72-50-02-990-804-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

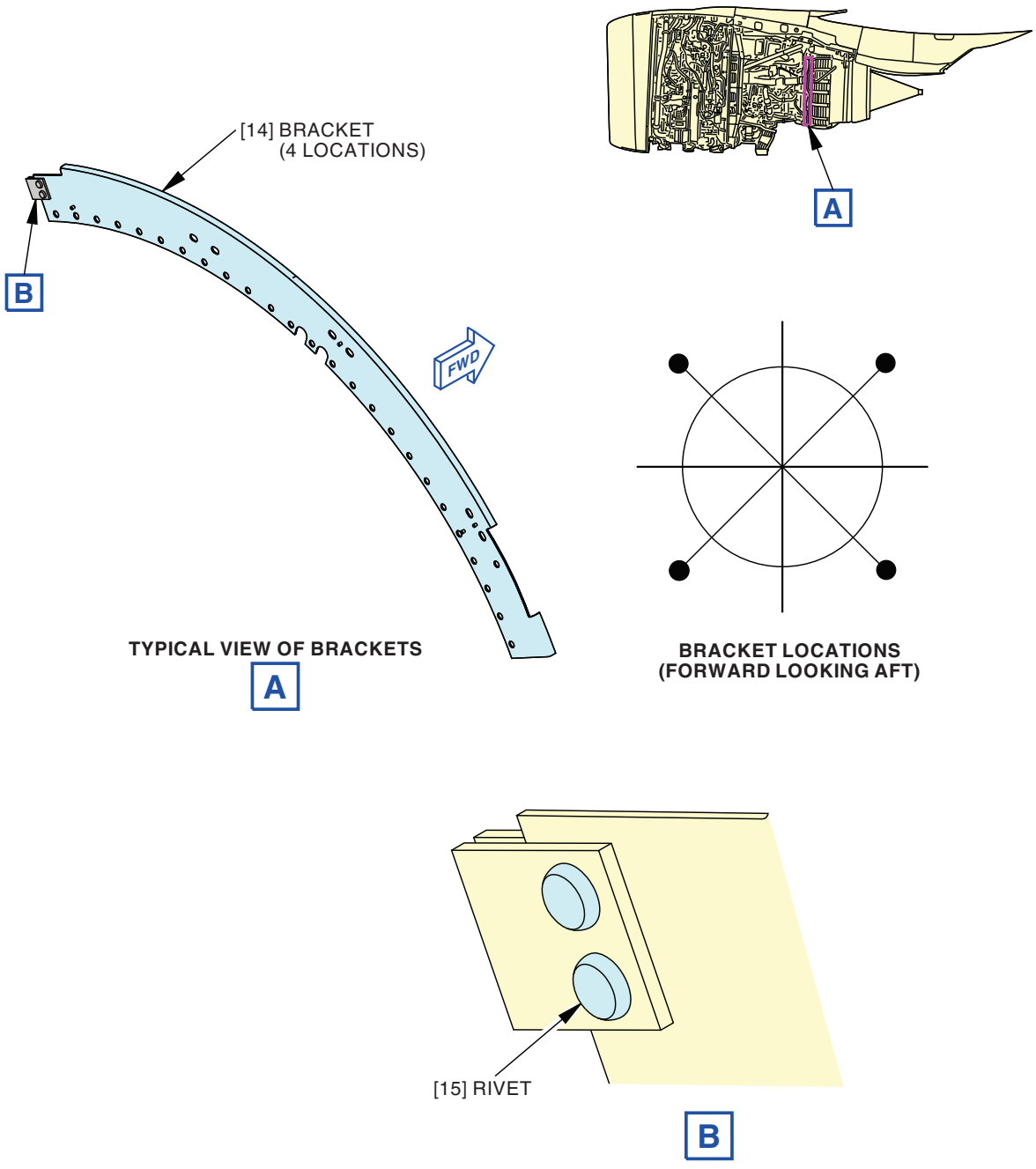


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2499348 S0000586303_V2

Air Duct Inspection
Figure 603/72-50-02-990-805-G00

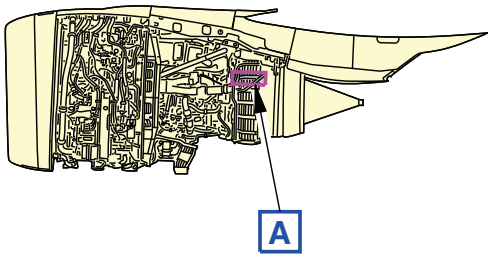
ECCN 9E991 BOEING PROPRIETARY - See title page for details

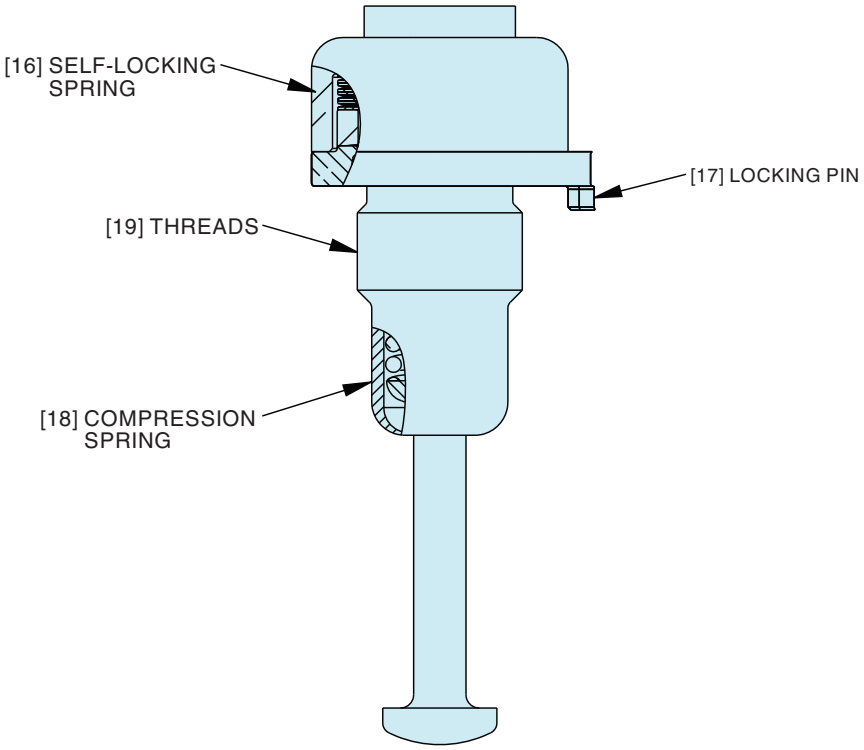


ICNLEAP1B725002D5882800005A00101
2553191 S0000605257_V1

Bracket Inspection
Figure 604/72-50-02-990-806-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details





TYPICAL VIEW OF LPT BOROSCOPE PLUGS

A

ICNLEAP1B725002D5882800001A00101

2553201 S0000605254_V1

LPT Borescope Plug Inspection
Figure 605/72-50-02-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

LPT COOLING MANIFOLD ASSEMBLY LOOP CLAMP AND BRACKET - REPAIR

1. General

A. This procedure has this task:

(1) A replacement of the loop clamp and bracket on the cooling manifold assembly.

TASK 72-50-02-960-801-G00

2. LPT Cooling Manifold Assembly Loop Clamp and Bracket - Replacement

(Figure 801 and Figure 802)

A. General

(1) This task gives instructions to replace the loop clamp and bracket on the cooling manifold assembly.

(2) The table below gives the list of the part numbers that are applicable to this repair.

Table 801/72-50-02-993-801-G00 Cooling Manifold Assembly Part Numbers

Part Number	Designation
364-098-830-0	MANIFOLD, COOLING-ASSY SUP
364-098-930-0	MANIFOLD, COOLING-ASSY INF

B. References

Reference	Title
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
B50190 [CP1040]	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B50290 [CP2559]	Solvent - degreasing, Ardrox 5503	
B50295 [CP2010]	Solvent - cleaning, White Spirit	
B50333 [CP2687]	Solvent - Degreasing, Diestone DLS	
B50344 [CP2726]	Solvent - Supersolve AS	
B50345 [CP2727]	Solvent - Lotoxane Fast	
B50347 [CP2947]	Solvent - Degreasing, DYNA 200	
D00601 [CP2101]	Vaseline - Graphite Mineral	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Repair

SUBTASK 72-50-02-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-50-02-010-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Loop Clamp and Bracket Replacement

SUBTASK 72-50-02-020-002-G00

(1) To remove the loop clamp [3], do these steps:

(a) Remove the nut [5] from the machine bolt [10].

(b) Remove the loop clamp [3], flat washer [6], and machine bolt [10].

SUBTASK 72-50-02-020-003-G00

(2) To remove the loop clamp [1], do these steps:

(a) Remove the nut [5] from the double hex bolt [4].

(b) Remove the loop clamp [1], spacer [7], flat washer [6], and double hex bolt [4].

SUBTASK 72-50-02-020-004-G00

(3) To remove the loop clamp [2], do these steps:

(a) Remove the nut [5] from the machine bolt [10].

(b) Remove the loop clamp [2], flat washer [6], and machine bolt [10].

SUBTASK 72-50-02-020-005-G00

(4) Remove the bracket [8] or bracket [9].

(a) Do the steps above to remove all the loop clamps from the bracket [8] or bracket [9].

SUBTASK 72-50-02-110-001-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THIS PRODUCT. IF NOT, IT CAN CAUSE INJURY.

WARNING DO NOT GET SOLVENT IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE SOLVENT. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME AND HEAT. SOLVENT IS POISONOUS AND FLAMMABLE WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(5) Clean all the removed parts and uncovered area of the cooling manifold (TASK 70-21-23-100-801-G00).

(a) Use alcohol, B00676 [CP1041], to clean the loop clamp [1], loop clamp [2], or loop clamp [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Use acetone solvent, B01058 [CP1039], solvent, B50190 [CP1040], alcohol, B00676 [CP1041], degreasing solvent, B50333 [CP2687], solvent, B50344 [CP2726], Ardrox 5503, B50290 [CP2559], solvent, B50345 [CP2727], White Spirit, B50295 [CP2010], or solvent, B50347 [CP2947], to clean the cooling manifold, bracket [8], bracket [9], nut [5], flat washer [6], double hex bolt [4], machine bolt [10], and spacer [7].

SUBTASK 72-50-02-230-001-G00

(6) Do a fluorescent penetrant inspection of the uncovered area of the cooling manifold (TASK 70-11-06-230-803-G00).

(a) Use sensitivity S2.

(b) Cracks are not permitted.

(c) Linear indications are not permitted.

(d) Indications more than 0.011 in. (0.3 mm) are not permitted.

SUBTASK 72-50-02-230-002-G00

(7) Do a fluorescent penetrant inspection of the applicable bracket (TASK 70-11-06-230-803-G00).

(a) Use sensitivity S2.

(b) Cracks are not permitted.

(c) Linear indications are not permitted.

(d) Indications more than 0.011 in. (0.3 mm) are not permitted.

SUBTASK 72-50-02-420-001-G00

(8) To install the loop clamp [3], do these steps:

(a) Lubricate the thread and pressure face of the machine bolt [10] with a 0.004 in. (0.1 mm) - 0.007 in. (0.2 mm) layer of graphite mineral vaseline, D00601 [CP2101] (View C-C, Figure 801).

(b) Install the loop clamp [3] with the correct orientation.

(c) Install the machine bolt [10] with the flat washer [6].

(d) Install the nut [5].

1) Tighten the nut [5] by hand.

a) Make sure that you can easily tighten the nut [5].

b) If the torque is too high or too low, replace the machine bolt [10] and nut [5].

SUBTASK 72-50-02-420-002-G00

(9) To install the loop clamp [1], do these steps:

(a) Lubricate the thread and pressure face of the double hex bolt [4] with a 0.004 in. (0.1 mm) - 0.007 in. (0.2 mm) layer of graphite mineral vaseline, D00601 [CP2101] (View C-C, Figure 801).

(b) Install the spacer [7] and loop clamp [1] with the correct orientation.

(c) Install the double hex bolt [4] with the flat washer [6].

(d) Install the nut [5].

1) Tighten the nut [5] by hand.

a) Make sure that you can easily tighten the nut [5].

b) If the torque is too high or too low, replace the double hex bolt [4] and nut [5].

SUBTASK 72-50-02-420-003-G00

(10) To install the loop clamp [2], do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Lubricate the thread and pressure face of the machine bolt [10] with a 0.004 in. (0.1 mm) - 0.007 in. (0.2 mm) layer of graphite mineral vaseline, D00601 [CP2101] (View C-C, Figure 801).

(b) Install the loop clamp [2] with the correct orientation.

(c) Install the machine bolt [10] with the flat washer [6].

(d) Install the nut [5].

1) Tighten the nut [5] by hand.

a) Make sure that you can easily tighten the nut [5].

b) If the torque is too high or too low, replace the machine bolt [10] and nut [5].

SUBTASK 72-50-02-420-004-G00

(11) To install the bracket [8] or bracket [9], do these steps:

(a) Put the bracket [8] or bracket [9] in the correct position on the cooling manifold assembly (Figure 802).

(b) Do the steps above to install all the loop clamps on the bracket [8] or bracket [9].

SUBTASK 72-50-02-420-005-G00

(12) Tighten the double hex bolt [4] or machine bolt [10] to 58.8 in-lb (6.64 N·m) - 65.1 in-lb (7.36 N·m).

(a) For installation of the bracket, tighten the double hex bolt [4] and machine bolt [10] in the correct sequence (View B-B, Figure 801).

(b) If it is necessary, remove the unwanted graphite mineral vaseline, D00601 [CP2101].

(c) Make sure that you can see a minimum of 1.5 thread of the double hex bolt [4] or machine bolt [10] that extends out of the nut [5].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-50-02-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-50-02-860-004-G00

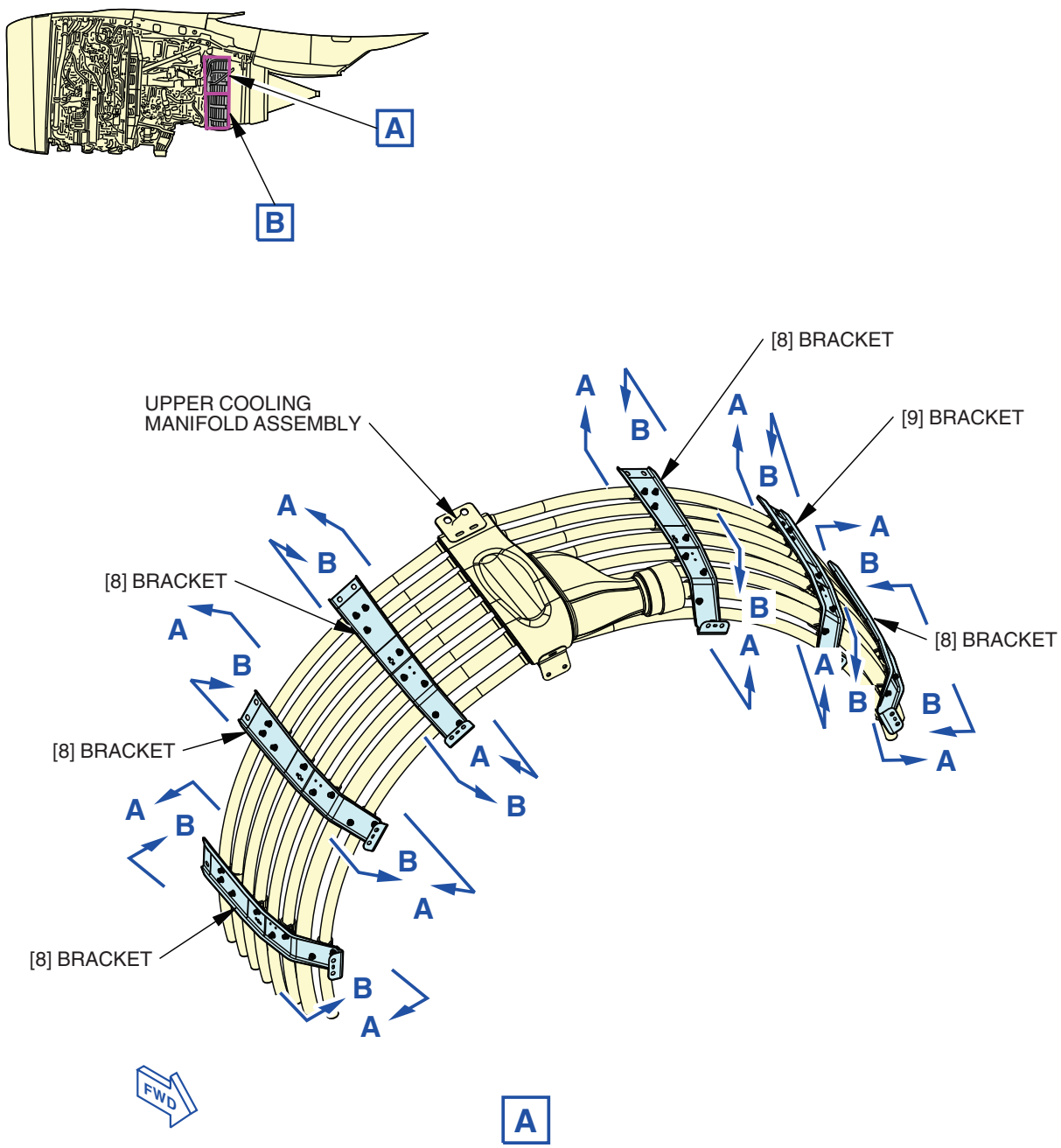
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

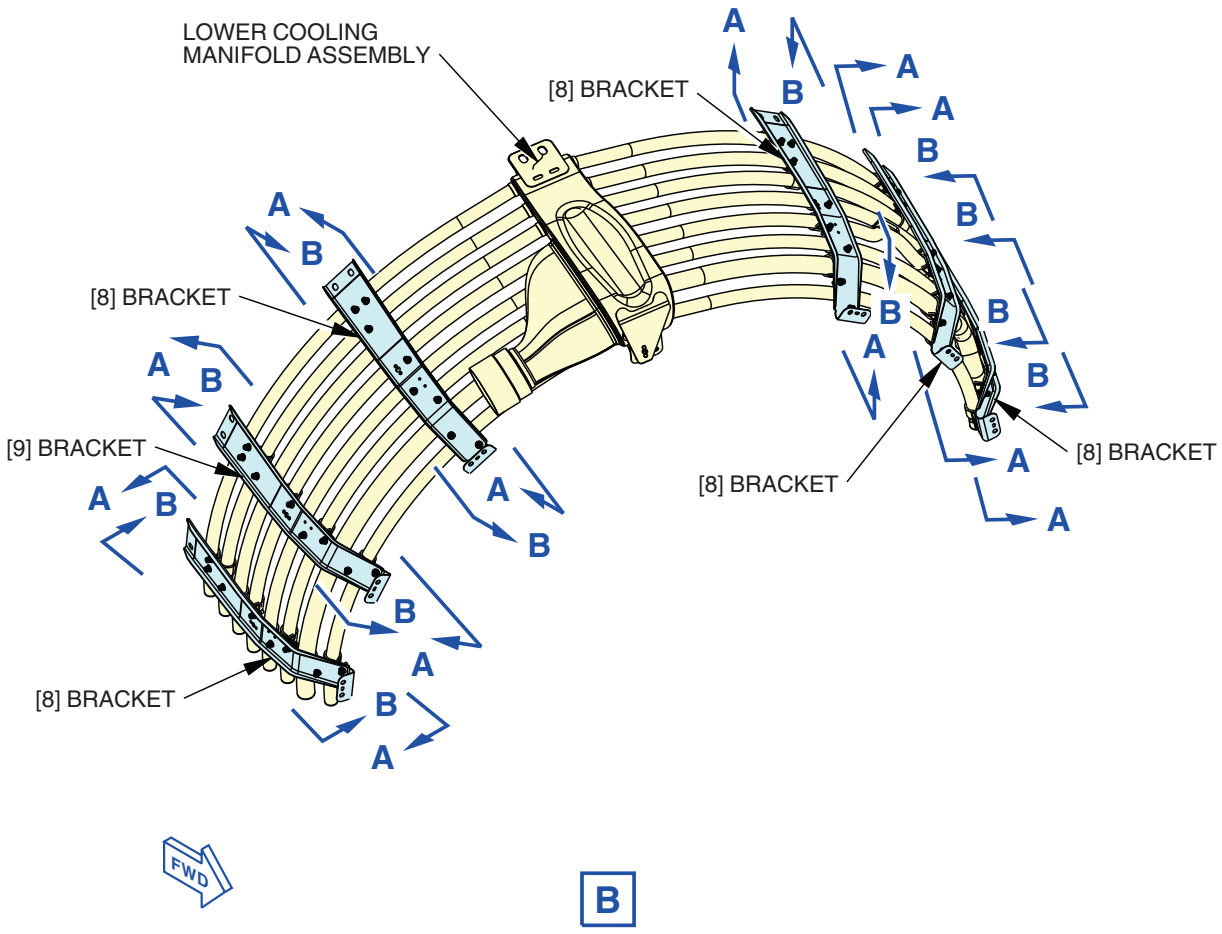
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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3133434 S0000862233_V1

Loop Clamp and Bracket Replacement
Figure 801/72-50-02-990-807-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

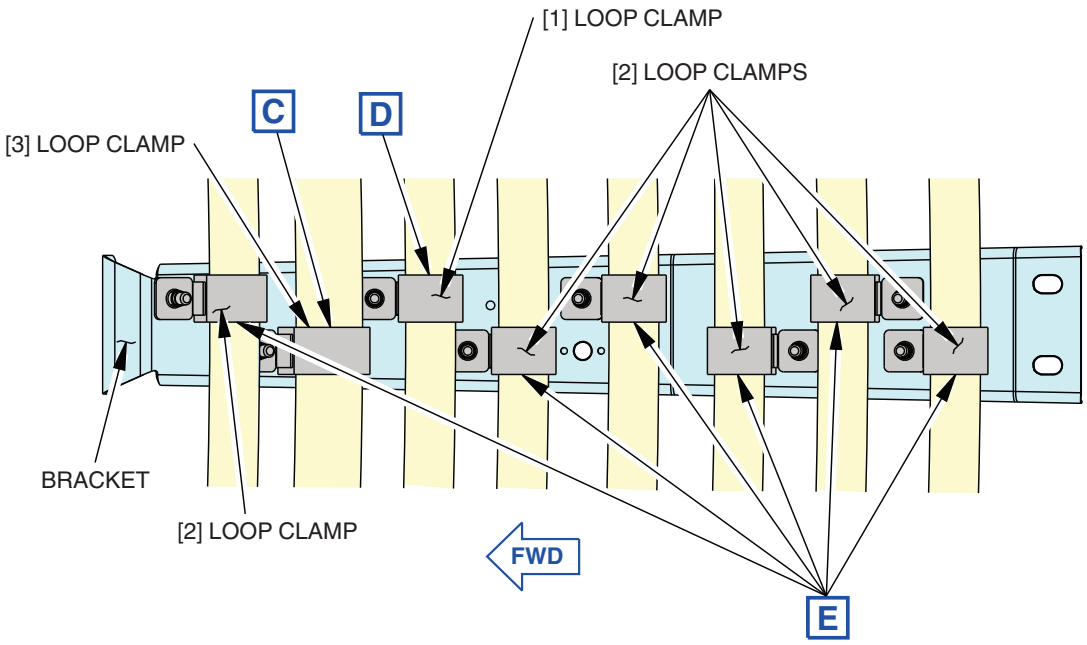


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3133435 S0000862234_V1

Loop Clamp and Bracket Replacement
Figure 801/72-50-02-990-807-G00 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



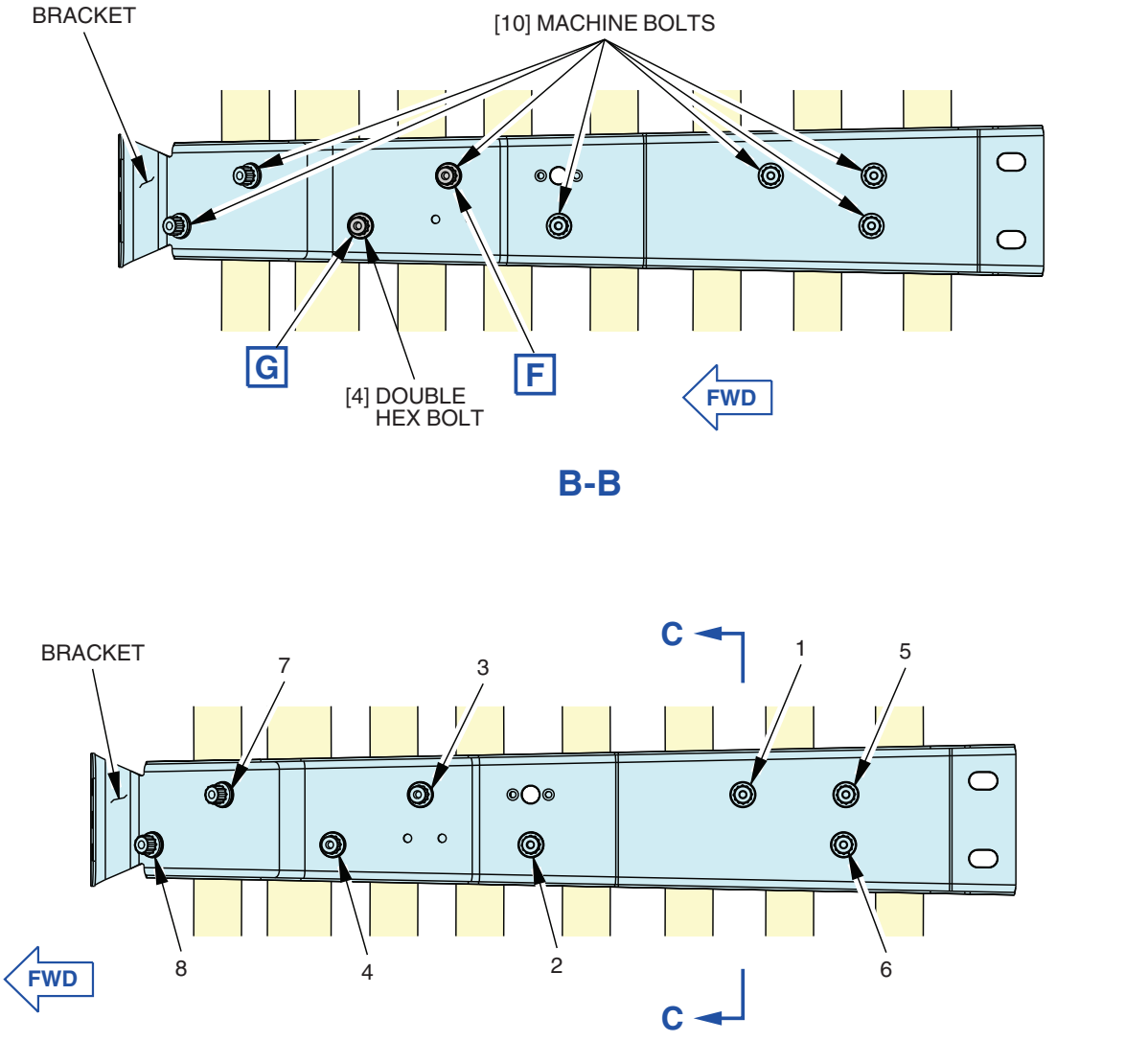
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3133444 S0000862235_V1

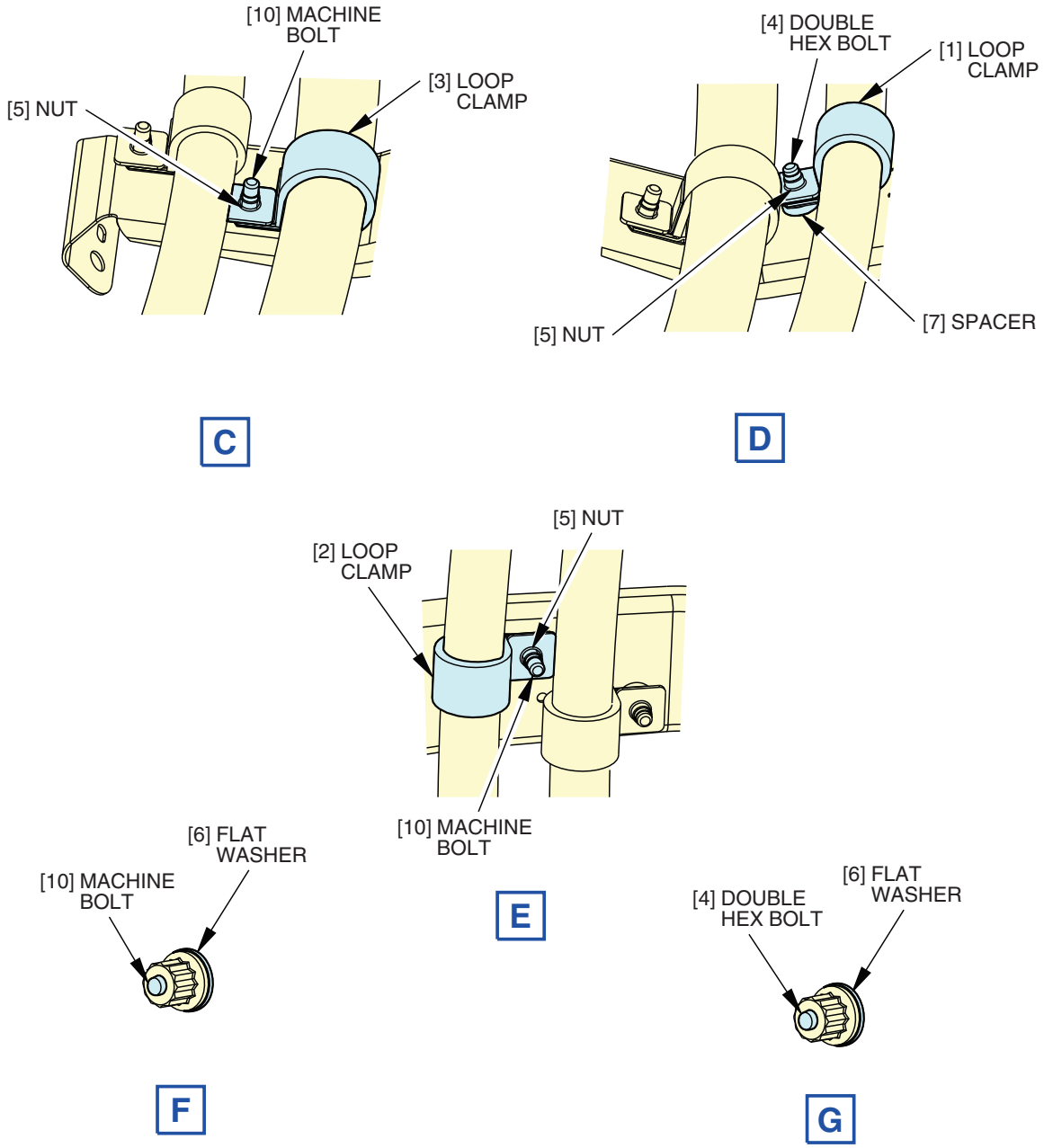
Loop Clamp and Bracket Replacement
Figure 801/72-50-02-990-807-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Loop Clamp and Bracket Replacement
Figure 801/72-50-02-990-807-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

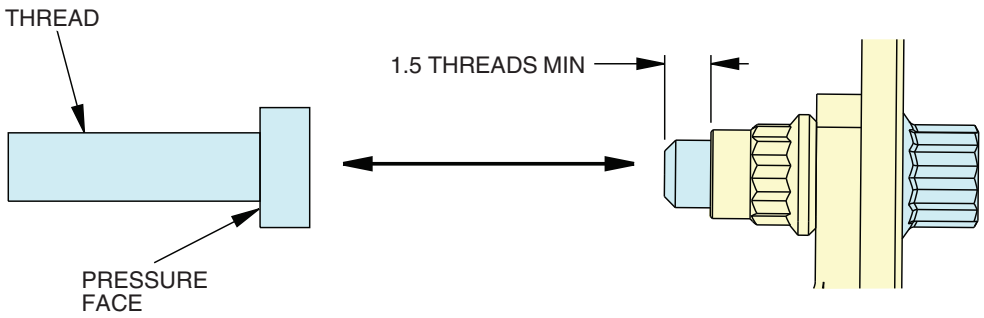


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3133449 S0000862237_V1

Loop Clamp and Bracket Replacement
Figure 801/72-50-02-990-807-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(MACHINE BOLT [10] OR DOUBLE HEX BOLT [4])

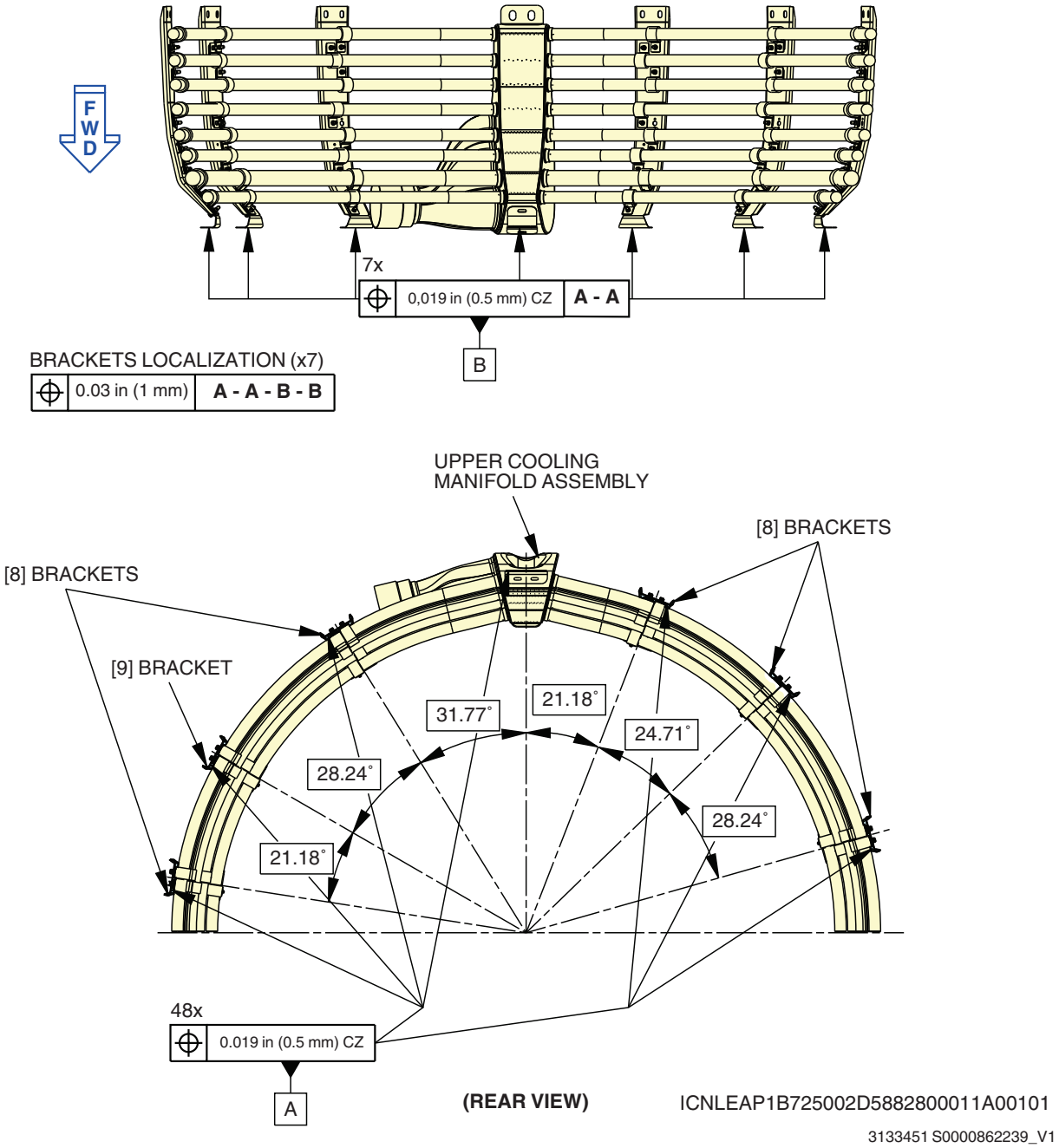
C-C

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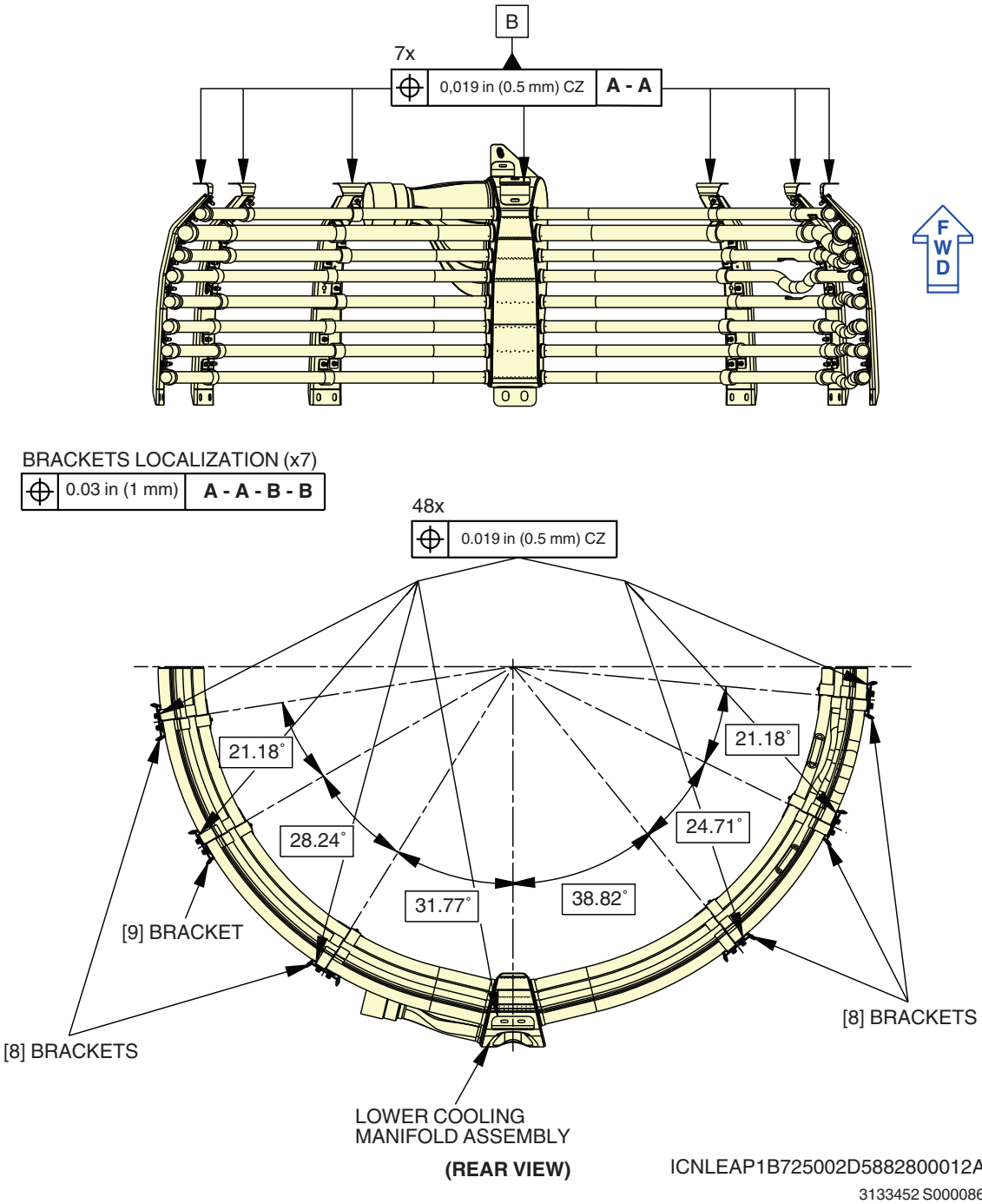
Loop Clamp and Bracket Replacement
Figure 801/72-50-02-990-807-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Bracket Position
Figure 802/72-50-02-990-808-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Bracket Position
Figure 802/72-50-02-990-808-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TURBINE REAR VANE MODULE- INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Turbine rear vane module inspection.

TASK 72-50-03-200-801-G00

2. Turbine Rear Vane Module Inspection

(Figure 601, Figure 602, Figure 603)

A. General

(1) This task gives the instructions to do the inspection of turbine rear vane module.

B. References

Reference	Title
71-00-00-800-802-G00	Engine Operation Limits (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
71-71-00-200-801-G00	Engine Vents and Drains Inspection (P/B 601)
72-00-00-100-811-G00	Exhaust Tube Support - Cleaning (P/B 701)
72-00-00-100-812-G00	Injection Nozzle - Cleaning (P/B 701)
72-00-00-100-813-G00	Exhaust Air Tube - Cleaning (P/B 701)
72-00-00-100-814-G00	Outer Exhaust Air Tube - Cleaning (P/B 701)
72-00-00-100-815-G00	Rear Ventilation Tube - Cleaning (P/B 701)
78-11-02-000-801-G00	Primary Plug Assembly Removal (P/B 401)
78-11-02-400-801-G00	Primary Plug Assembly Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-25834	Coking Checking Inspection Equipment - Vent Tube Part #: 956A8462G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-50-03-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-50-03-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-50-03-010-002-G00

(3) Do this task: Primary Plug Assembly Removal, TASK 78-11-02-000-801-G00.

G. Turbine Rear Vane Module Inspection

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0075 AND PRE SB CFM-LEAP-72-0241 AND PRE SB CFM-LEAP-72-0359

SUBTASK 72-50-03-211-001-G00

(1) Do these steps to examine the turbine rear vane [1] (Figure 601 (Sheet 1) and Figure 601 (Sheet 5)):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

2) Cracks in weld.

a) Not permitted.

(b) Examine the Hub for these conditions (Figure 601 (Sheet 5)):

1) Oil marks on inner face

a) Any amount is serviceable.

2) Scratches and scores

a) Any number up to 0.059 in. (1.5 mm) long and 0.00295 in. (0.075 mm) depth.

3) Nicks, dents

a) Any number up to 0.007 in. (0.2 mm) depth.

(c) Examine the Outer Casing for these conditions (Figure 601 (Sheet 5)):

1) Scratches and scores

a) Any number up to 0.11 in. (3 mm) long and 0.0059 in. (0.15 mm) depth.

2) Nicks, dents

a) Any number up to 0.019 in. (0.5 mm) depth.

(d) Examine the Struts for these conditions (Figure 601 (Sheet 5)):

1) Scratches and scores

a) number up to 0.98 in. (25 mm) long and 0.015 in. (0.4 mm) depth.

2) Nicks, dents

a) Any number up to 0.015 in. (0.4 mm) depth.

(e) Examine the Outer platform for these conditions (Figure 601 (Sheet 5)):

1) Scratches and scores

a) Not permitted.

2) Nicks, dents

a) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0075 AND PRE SB CFM-LEAP-72-0241 AND PRE SB CFM-LEAP-72-0359 (Continued)

(f) Examine the Inner platform for these conditions (Figure 601 (Sheet 5)):

1) Scratches and scores

a) Not permitted.

2) Nicks, dents

a) Not permitted.

(g) Do these steps to examine the Outer Aft flange (Figure 601 (Sheet 5)):

1) Examine all areas for these conditions:

a) Scratches and scores

<1> Any number up to 0.043 in. (1.1 mm) long and 0.0019 in. (0.05 mm) depth.

b) Nicks, dents

<1> Any number up to 0.007 in. (0.2 mm) depth.

2) Examine mating surface (Surf. J) for these conditions (Figure 601 (Sheet 5)):

a) Fretting

<1> Any amount if dimension H is between 0.12 in. (3 mm) and 0.137 in. (3.5 mm).

(h) Do these steps to examine the Inner Aft flange (Figure 601 (Sheet 5)):

1) Examine all areas for these conditions:

a) Scratches and scores

<1> Not permitted.

b) Nicks, dents

<1> Not permitted.

2) Examine mating surface (Surf. M) for these conditions (Figure 601 (Sheet 5)):

a) Fretting

<1> Any amount if dimension N is between 0.12 in. (3 mm) and 0.133 in. (3.4 mm).

(i) Examine the Inner Forward flange for these conditions (Figure 601 (Sheet 5)):

1) Scratches and scores

a) Any number up to 0.074 in. (1.9 mm) in length and 0.003 in. (0.1 mm) in depth.

2) Nicks and dents

a) Any number up to 0.009 in. (0.23 mm) in depth.

(j) Examine the Outer Forward flange for these conditions (Figure 601 (Sheet 5)):

1) Scratches and scores

a) Not serviceable.

2) Nicks and dents

a) Not serviceable.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-CFM-LEAP-1B-72-0075 OR POST SB CFM-LEAP-72-0241 OR POST SB CFM-LEAP-72-0359

SUBTASK 72-50-03-211-012-G00

(2) Do these steps to examine the turbine rear vane [1] (Figure 601 (Sheet 12), Figure 601 (Sheet 13), Figure 601 (Sheet 14), and Figure 601 (Sheet 15)):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

2) Cracks in weld

a) Not permitted.

(b) Examine the Hub for these conditions Area 2 (Figure 601 (Sheet 12)):

1) Scratches and scores

a) Any number up to 0.059 in. (1.5 mm) in length and 0.00295 in. (0.075 mm) in depth.

2) Nicks and dents

a) Any number up to 0.007 in. (0.2 mm) in depth.

(c) Examine the Outer Casing for these conditions Area 4 (Figure 601 (Sheet 13)):

1) Scratches and scores

a) Any number up to 0.11 in. (3 mm) in length and 0.0059 in. (0.15 mm) in depth.

2) Nicks and dents

a) Any number up to 0.019 in. (0.5 mm) in depth.

(d) Examine the Struts for these conditions (Figure 601 (Sheet 13)):

1) Scratches and scores

a) Any number up to 0.98 in. (25 mm) in length and 0.015 in. (0.4 mm) in depth.

2) Nicks and dents

a) Any number up to 0.015 in. (0.4 mm) in depth.

(e) Examine the Inner Aft flange (Figure 601 (Sheet 14)):

1) Scratches and scores

a) Not permitted.

2) Nicks and dents

a) Not permitted.

3) Wear on the mating surface (surface M)

a) Any number up to 0.007 in. (0.2 mm) in depth if dimension N is between 0.118 in. (3 mm) and 0.133 in. (3.4 mm).

(f) Examine the Inner Forward flange for these conditions (Figure 601 (Sheet 14)):

1) Scratches and scores

a) Not permitted.

2) Nicks and dents

a) Not permitted.

(g) Examine the Outer Forward flange for these conditions (Figure 601 (Sheet 15)):

1) Scratches and scores

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-CFM-LEAP-1B-72-0075 OR POST SB CFM-LEAP-72-0241 OR POST SB CFM-LEAP-72-0359 (Continued)

a) Not permitted.

2) Nicks and dents

a) Not permitted.

(h) Examine the shield (Figure 601 (Sheet 15)):

1) Scratches and scores

a) Any number up to 5.51 in. (140 mm) in length and 0.0015 in. (0.04 mm) in depth.

2) Nicks and dents

a) Not permitted.

3) Discoloration

a) Not permitted.

(i) Examine the drain (Figure 601 (Sheet 16), Figure 601 (Sheet 17), or Figure 601 (Sheet 18)):

1) Scratches and scores

a) Not permitted.

2) Nicks and dents

a) Not permitted.

(j) Examine the Outer aft flange for these conditions (Figure 601 (Sheet 19)):

1) Wear on the mating surface (surface J)

a) Any number up to 0.007 in. (0.2 mm) in depth if dimension H is between 0.118 in. (3 mm) and 0.141 in. (3.6 mm).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0175

SUBTASK 72-50-03-211-009-G00

(3) Do these steps to examine TBV duct [2] (Figure 602):

(a) Examine the outer surface of an elbow for these conditions (Figure 602 (Sheet 1) and Figure 602 (Sheet 4)):

1) Scratches

a) Permitted if depth is less than to 0.002 in. (0.05 mm).

2) Flat area

a) Permitted if the values are as defined in table 1 (Figure 602 (Sheet 4)).

b) A maximum of 1 flat area on a duct is permitted.

3) Wear

a) Permitted if depth is less than to 0.002 in. (0.05 mm).

4) Corrosion

a) Not permitted.

(b) Examine the inner surface of an elbow for these conditions (Figure 602 (Sheet 1) and Figure 602 (Sheet 4)):

1) Scratches

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0175 (Continued)

a) Permitted if depth is less than to 0.004 in. (0.10 mm).

2) Flat area

a) Permitted if the values are as defined in table 1 (Figure 602 (Sheet 4)).

b) A maximum of 1 flat area on a duct is permitted.

3) Wear

a) Permitted if depth is less than to 0.004 in. (0.10 mm).

4) Corrosion

a) Permitted if depth is less than to 0.004 in. (0.10 mm).

(c) Examine the straight duct for these conditions (Figure 602 (Sheet 1) and Figure 602 (Sheet 4)):

1) Scratches

a) Permitted if depth is less than to 0.004 in. (0.10 mm).

2) Flat area

a) Permitted if the values are as defined in table 1 (Figure 602 (Sheet 4)).

b) A maximum of 1 flat area on a duct is permitted.

3) Wear

a) Permitted if depth is less than to 0.004 in. (0.10 mm).

4) Corrosion

a) Permitted if depth is less than to 0.004 in. (0.10 mm).

(d) Examine the bending for these conditions:

1) Wrinkles (longitudinal waviness)

a) Permitted if waviness depth is less than to 2% of nominal diameter.

- 0.05 in. (1.27 mm) for a DN of 2.5 in. (63.50 mm).

2) Distortion

a) Permitted if distortion is less than to 5% of nominal diameter.

- 0.15 in. (3.81 mm) for a diameter of 3.00 in. (76.20 mm)
- 0.125 in. (3.175 mm) for a diameter of 2.5 in. (63.5 mm).

(e) Examine the oblong duct for these conditions:

1) Orange peel effect (Figure 602 (Sheet 3)):

a) Permitted.

2) Distortion

a) Permitted for the external area only if value does not exceed (Figure 602 (Sheet 2)).

(f) Examine the welded joint for these conditions (Figure 602 (Sheet 3)):

1) Scratches, shocks and dents

a) Not permitted.

(g) Examine TBV flange for these conditions (Figure 602 (Sheet 2)):

1) Nick, dent

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0175 (Continued)

a) Not permitted.

2) Discoloration

a) Permitted if the material is not crazed (No appearance of the granular structure of the material).

(h) Examine TBV flange sealing face for these conditions (Figure 602 (Sheet 2)):

1) Scratch

a) Not permitted.

(i) Examine TBV flange other face for these conditions (Figure 602 (Sheet 2)):

1) Scratch

a) Permitted if:

- Depth is less than to 0.0060 in. (0.15 mm)
- And if length is less than to 0.1575 in. (4.00 mm).

(j) Examine V-band flange for these conditions (Figure 602 (Sheet 2)):

1) Nick, dent

a) Not permitted.

(k) Examine V-band flange sealing face for these conditions (Figure 602 (Sheet 2)):

1) Scratch

a) Not permitted.

(l) Examine V-band flange other face for these conditions (Figure 602 (Sheet 2)):

1) Scratch

a) Permitted if:

- Depth is less than to 0.0060 in. (0.15 mm)
- And if length is less than to 0.1575 in. (4.00 mm).

(m) Examine all other areas:

1) Flat area

a) Not permitted.

2) Corrosion

a) Permitted if depth is less than to 0.004 in. (0.10 mm).

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0075

SUBTASK 72-50-03-211-002-G00

(4) Do these steps to examine the drain tubes [3] (Figure 601 (Sheet 1) and Figure 601 (Sheet 2)):

(a) Examine all areas except for the standard connecting part for these conditions:

1) Cracks, break, and torn

a) Not permitted.

2) Flat areas, Dents

a) Not permitted.

3) Nicks, scratches, and wear

a) Not permitted.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL POST SB 737-CFM-LEAP-1B-72-0075 AND PRE SB CFM-LEAP-72-0241

SUBTASK 72-50-03-211-010-G00

(5) Do these steps to examine the drain tubes [3] (Figure 601 (Sheet 3) and Figure 601 (Sheet 16)):

(a) Examine all areas except for the standard connecting part for these conditions:

1) Cracks, break, and torn

a) Not permitted.

2) Flat areas, Dents

a) Not permitted.

3) Nicks, scratches, and wear

a) Not permitted.

PTF ALL POST SB CFM-LEAP-72-0241 AND POST SB CFM-LEAP-72-0359

SUBTASK 72-50-03-211-011-G00

(6) Do these steps to examine the rear drain tubes [3] (Figure 601 (Sheet 4), Figure 601 (Sheet 17), and Figure 601 (Sheet 18)):

(a) Examine all areas for these conditions:

1) Cracks, break, and torn

a) Not permitted.

2) Flat areas, Dents

a) Not permitted.

3) Nicks, scratches, and wear

a) Not permitted.

PTF ALL

SUBTASK 72-50-03-211-003-G00

(7) Do these steps to examine the exhaust tube support [4] (Figure 601 (Sheet 1) and Figure 601 (Sheet 7)):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

2) Scores and Scratches

a) Any number up to 0.019 in. (0.5 mm) in depth, 0.39 in. (9.9 mm) in length.

3) Nicks and dents

a) Any number up to 0.059 in. (1.5 mm) in depth.

4) Coking

a) Permitted if the section of the exhaust tube support [4] is not obstructed over 28% of area (minimum inner diameter of 2.0 in. (50.8 mm)).

<1> Use the coking checking equipment, SPL-25834, in the rear vent tube (Figure 601 (Sheet 10)).

b) If the section of exhaust tube is obstructed over 28% of area, clean the exhaust tube support [4], do this task: Exhaust Tube Support - Cleaning, TASK 72-00-00-100-811-G00.

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<1> Cleaning must be performed within Continue-In-Service (CIS) limit:

- Within 150 flight cycles, if oil consumption is below the serviceable limit (TASK 71-00-00-800-802-G00).
- Within the remaining flight cycles of the 150 flight cycles CIS limit, if oil leakages are seen at drain tubes and are within CIS limit (TASK 71-71-00-200-801-G00).

(b) Examine the shank nuts for these conditions (Figure 601 (Sheet 7)):

1) Missing

a) Not permitted.

(c) Examine the aft flange for these conditions (Figure 601 (Sheet 7)):

1) Fretting and wear

a) Not permitted.

SUBTASK 72-50-03-211-004-G00

(8) Do these steps to examine the injection nozzle [5] (Figure 601 (Sheet 1) and Figure 601 (Sheet 7)):

(a) Examine all areas except for the standard connecting part for these conditions (Figure 601 (Sheet 7)):

1) Cracks

a) Not permitted.

2) Flat areas

a) Not permitted.

3) Nicks, scratches, chafing, or scores in straight section bends

a) Not permitted.

4) Coking

a) Outer part of the tube: any amount is permitted.

b) Inner part of the tube:

<1> Coking is not permitted, clean coking of the injection nozzle [5], do this task: Injection Nozzle - Cleaning, TASK 72-00-00-100-812-G00 (Figure 601 (Sheet 20)).

<2> Varnish is permitted (Figure 601 (Sheet 20)).

SUBTASK 72-50-03-211-005-G00

(9) Do these steps to examine the air exhaust tubes [6] (Figure 601 (Sheet 1) and Figure 601 (Sheet 8)):

(a) Examine all areas except for the standard connecting part for these conditions (Figure 601 (Sheet 8)):

1) Cracks

a) Not permitted.

2) Flat areas

a) Not permitted.

3) Nicks, scratches, chafing, or scores in straight section bends

a) Not permitted.

4) Coking

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a) The oil coke build-up inside the air exhaust tubes [6] is not permitted and must be cleaned if:

- At least one section of any length of one tube is covered 360° by oil coke build-up (brown surface varnish is acceptable) (Figure 601 (Sheet 21))
- At least one section of any length of one tube is not circular anymore because of one-sided heavy oil coke build-up (Figure 601 (Sheet 21)).

b) If the coking is above limit, clean coking of the air exhaust tube [6] and outer air exhaust tube [6] within CIS limit, do these tasks: Exhaust Air Tube - Cleaning, TASK 72-00-00-100-813-G00 and Outer Exhaust Air Tube - Cleaning, TASK 72-00-00-100-814-G00.

- Within 150 flight cycles, if oil consumption is below the serviceable limit (TASK 71-00-00-800-802-G00).
- Within the remaining flight cycles of the 150 flight cycles CIS limit, if oil leakages are seen at drain tubes and are within CIS limit (TASK 71-71-00-200-801-G00).

SUBTASK 72-50-03-211-013-G00

(10) Do these steps to examine the rear ventilation tube [10] (Figure 603):

(a) Examine the internal section for these conditions:

1) Coking

a) Permitted any amount, if the section of the rear ventilation tube [10] is not obstructed over 20% of area (minimum inner diameter of 1.4 in. (35.6 mm)).

<1> Use the coking checking equipment, SPL-25834, in the rear vent tube (Figure 601 (Sheet 11)).

b) If the section of the rear ventilation tube [10] is obstructed over 20% of area (minimum inner diameter of 1.4 in. (35.6 mm)), clean the coking of the rear ventilation tube [10] within CIS limit, do this task: Rear Ventilation Tube - Cleaning, TASK 72-00-00-100-815-G00.

- Within 150 flight cycles, if oil consumption is below the serviceable limit (TASK 71-00-00-800-802-G00).
- Within the remaining flight cycles of the 150 flight cycles CIS limit, if oil leakages are seen at drain tubes and are within CIS limit (TASK 71-71-00-200-801-G00).

SUBTASK 72-50-03-211-006-G00

(11) Do these steps to examine the spacer [7] (Figure 601 (Sheet 1) and Figure 601 (Sheet 9)):

(a) Examine all areas for these conditions:

1) Scratches and scores

a) Not permitted.

2) Nicks, dents and cracks

a) Not permitted.

(b) Examine the aft flange mating surface (Surf D) for these conditions (Figure 601 (Sheet 9)):

1) Fretting

a) Any amount if dimension B is between 0.12 in. (3 mm) and 0.133 in. (3.4 mm)

(c) Examine the inner faces for these conditions (Figure 601 (Sheet 9)):

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1) Oil marks
   a) Permitted

SUBTASK 72-50-03-211-007-G00

(12) Do these steps to examine the handling mounts [8] (Figure 601 (Sheet 1)):
   (a) Examine all areas for these conditions:
      1) Scratches and scores
         a) Not permitted.
      2) Nicks, dents and cracks
         a) Not permitted.

PTF ALL PRE SB 737-CFM-LEAP-1B-72-0075 AND PRE SB CFM-LEAP-72-0241 AND PRE SB CFM-LEAP-72-0359

SUBTASK 72-50-03-211-008-G00

(13) Do these steps to examine the shield (Figure 601 (Sheet 1)):
   (a) Examine all areas for these conditions:
      1) Discoloration
         a) Any amount.
      2) Scratches
         a) Any amount less than 5.51 in. (140 mm) in length and 0.0015 in. (0.04 mm) in depth.

PTF ALL

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-50-03-410-001-G00

(1) Do this task: Primary Plug Assembly Installation, TASK 78-11-02-400-801-G00.

SUBTASK 72-50-03-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the applicable left or right thrust reversers:
   (a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.
      1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

   (b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.
      1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-50-03-860-004-G00

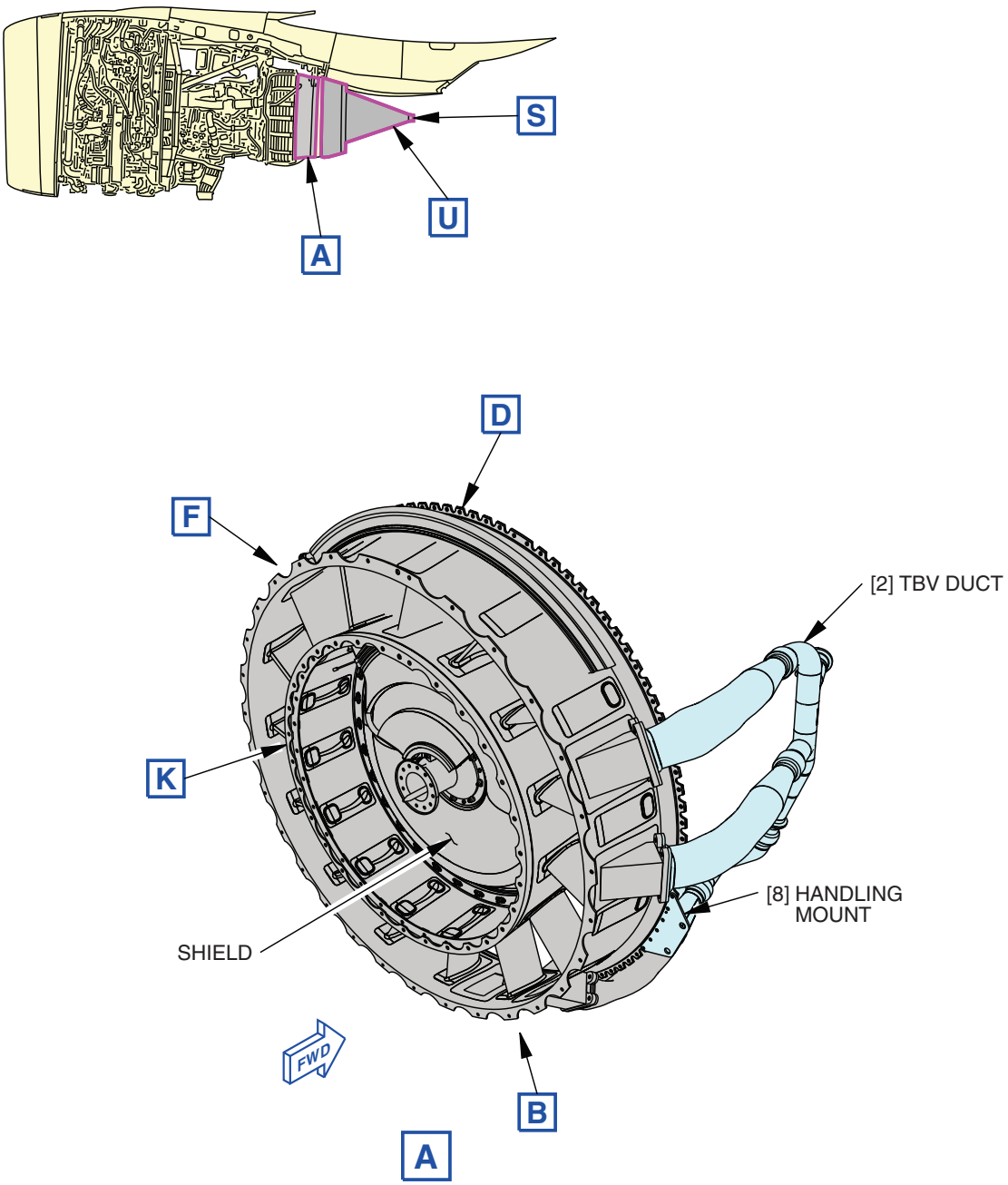
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

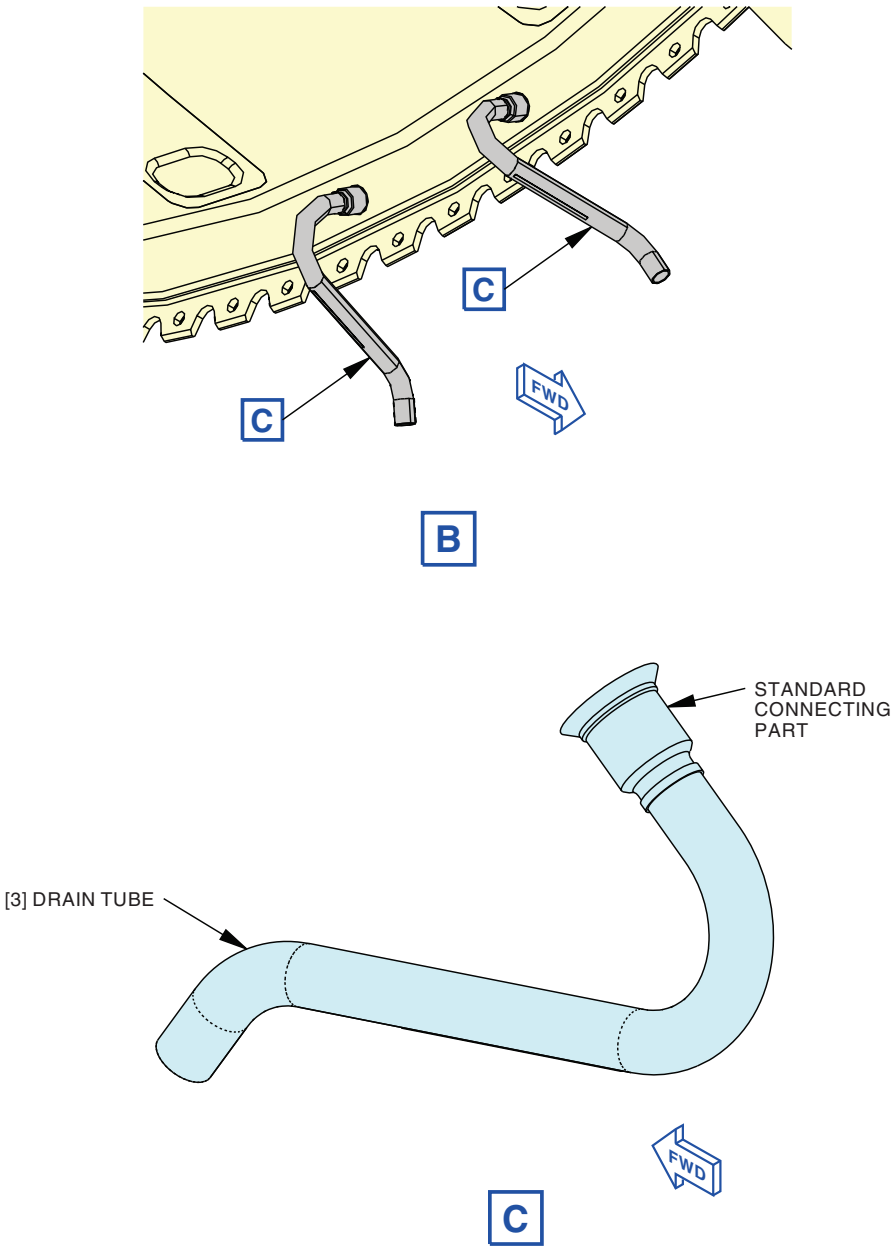


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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 1 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

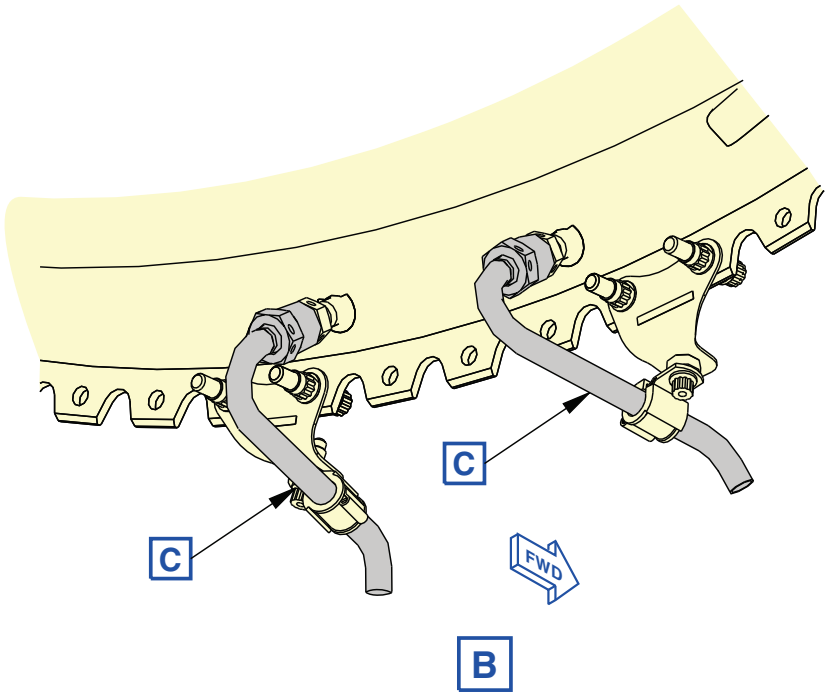


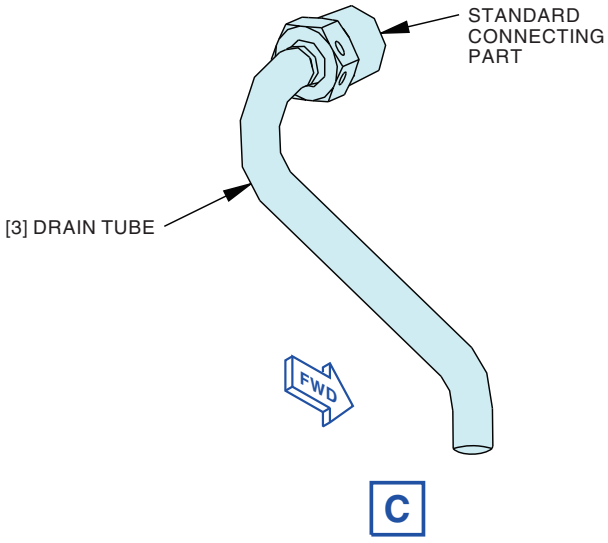
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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 2 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



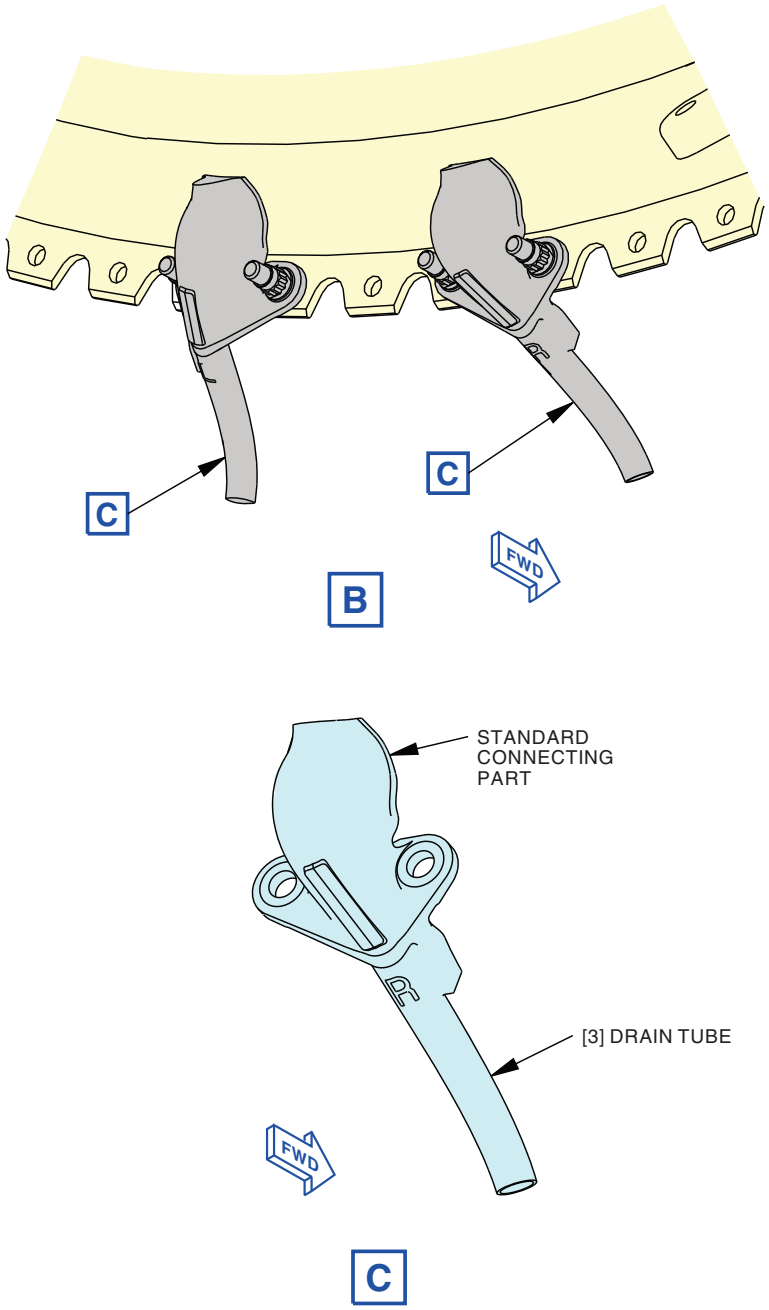


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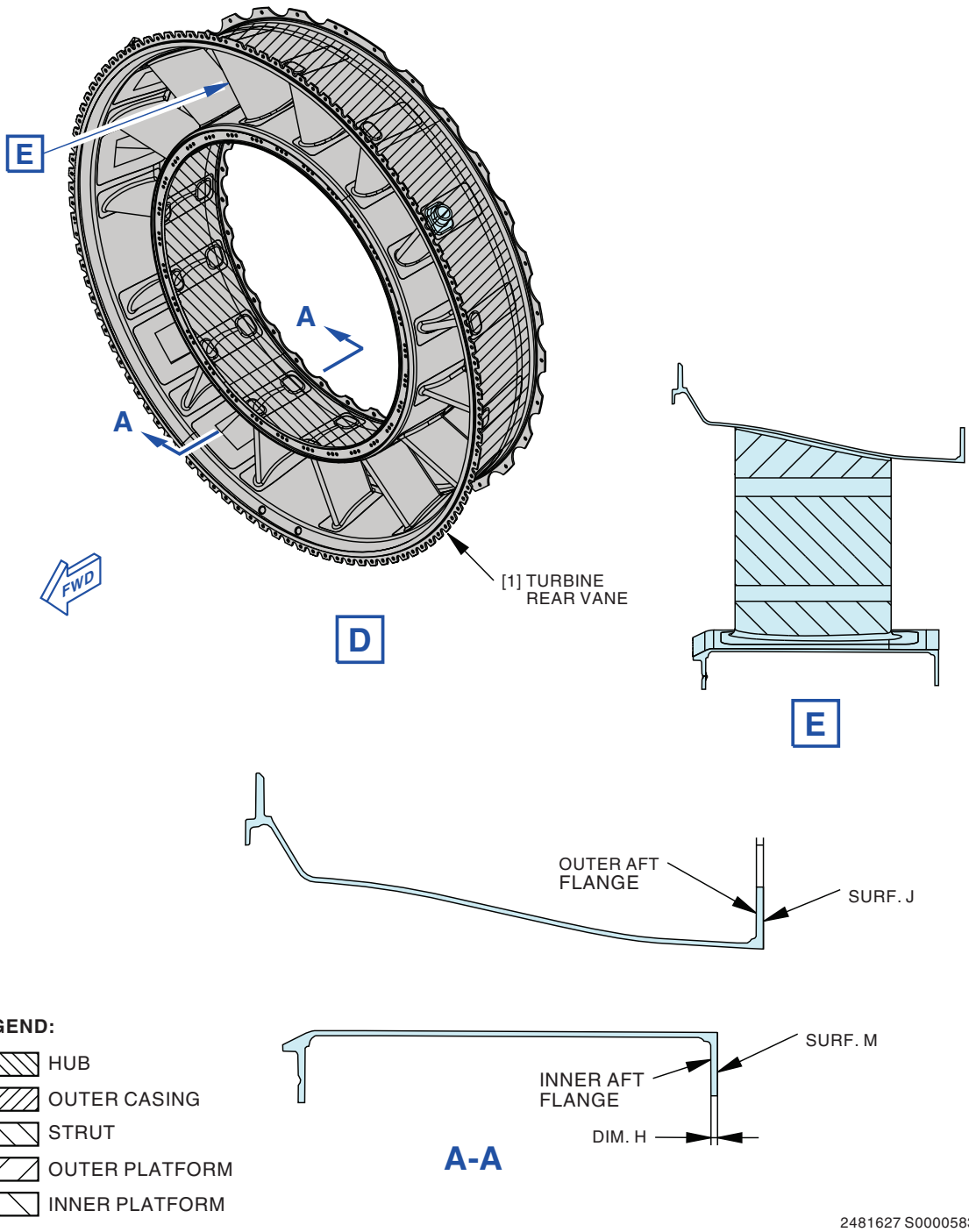
Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 3 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



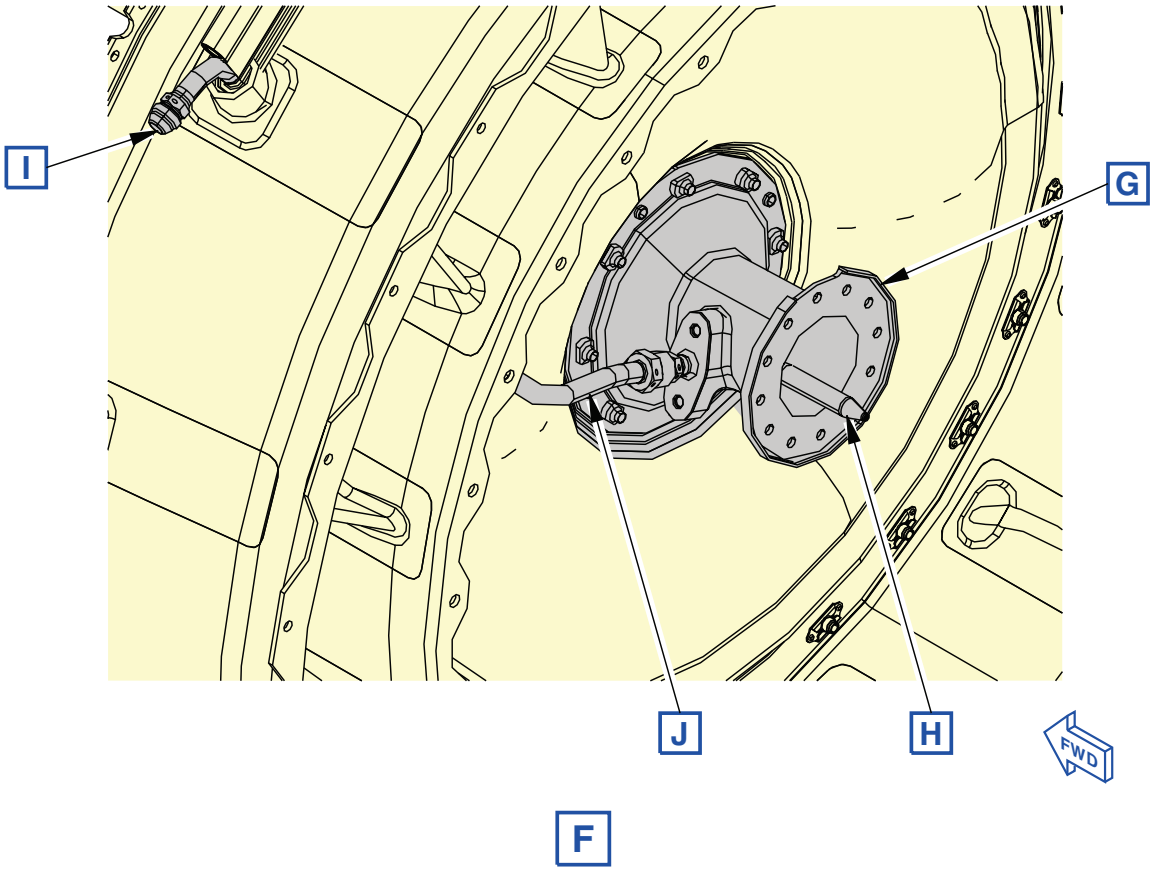
Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 4 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 5 of 21)

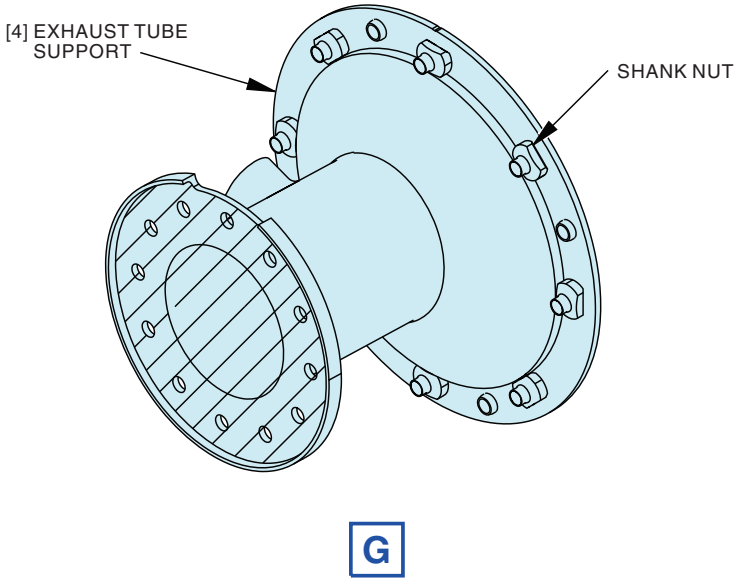
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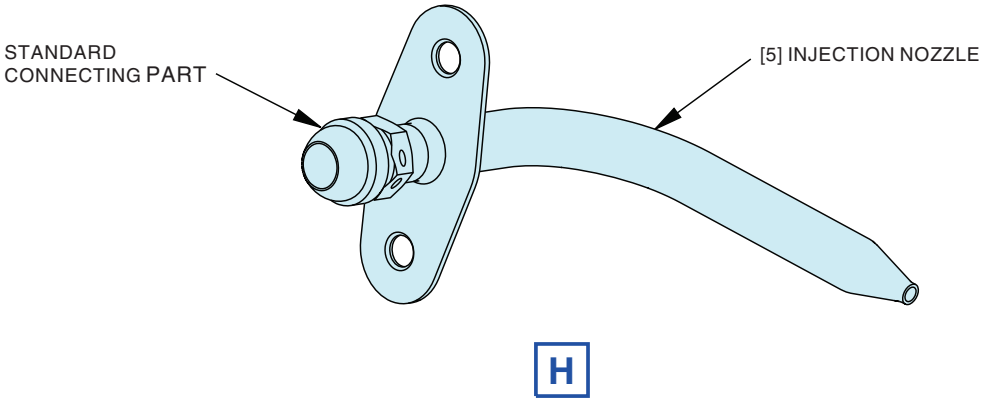


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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 6 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





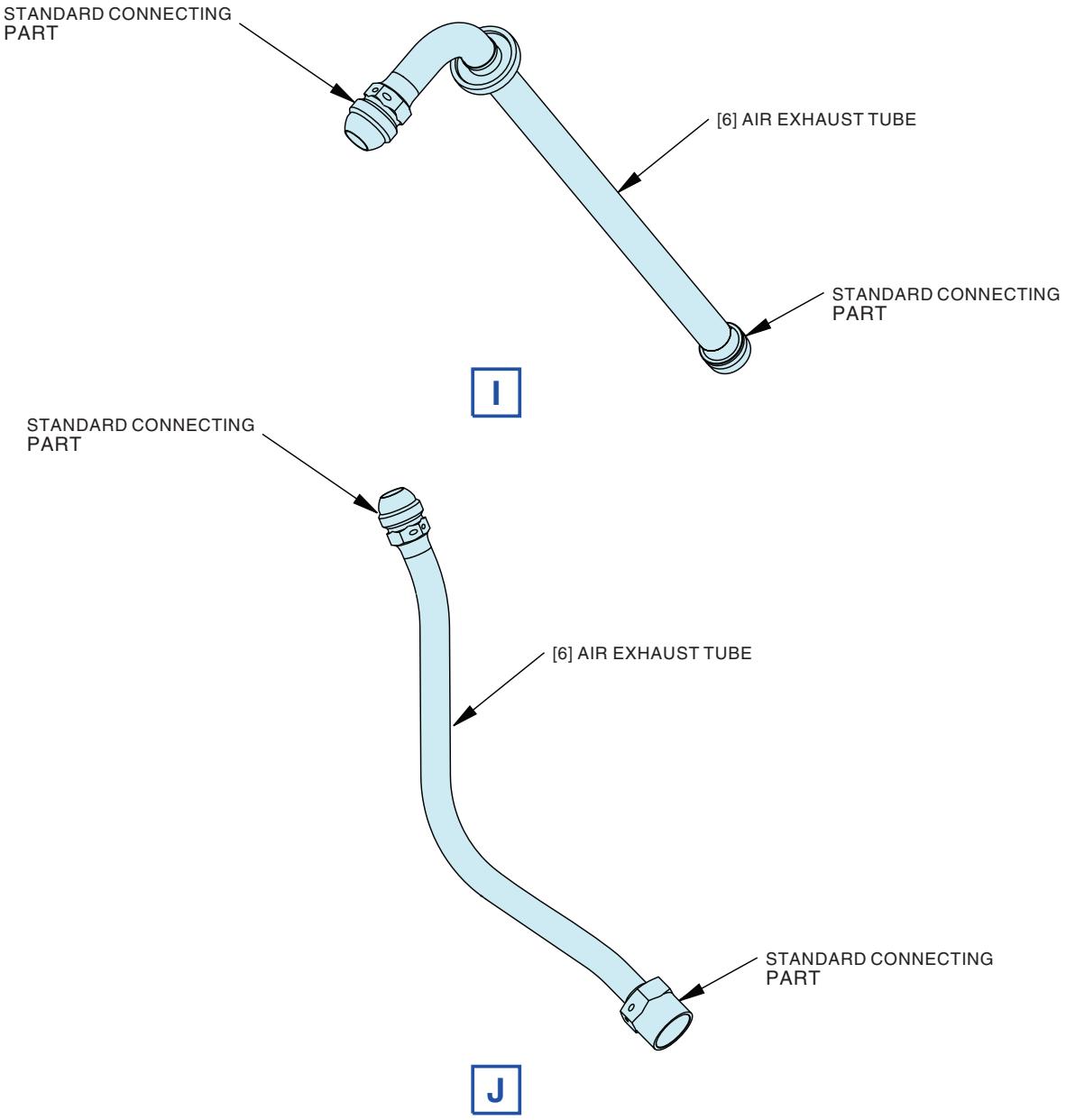
LEGEND:

AFT FLANGE

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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 7 of 21)

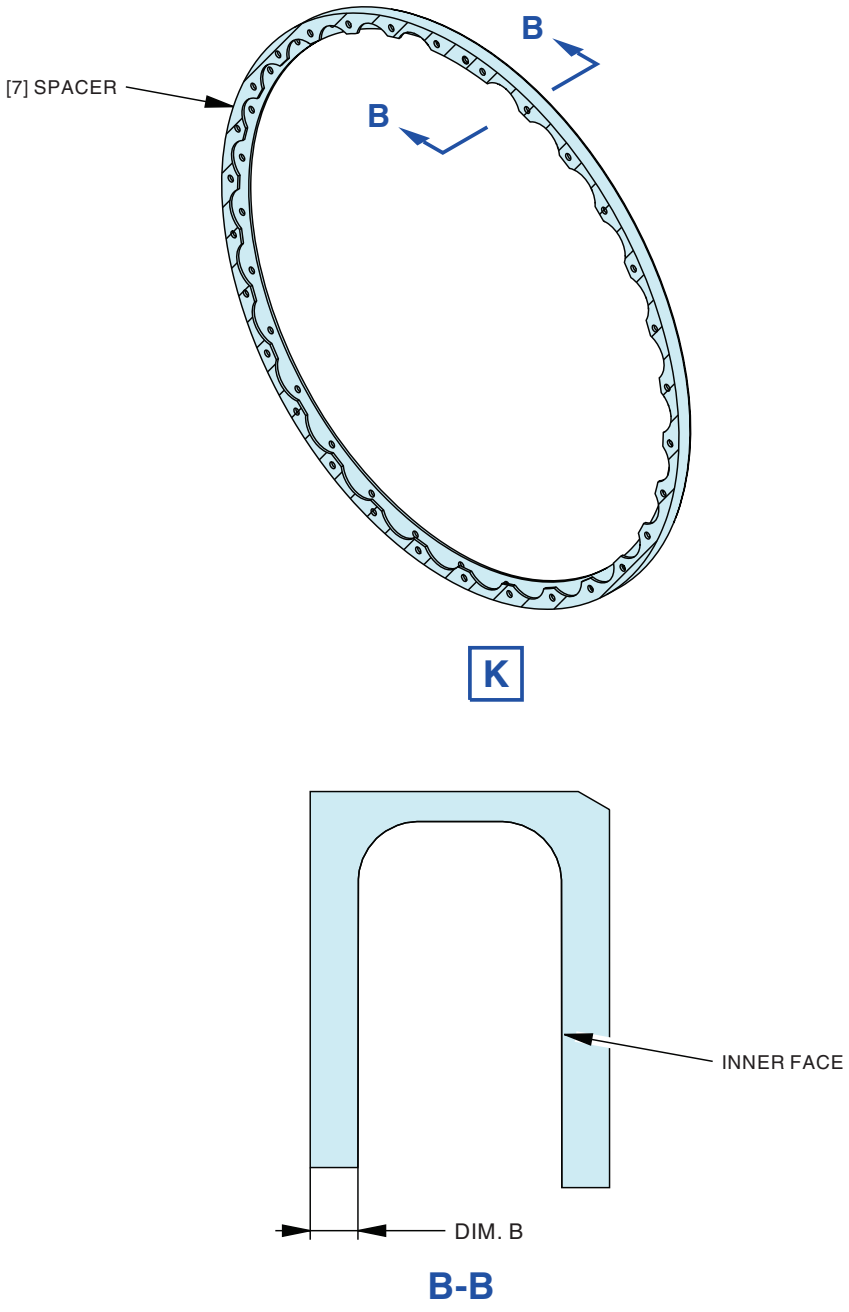
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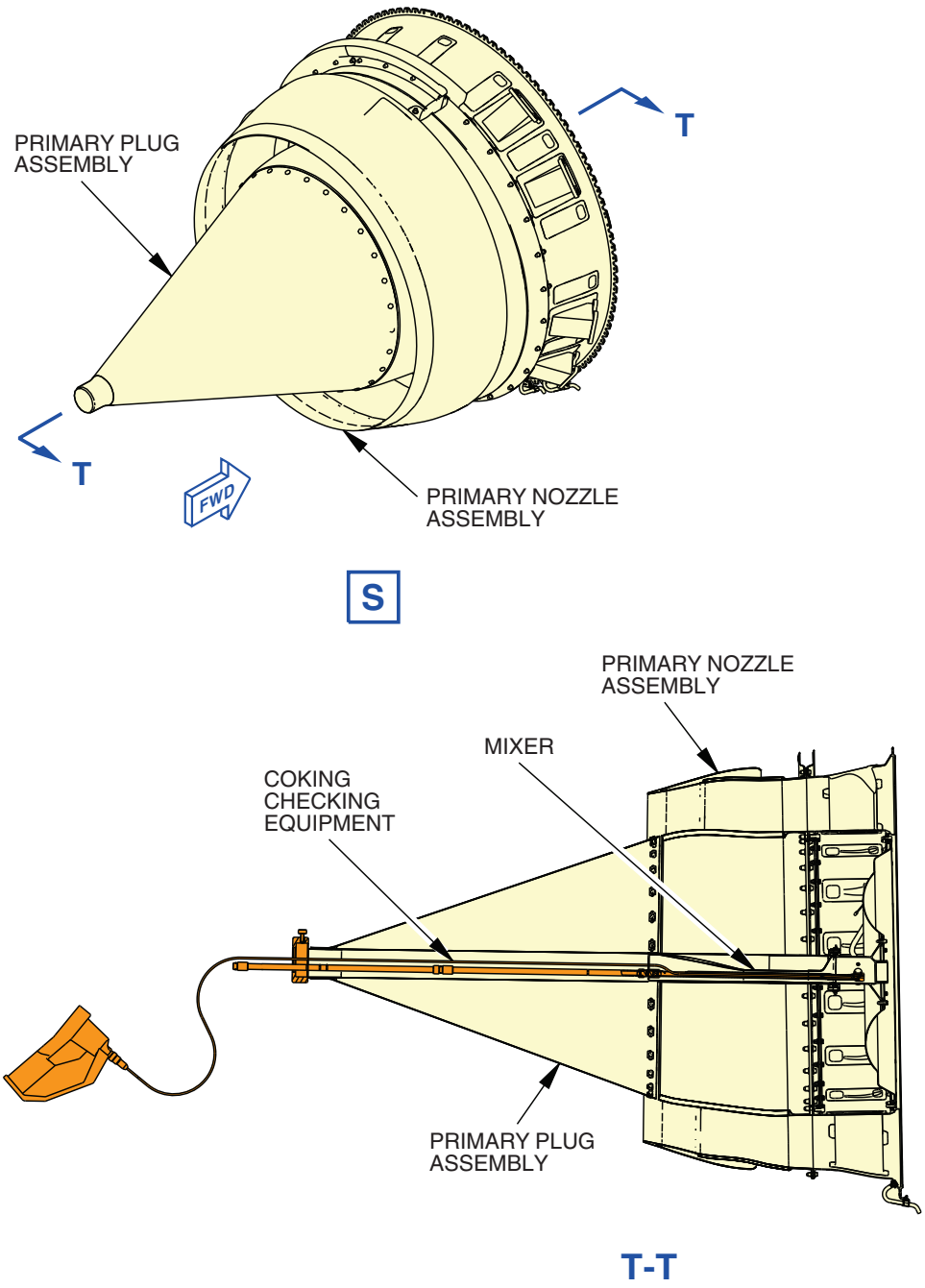
Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 8 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 9 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

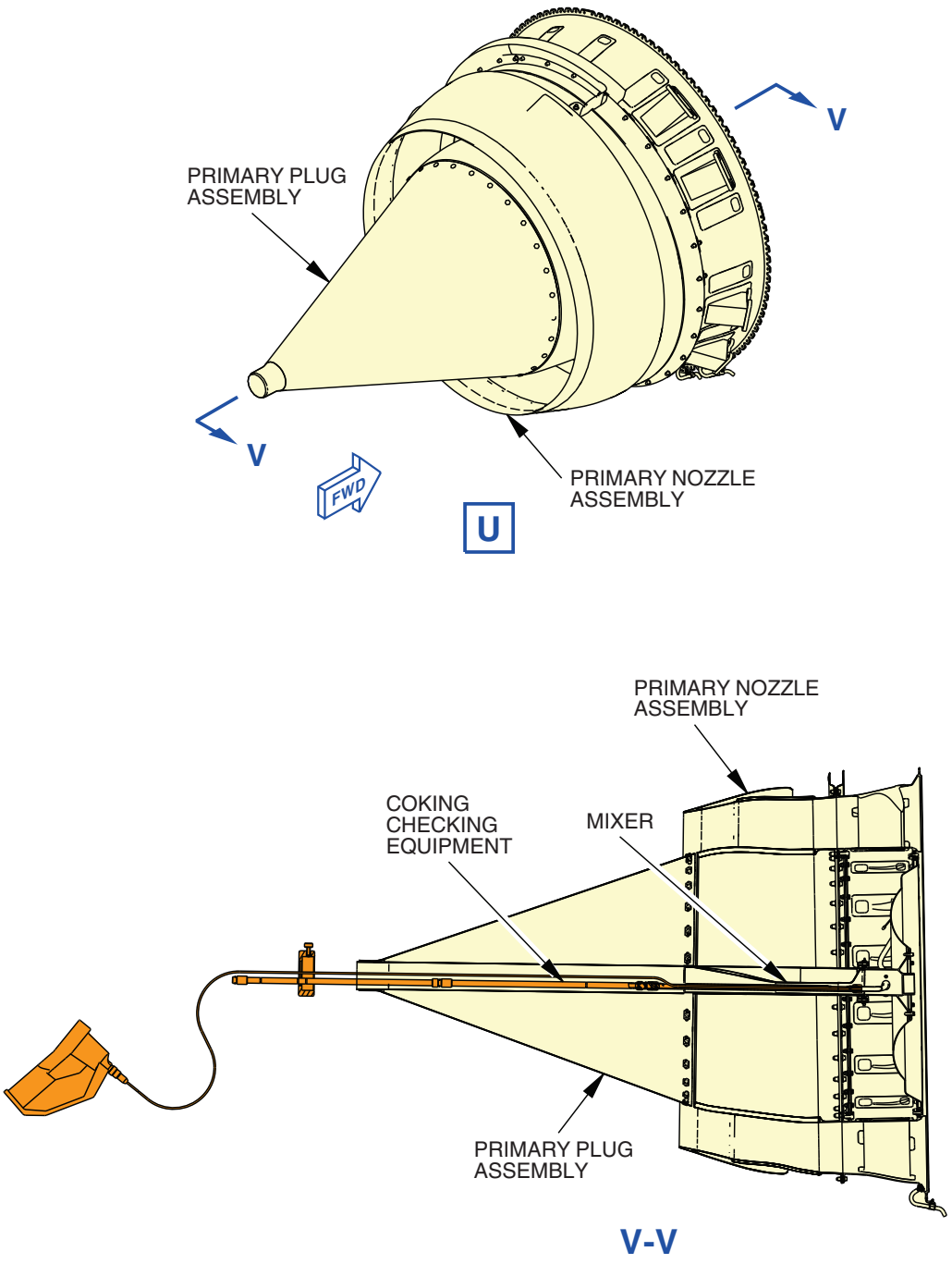


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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 10 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

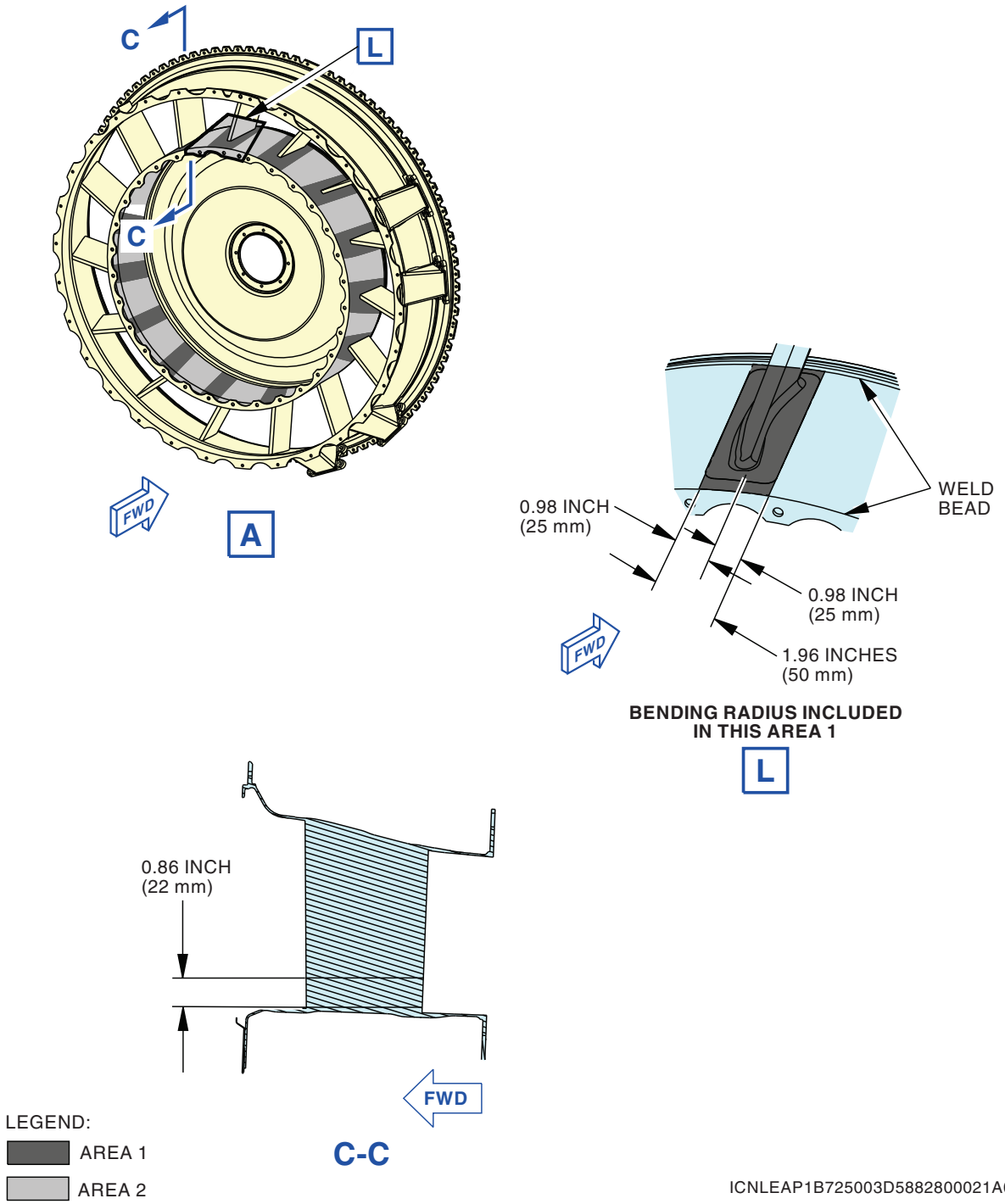


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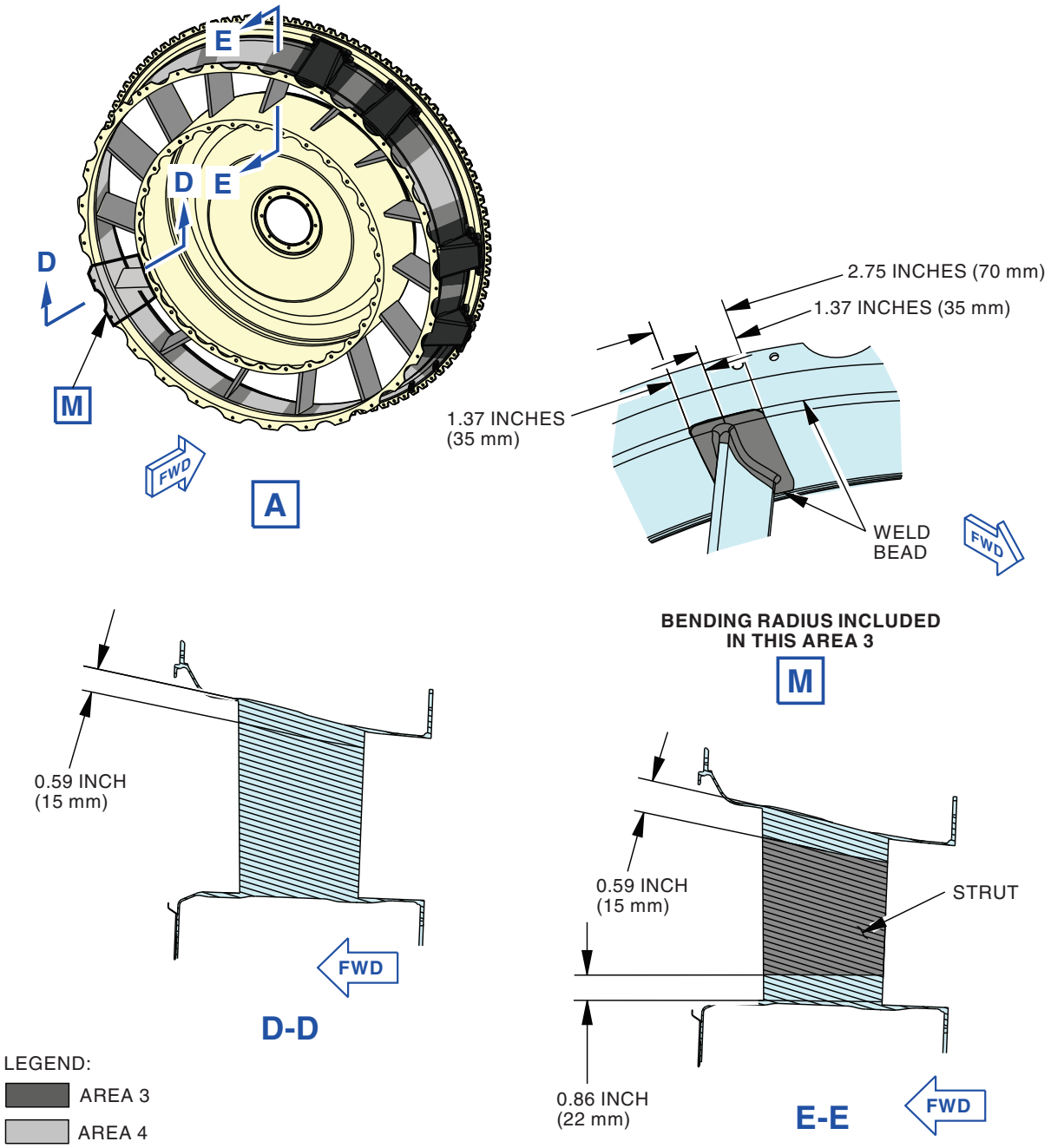
Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 11 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 12 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

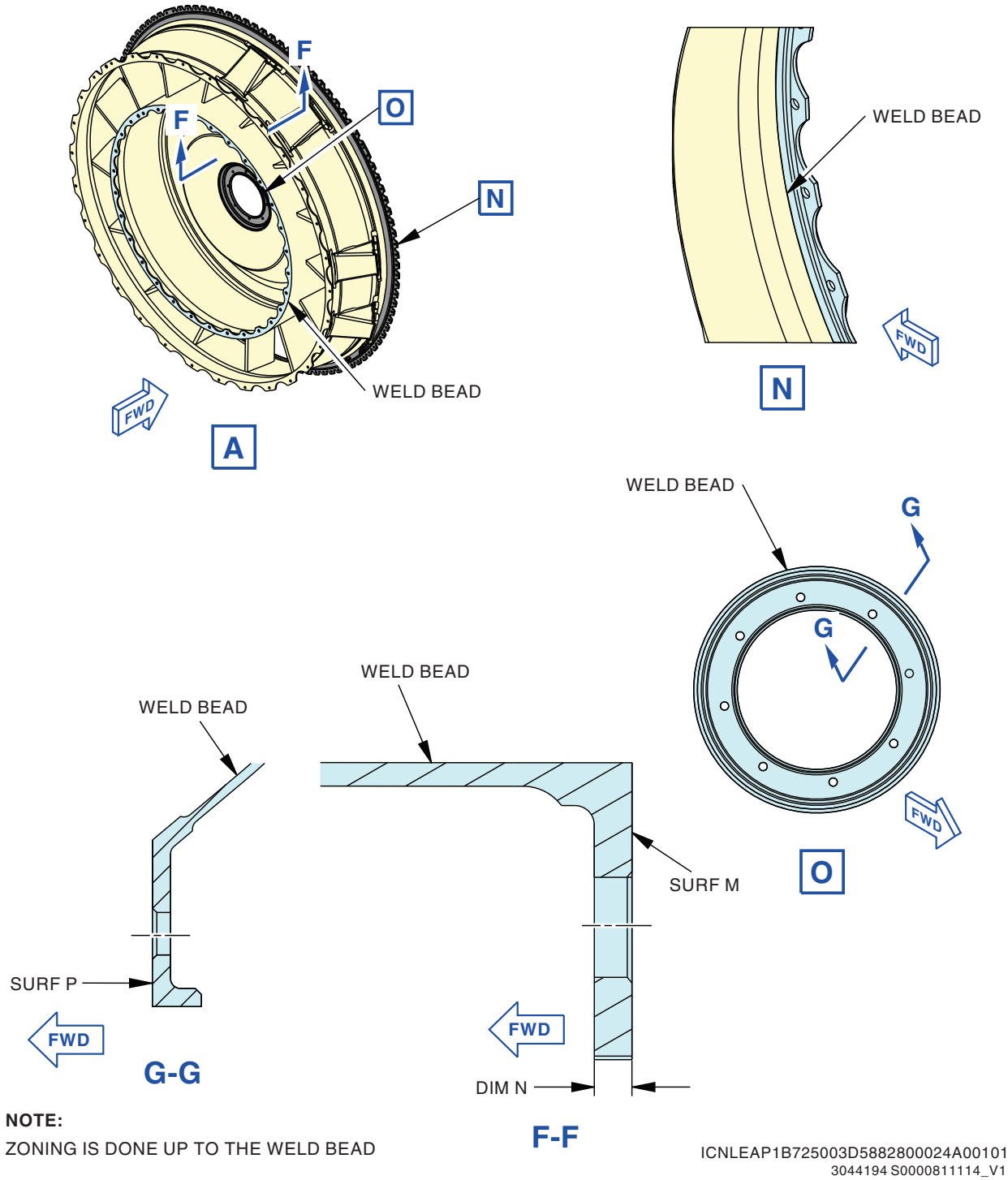


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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 13 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

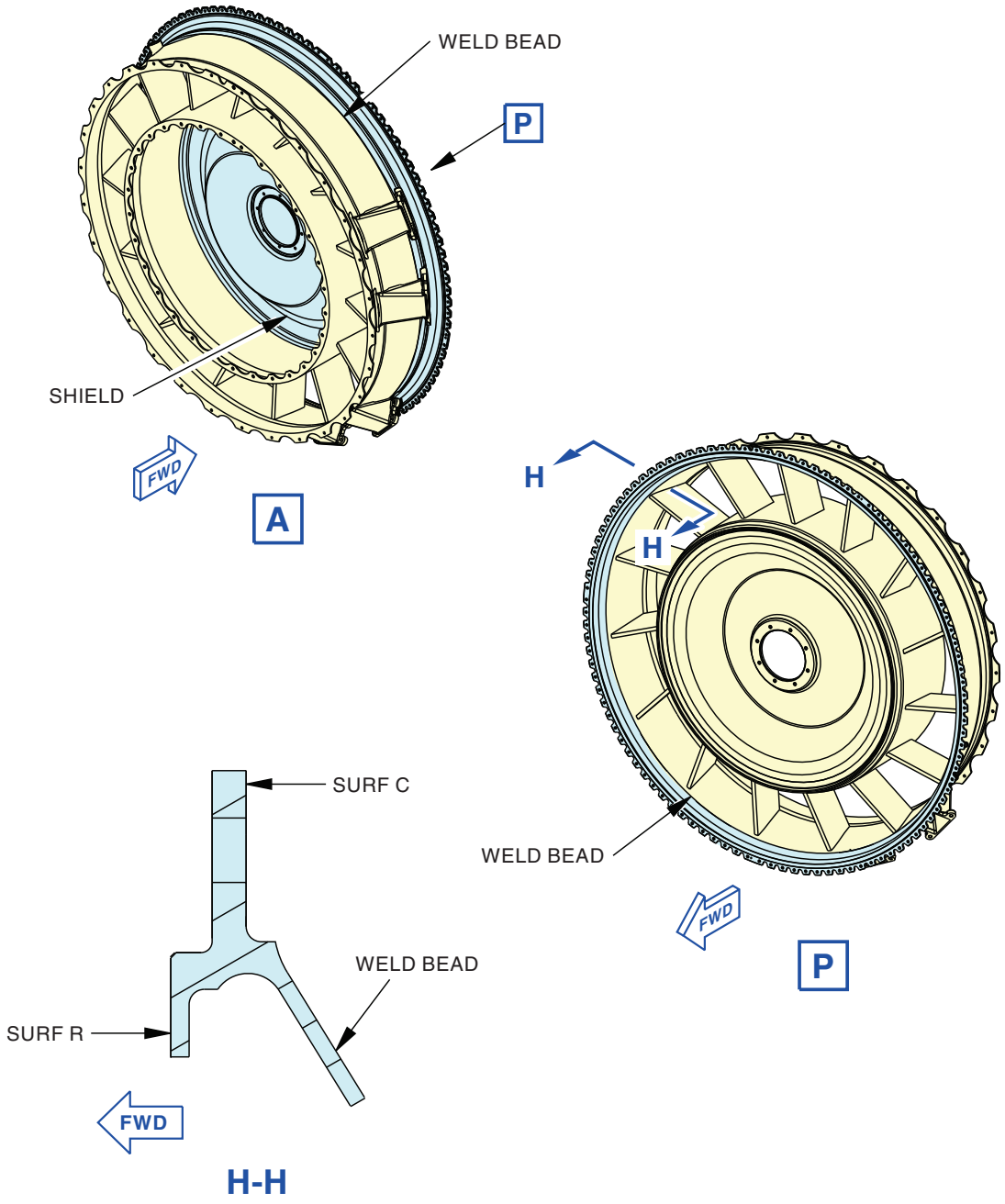


NOTE:
ZONING IS DONE UP TO THE WELD BEAD

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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 14 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

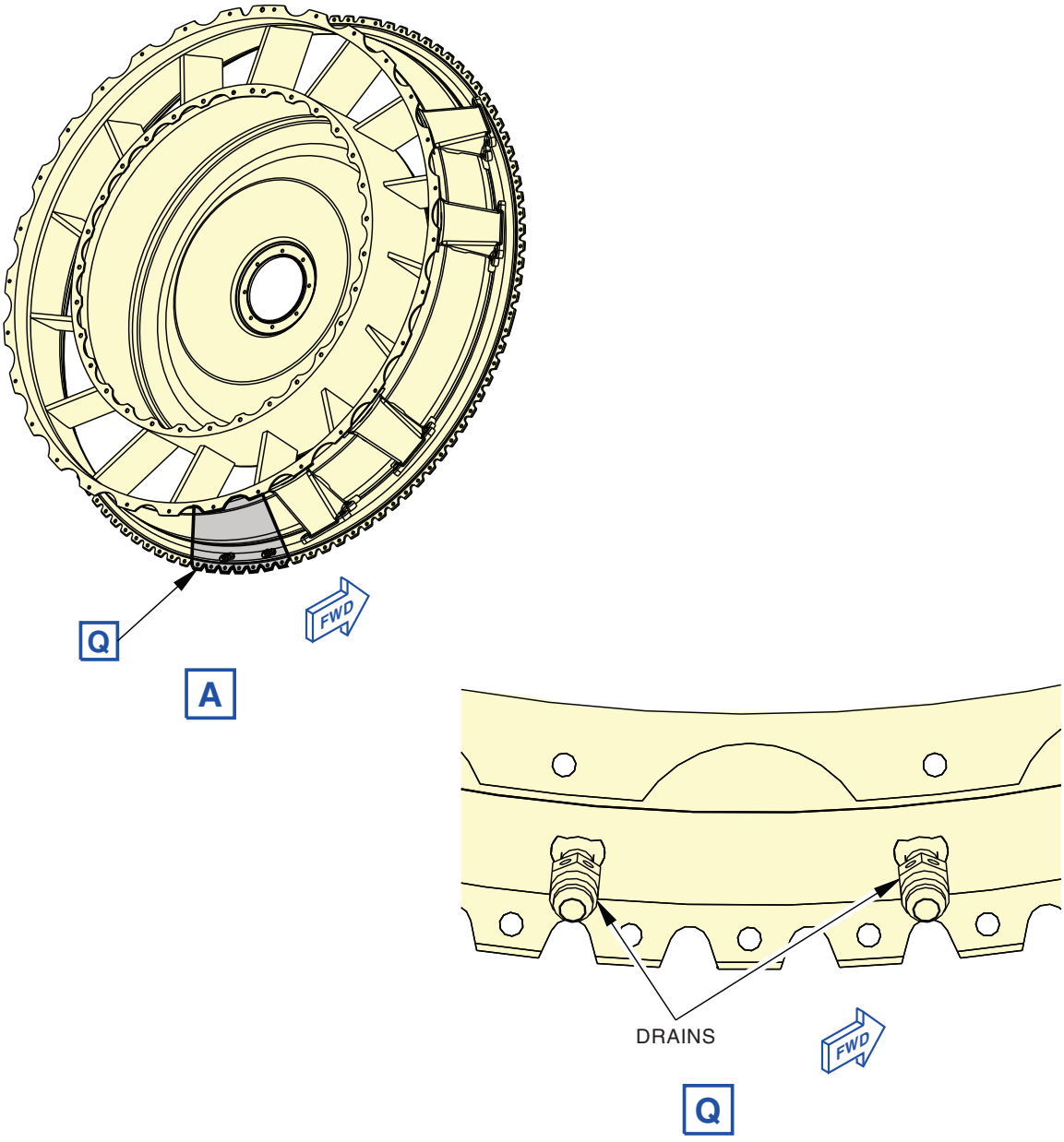
ZONING IS DONE UP TO THE WELD BEAD

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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 15 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

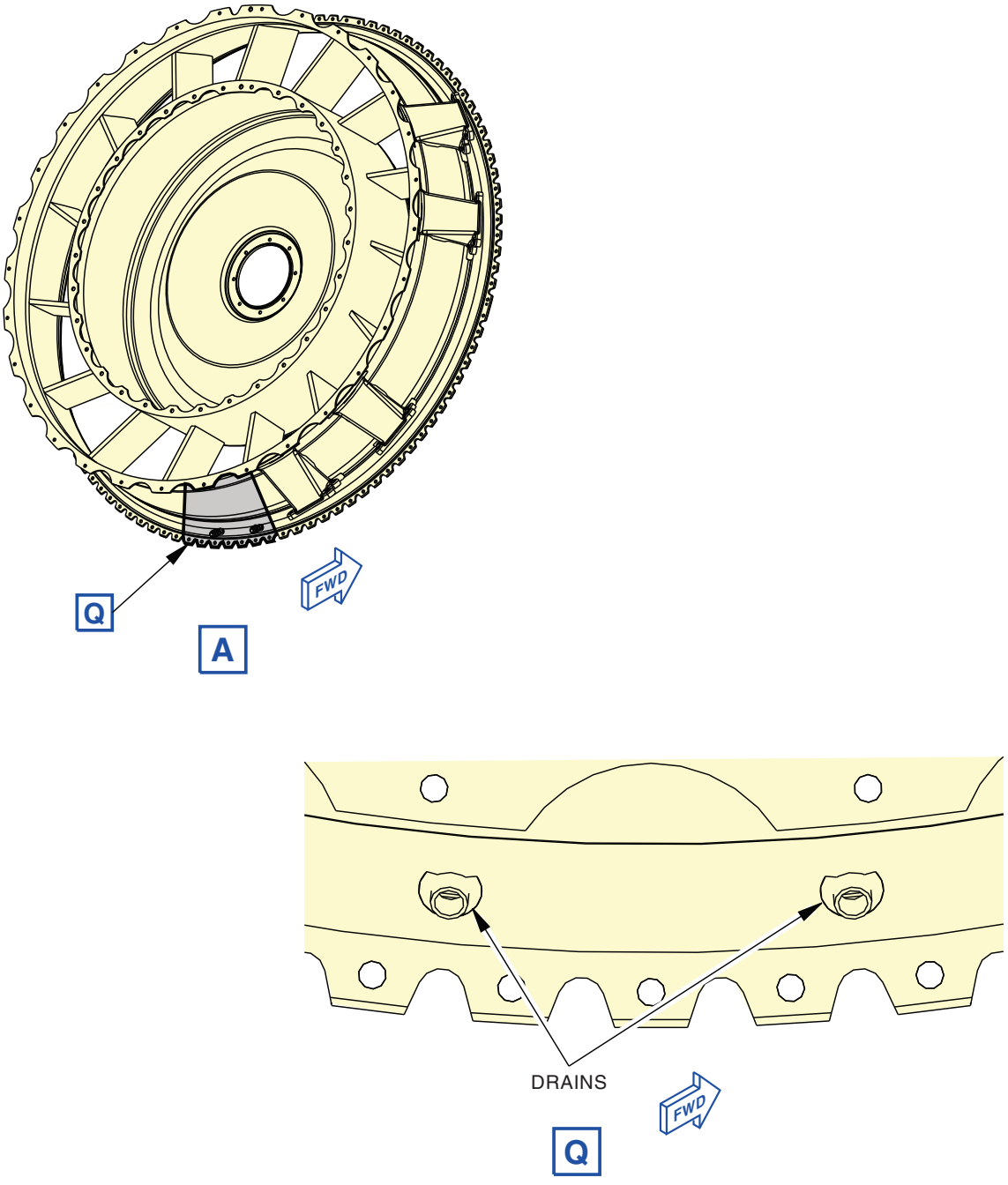


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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 16 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

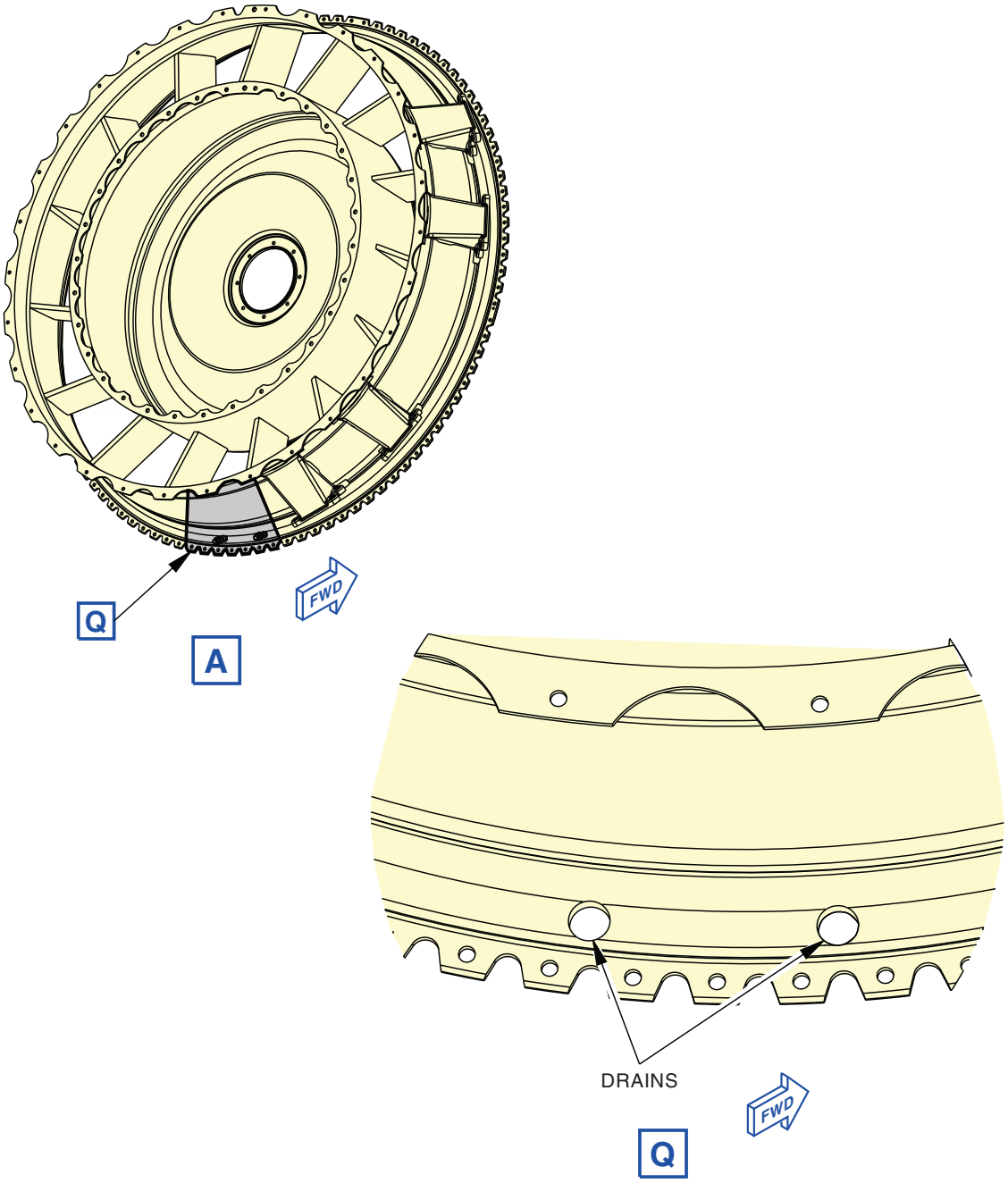


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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 17 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

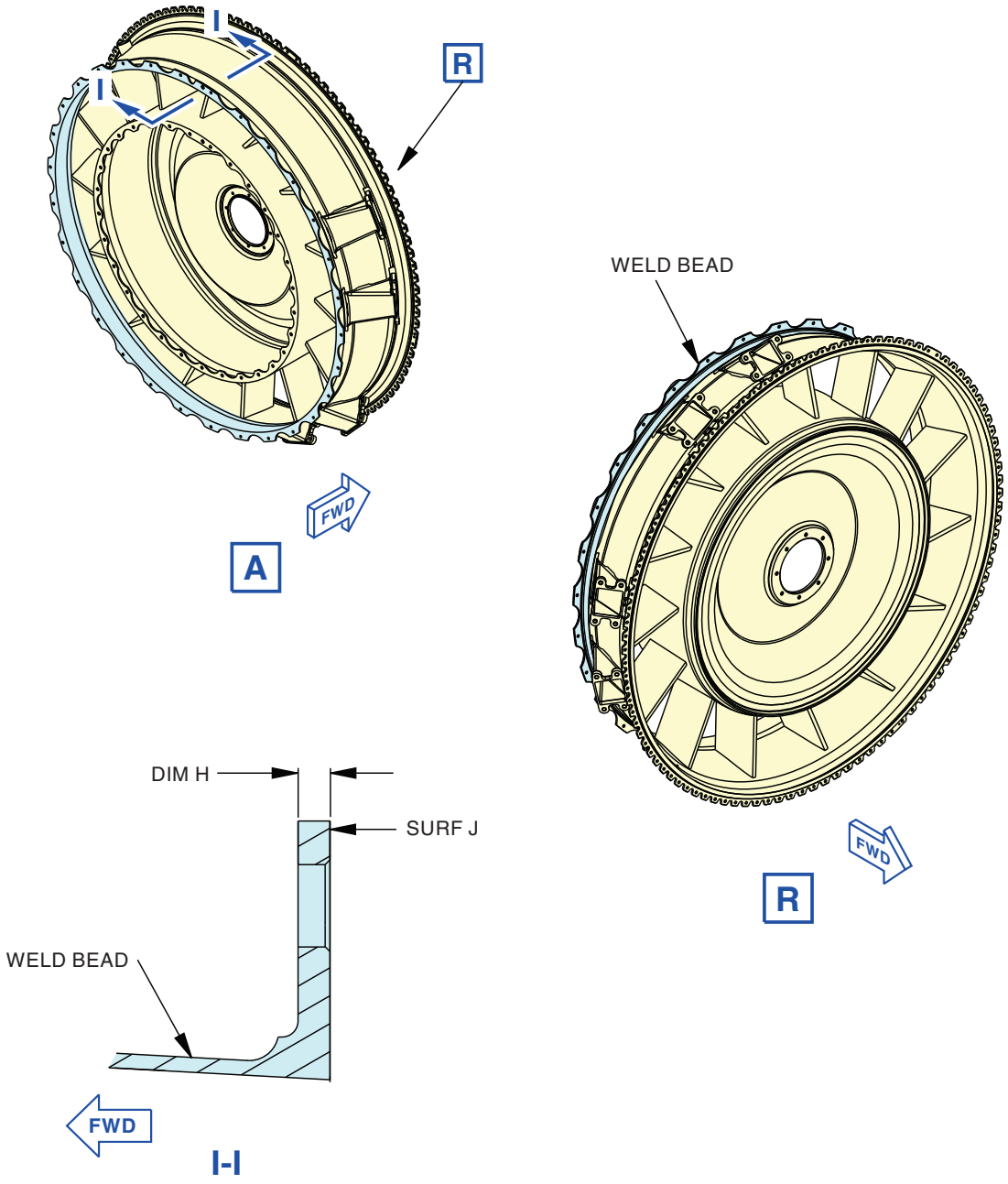


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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 18 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:
ZONING IS DONE UP TO THE WELD BEAD

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3044199 S0000811119_V1

Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 19 of 21)

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VARNISH



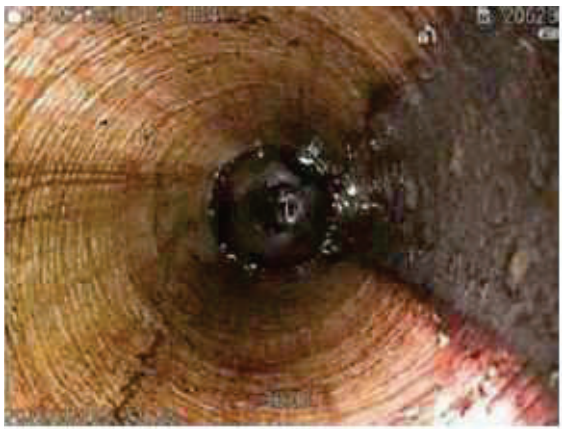
COKING

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Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 20 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ONE-SIDE COKE BUILD-UP EXAMPLE
(ACCEPTABLE CONDITION)



ONE-SIDED HEAVY COKE BUILD-UP EXAMPLES
(NON-ACCEPTABLE CONDITION)

ONE-SIDED BUILD-UP INSPECTION



BROWN SURFACE VARNISH EXAMPLE
(ACCEPTABLE CONDITION)



BROWN SURFACE VARNISH EXAMPLE
(ACCEPTABLE CONDITION)

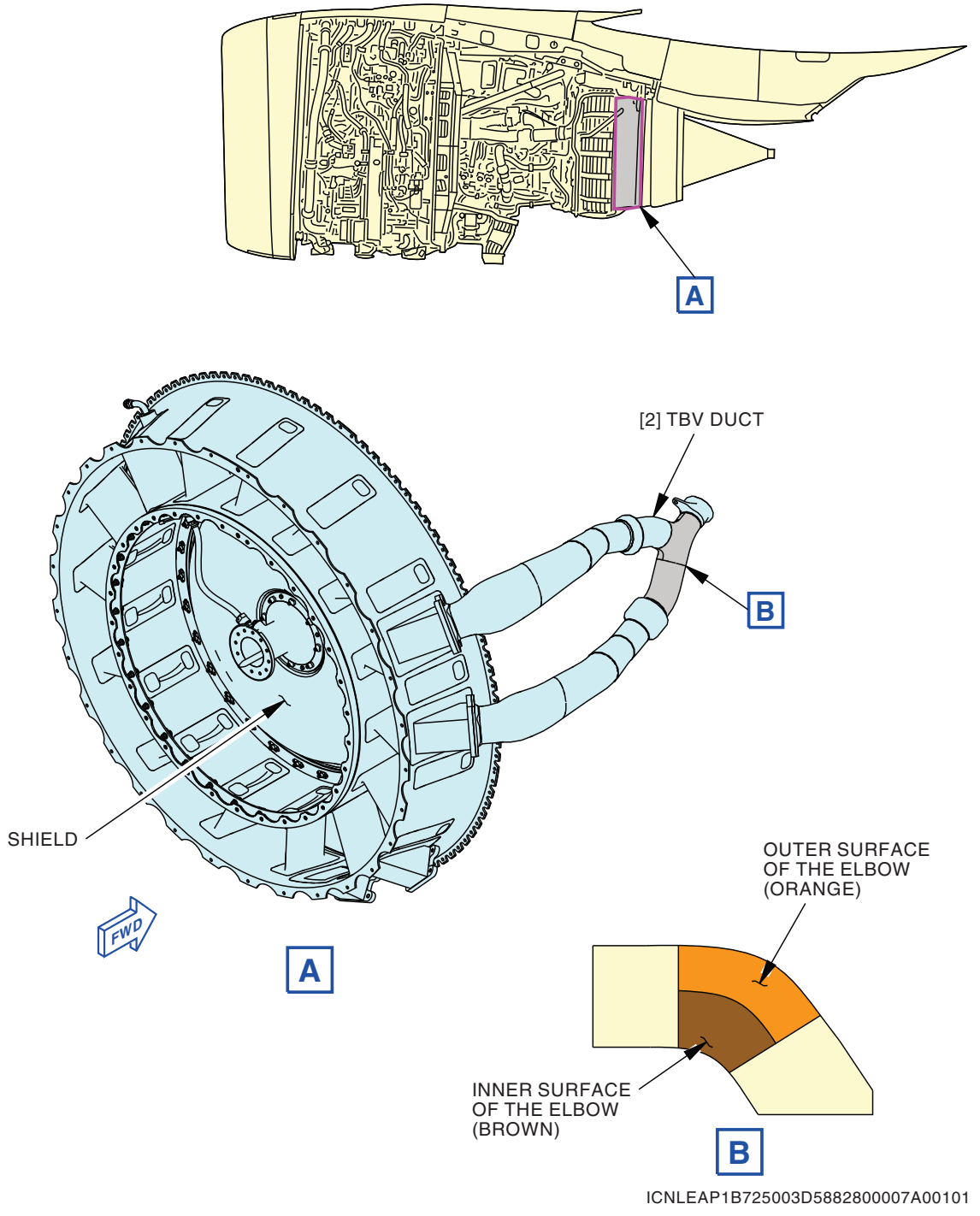
BROWN SURFACE VARNISH INSPECTION

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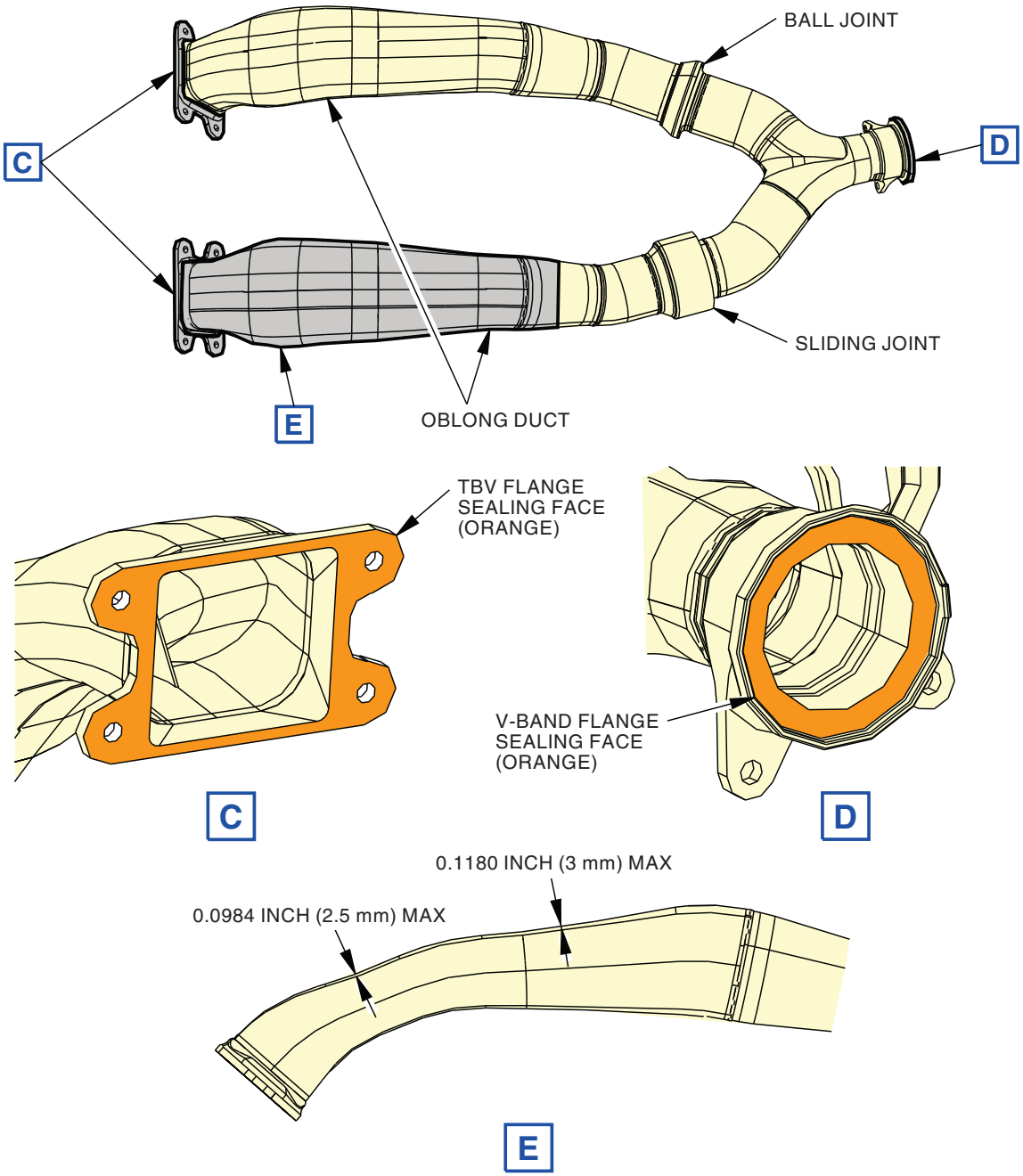
Turbine Rear Vane Module Inspection
Figure 601/72-50-03-990-801-G00 (Sheet 21 of 21)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TBV Duct Inspection
Figure 602/72-50-03-990-802-G00 (Sheet 1 of 4)

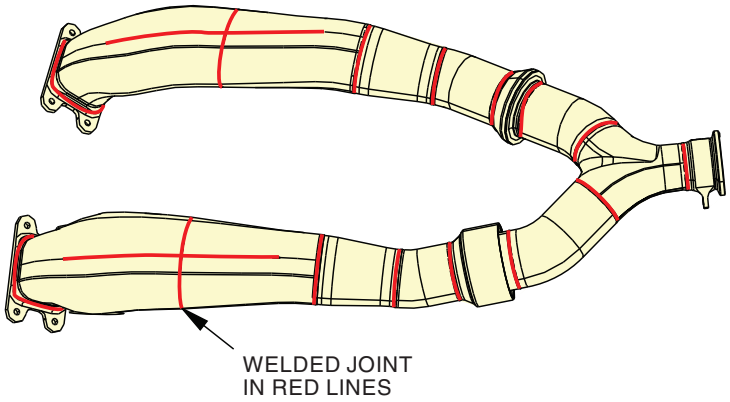
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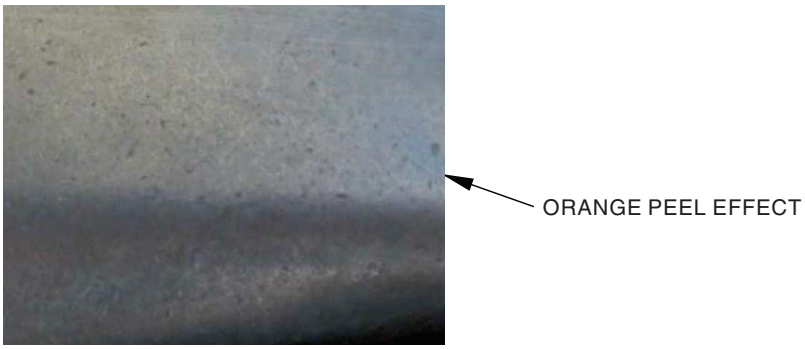


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TBV Duct Inspection
Figure 602/72-50-03-990-802-G00 (Sheet 2 of 4)

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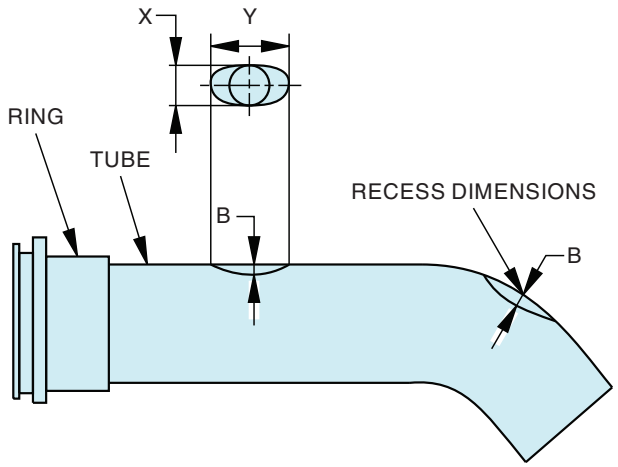


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TBV Duct Inspection
Figure 602/72-50-03-990-802-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



FLAT AREA PARAMETERS FOR TBV DUCT

OUTER SURFACE OF AN ELBOW			
EXTERNAL DIAMETER	≤B	X	Y
1.75 INCH (44.45 mm)	0.087 INCH (2.2 mm)	0.433 INCH (11 mm)	0.433 INCH≤ y≤ 0.866 INCH (11 mm ≤ y≤ 22 mm)
2 INCH (50.8 mm) or up	0.1 INCH (2.5 mm)	0.5 INCH (12.5 mm)	0.5 INCH≤ y≤ 1 INCH (12.5 mm ≤ y≤ 25 mm)
STRAIGHT OF DUCT AND INNER SURFACE OF AN ELBOW			
1.75 INCH (44.45 mm)	0.173 INCH (4.4 mm)	0.866 INCH (22 mm)	0.866 INCH ≤ y≤ 1.732 INCH (22 mm ≤ y≤ 44 mm)
2 INCH (50.8 mm) or up	0.2 INCH (5 mm)	0.98 INCH (25 mm)	0.98 INCH≤ y≤ 1.97 INCH (25 mm ≤ y≤ 50 mm)

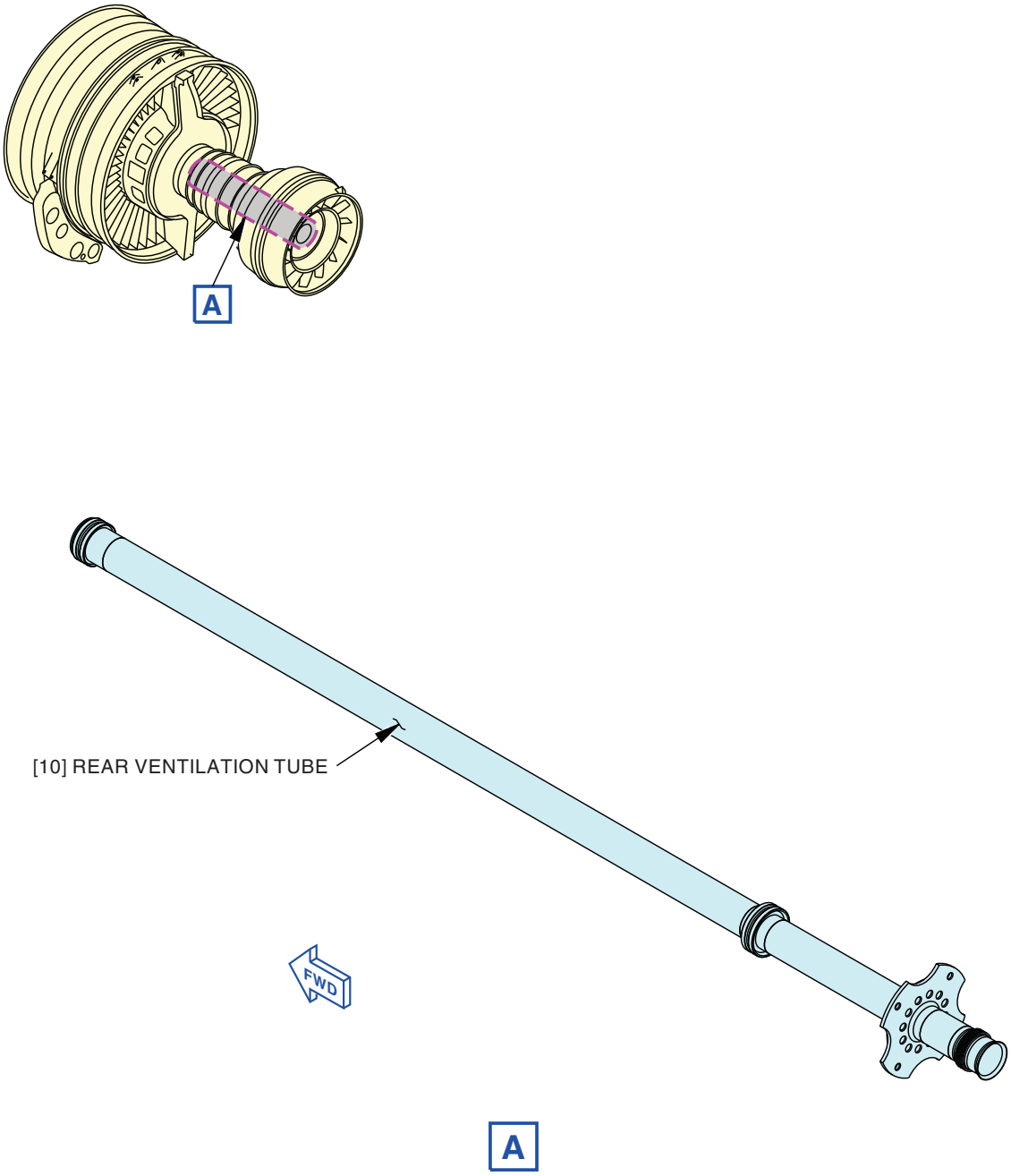
FLAT AREA ACCEPTABLE DIMENSIONS FOR TBV DUCT

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TBV Duct Inspection
Figure 602/72-50-03-990-802-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Rear Ventilation Tube Inspection
Figure 603/72-50-03-990-804-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INJECTION NOZZLE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the injection nozzle

(2) An installation of the injection nozzle.

TASK 72-59-00-000-801-G00

2. Injection Nozzle Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the injection nozzle.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-11-01-000-801-G00	Primary Nozzle Assembly Removal (P/B 401)
78-11-02-000-801-G00	Primary Plug Assembly Removal (P/B 401)
78-11-03-000-801-G00	Center Vent Tube Extension Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 72-59-00-860-001-G00

(1) Make sure that the ENGINE START switch and the START LEVEL switch are not operated.

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-59-00-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task:Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable fan cowl panels:

(TASK 71-11-04-010-801-G00)

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the thrust reversers:

(TASK 78-31-00-010-801-G00)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Injection Nozzle Removal

SUBTASK 72-59-00-020-001-G00

(1) Remove the primary nozzle assembly (TASK 78-11-01-000-801-G00).

SUBTASK 72-59-00-020-002-G00

(2) Remove the primary plug assembly (TASK 78-11-02-000-801-G00).

SUBTASK 72-59-00-020-003-G00

(3) Remove the center vent tube extension (TASK 78-11-03-000-801-G00).

SUBTASK 72-59-00-020-004-G00

WARNING BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER THE ENGINE IS STOPPED. THE ENGINE PARTS CAN STAY HOT FOR ALMOST ONE HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

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(WARNING PRECEDES)

WARNING DO NOT OPEN THE OIL SYSTEM UNTIL THE PRESSURE GOES TO ZERO. THE PRESSURE GOES TO ZERO APPROXIMATELY 5 MINUTES AFTER AN ENGINE IS STOPPED. A PRESSURIZED OIL SYSTEM CAN RELEASE A SPRAY OF HOT OIL THAT CAN BURN YOU.

WARNING DO NOT LET HOT OIL GET ON YOU. PUT ON CLOTHES, GOGGLES, AND OTHER EQUIPMENT FOR PROTECTION, OR LET THE ENGINE BECOME COOL. HOT OIL CAN BURN YOU.

WARNING DO NOT LET THE OIL STAY ON YOUR SKIN. YOU CAN ABSORB POISONOUS MATERIALS FROM THE OIL THROUGH YOUR SKIN.

CAUTION DO NOT LET OIL GET ON THE ENGINE OR OTHER COMPONENTS. IMMEDIATELY CLEAN THE OIL WHEN IT FALLS ON THEM. OIL CAN CAUSE DAMAGE TO EQUIPMENT.

(4) Remove the injection nozzle [3], do these steps:

(a) Remove the safety wire from the two ends of the exhaust air tube [1].

CAUTION USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(b) Disconnect the exhaust air tube [1] from the air tube [2].

CAUTION USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(c) Disconnect the exhaust air tube [1] from the injection nozzle [3].

(d) Remove the exhaust air tube [1].

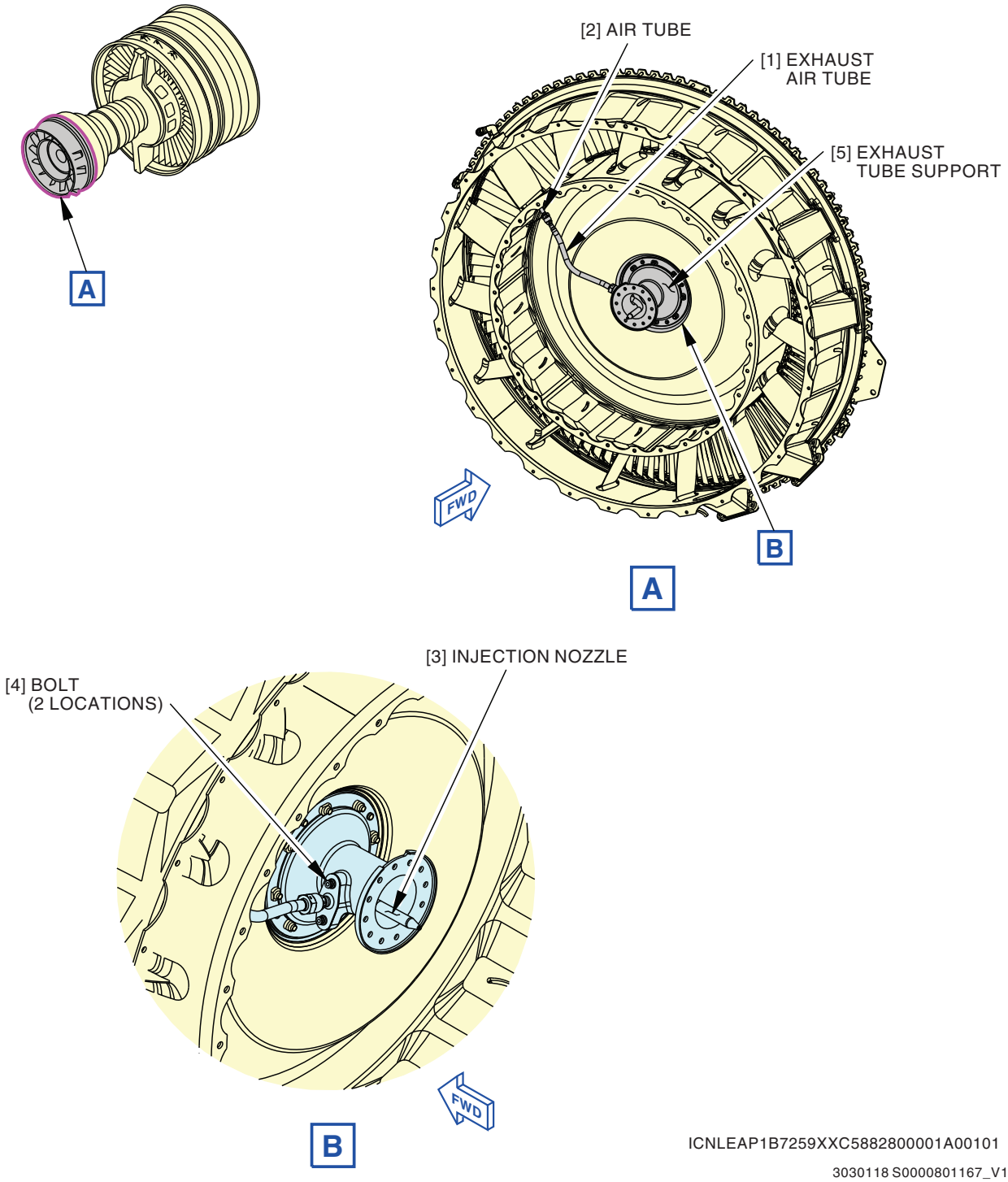
(e) Remove the two bolts [4] that attach the injection nozzle [3] to the exhaust tube support [5].

(f) Remove the injection nozzle [3] from the exhaust tube support [5].

(g) Seal all tubes and openings with caps or plugs.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Injection Nozzle Installation
Figure 401/72-59-00-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-59-00-400-801-G00

3. Injection Nozzle Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the injection nozzle.

B. References

Reference	Title
78-11-01-400-801-G00	Primary Nozzle Assembly Installation (P/B 401)
78-11-02-400-801-G00	Primary Plug Assembly Installation (P/B 401)
78-11-03-400-801-G00	Center Vent Tube Extension Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Injection Nozzle Installation

SUBTASK 72-59-00-420-001-G00

(1) Install the injection nozzle [3], do these steps:

(a) Remove caps or plugs from tubes and openings.

(b) Make sure that the mating surfaces are clean and clear of unwanted materials.

(c) Apply graphite mineral vaseline, D00601 [CP2101], on the threads and on the mating surfaces of the two bolts [4].

(d) Put and hold in position the injection nozzle [3] on the exhaust tube support [5].

(e) Attach the injection nozzle [3] to the exhaust tube support [5] with the two bolts [4].

(f) Tighten the two bolts [4] to 92.50 in-lb (10.45 N·m) - 102.20 in-lb (11.55 N·m).

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(g) Apply oil, D00599 [CP2442], on the thread of the exhaust air tube [1] and on the mating surface between the coupling nut and air tube [2].

(h) Apply oil, D00599 [CP2442], on the thread of the injection nozzle [3] and on the mating surface between the exhaust air tube [1] and its coupling nut.

(i) Put and hold the exhaust air tube [1] in position between the injection nozzle [3] and air tube [2].

(j) Connect theexhaust air tube [1] to the coupling nut of the air tube [2].

1) Tighten with hand.

(k) Connect the exhaust air tube [1] to the nipple of the injection nozzle [3].

1) Tighten with hand.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0075

CAUTION USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(l) Tighten the coupling nut of the air tube [2] to 450 in-lb (51 N·m) - 550 in-lb (62 N·m).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0075

CAUTION USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(m) Tighten the coupling nut of the air tube [2] to 270 in-lb (31 N·m) - 300 in-lb (34 N·m).

PTF ALL

CAUTION USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(n) Tighten the coupling nut of the exhaust air tube [1] to 450 in-lb (51 N·m) - 550 in-lb (62 N·m).

(o) If it is necessary, remove the excess of lubrication on the two bolts [4] and on the coupling nuts.

(p) Safety the coupling nut of the air tube [2] and the nipple of the exhaust air tube [1] with safety wire, G02345 [CP8001], or safety cable, G50065 [CP8006].

(q) Safety the coupling nut of the exhaust air tube [1] to the nipple of the injection nozzle [3] with safety wire, G02345 [CP8001], or safety cable, G50065 [CP8006].

SUBTASK 72-59-00-420-002-G00

(2) Install the center vent tube extension (TASK 78-11-03-400-801-G00).

SUBTASK 72-59-00-420-003-G00

(3) Install the primary plug assembly (TASK 78-11-02-400-801-G00).

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SUBTASK 72-59-00-420-004-G00

(4) Install the primary nozzle assembly (TASK 78-11-01-400-801-G00).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 72-59-00-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Close applicable thrust reversers:

(TASK 78-31-00-010-802-G00)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Close the applicable fan cowl panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-59-00-020-005-G00

(2) Remove the DO NOT OPERATE tags, from the applicable ENGINE START switch and START LEVER switch.

(a) Remove the DO NOT OPERATE tag, from the applicable ENGINE START switch on the P5 overhead panel.

(b) Remove the DO NOT OPERATE tag, from the from the applicable START LEVER switch on the P5 overhead panel.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

INTERNAL EXHAUST AIR TUBE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the internal exhaust air tube

(2) An installation of the internal exhaust air tube.

TASK 72-59-01-000-801-G00

2. Internal Exhaust Air Tube Removal

(Figure 401, Figure 402, Figure 403, and Figure 404)

A. General

(1) This task gives the instructions to remove the exhaust air tube.

(2) The internal exhaust air tube is referred to as the exhaust air tube.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-59-00-000-801-G00	Injection Nozzle Removal (P/B 401)
78-11-01-000-801-G00	Primary Nozzle Assembly Removal (P/B 401)
78-11-02-000-801-G00	Primary Plug Assembly Removal (P/B 401)
78-11-03-000-801-G00	Center Vent Tube Extension Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-25055	Kit Fixture - Assembly Locking Ring Part #: 956A8609G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

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(Continued)

Number	Name/Location
426	Right Thrust Reverser, Engine 2

F. Prepare for the Exhaust Air Tube Removal

SUBTASK 72-59-01-860-001-G00

(1) Make sure that the ENGINE START switch and the START LEVEL switch are not operated.

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put the DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put the DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-59-01-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable fan cowl panels:

(TASK 71-11-04-010-801-G00)

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the thrust reversers:

(Open the Thrust Reverser (Selection), TASK 78-31-00-010-801-G00)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-59-01-020-003-G00

(3) Remove the primary nozzle assembly (TASK 78-11-01-000-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-59-01-020-004-G00

(4) Remove the primary plug assembly (TASK 78-11-02-000-801-G00).

SUBTASK 72-59-01-020-005-G00

(5) Remove the center vent tube extension (TASK 78-11-03-000-801-G00).

G. Exhaust Air Tube Removal

WARNING

BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER YOU STOP THE ENGINE. THE ENGINE PARTS CAN STAY HOT FOR ALMOST 1 HOUR. DO NOT TOUCH HOT PARTS WITHOUT APPLICABLE GLOVES. HOT PARTS CAN CAUSE INJURIES TO PERSONNEL.

SUBTASK 72-59-01-020-001-G00

CAUTION

MAKE SURE THAT YOU DO NOT HIT THE LOW PRESSURE TURBINE (LPT) REAR CASE DURING THE REMOVAL OF THE EXHAUST AIR TUBE. IF YOU DO NOT OBEY, DAMAGE TO THE LPT REAR CASE CAN OCCUR.

(1) Remove the exhaust air tube [2] from the Low Pressure Turbine (LPT) rear case as follows (Figure 401):

(a) Remove the exhaust air tube [1] (TASK 72-59-00-000-801-G00).

(b) Remove and discard the removable retaining ring [5] from the coupling nut [3] of the exhaust air tube [2] with the locking ring fixture, SPL-25055 (Figure 402).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0075

1) Use the support [6] to remove the removable retaining ring [5].
2) If it is necessary, turn the nut [7] clockwise until the axis [9] is fully extended.
3) Remove the cap [8] from its storage position and install it on the coupling nut [3].
4) Engage the hook of the axis [9] on the removable retaining ring [5].
5) Hold the support [6] in position on the coupling nut [3] and turn the nut [7] counterclockwise to remove the removable retaining ring [5].
6) Remove the support [6] and coupling nut [3] from the LPT rear case.
7) Loosen the nut [7].
   a) Remove and discard the removable retaining ring [5].
8) Remove the cap [8] from the coupling nut [3].
9) Install the cap [8] in its storage position.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0075

10) Use the support [10] to remove the removable retaining ring [5].
11) If it is necessary, turn the nut [11] clockwise until the axis [13] is fully extended.
12) Remove the cap [12] from its storage position and install it on the coupling nut [3].
13) Engage the hook of the axis [13] on the removable retaining ring [5].
14) Hold the support [10] in position on the coupling nut [3] and turn the nut [11] counterclockwise to remove the removable retaining ring [5].
15) Remove the support [10] and coupling nut [3] from the LPT rear case.
16) Loosen the nut [11].
   a) Remove and discard the removable retaining ring [5].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0075 (Continued)

17) Remove the cap [12] from the coupling nut [3].

18) Install the cap [12] in its storage position.

PTF ALL

(c) Remove the coupling nut [3] from the exhaust air tube [2].

(d) Remove and discard the lockwire from the coupling nut [21] of the air tube [4].

CAUTION

USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

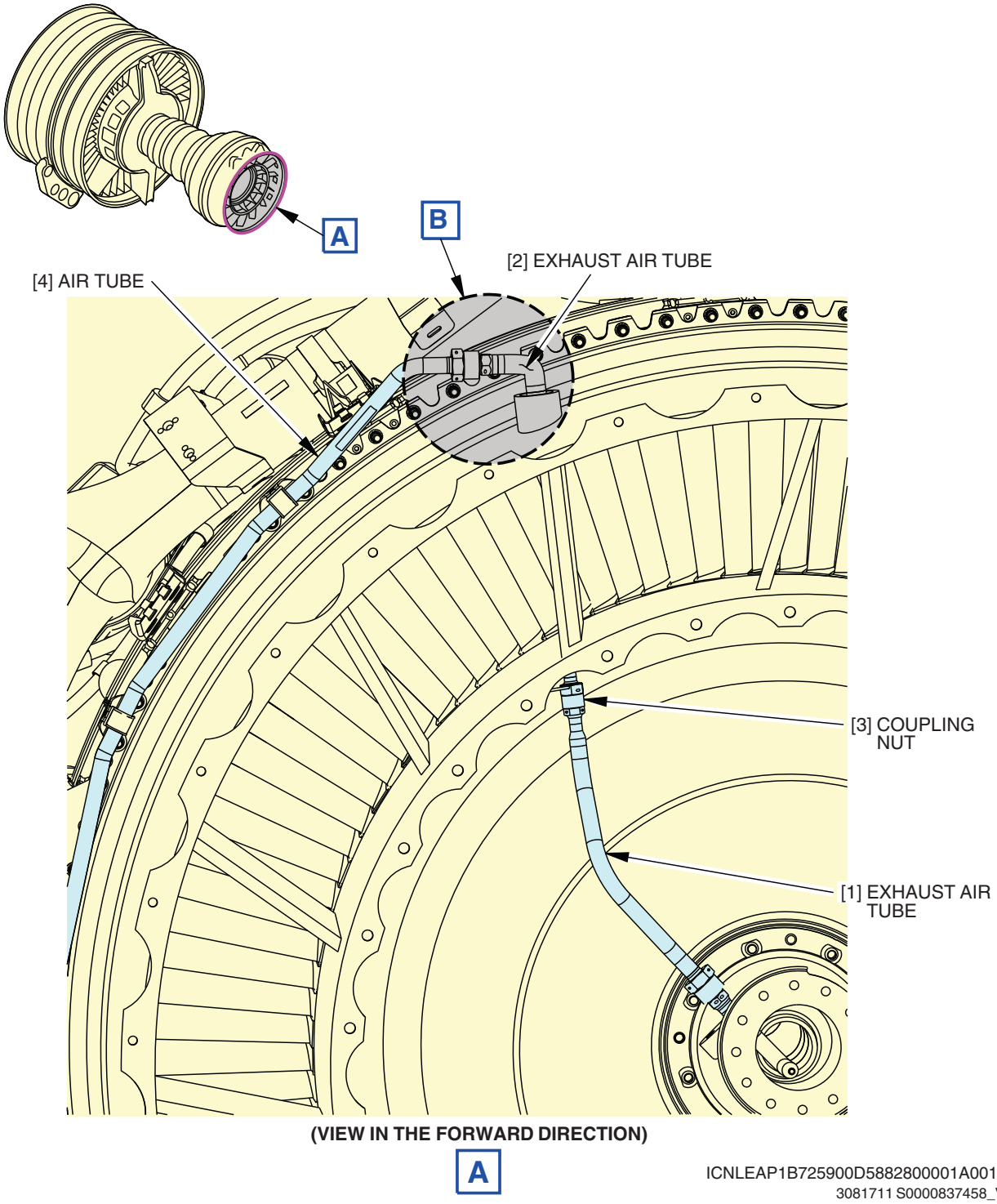
(e) Loosen the coupling nut [21] of the air tube [4] to disconnect it from the nipple of the exhaust air tube [2].

(f) Remove the exhaust air tube [2].

(g) Install the caps or plugs on the tubes and openings.

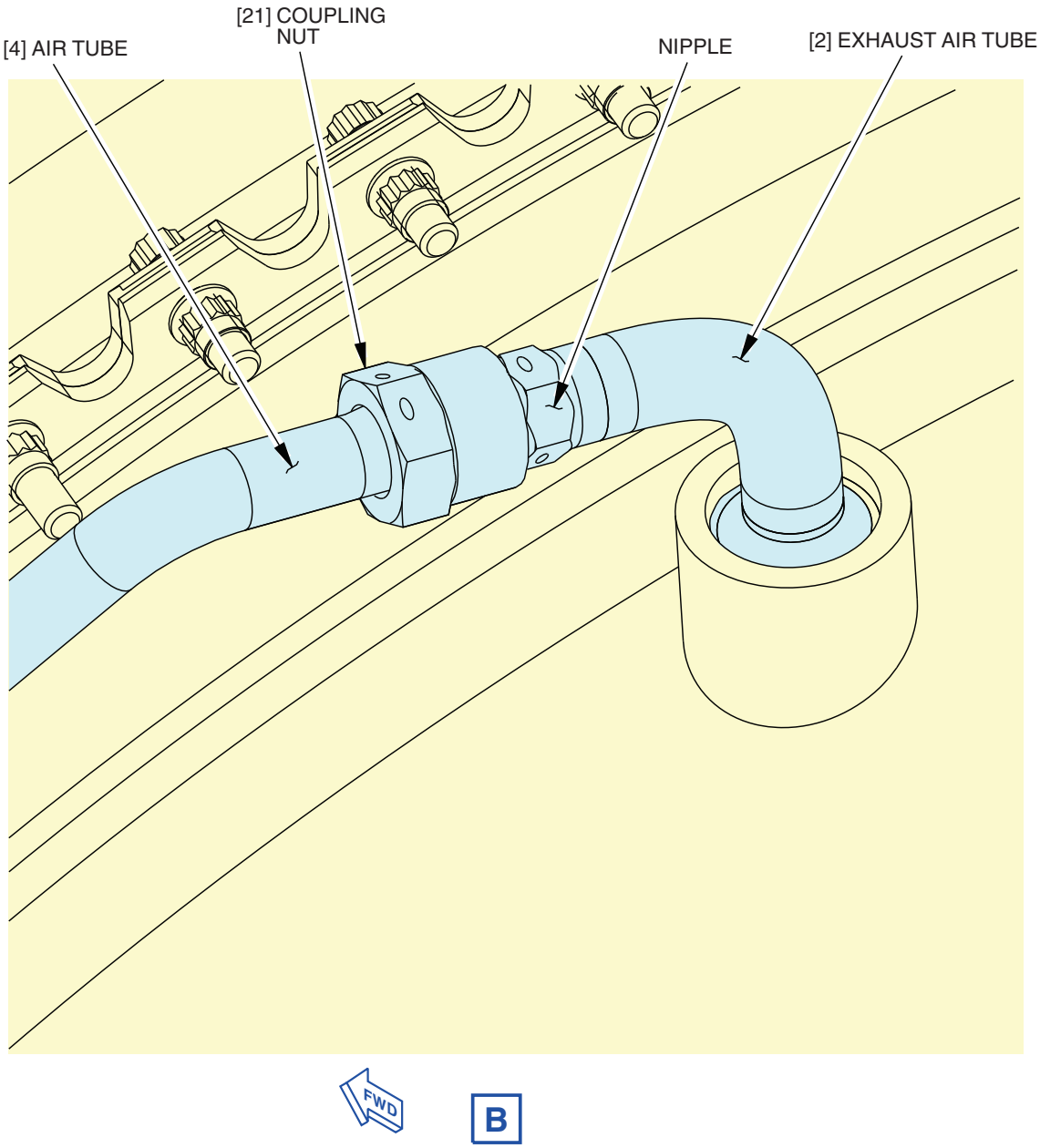
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Internal Exhaust Air Tube Installation
Figure 401/72-59-01-990-801-G00 (Sheet 1 of 2)

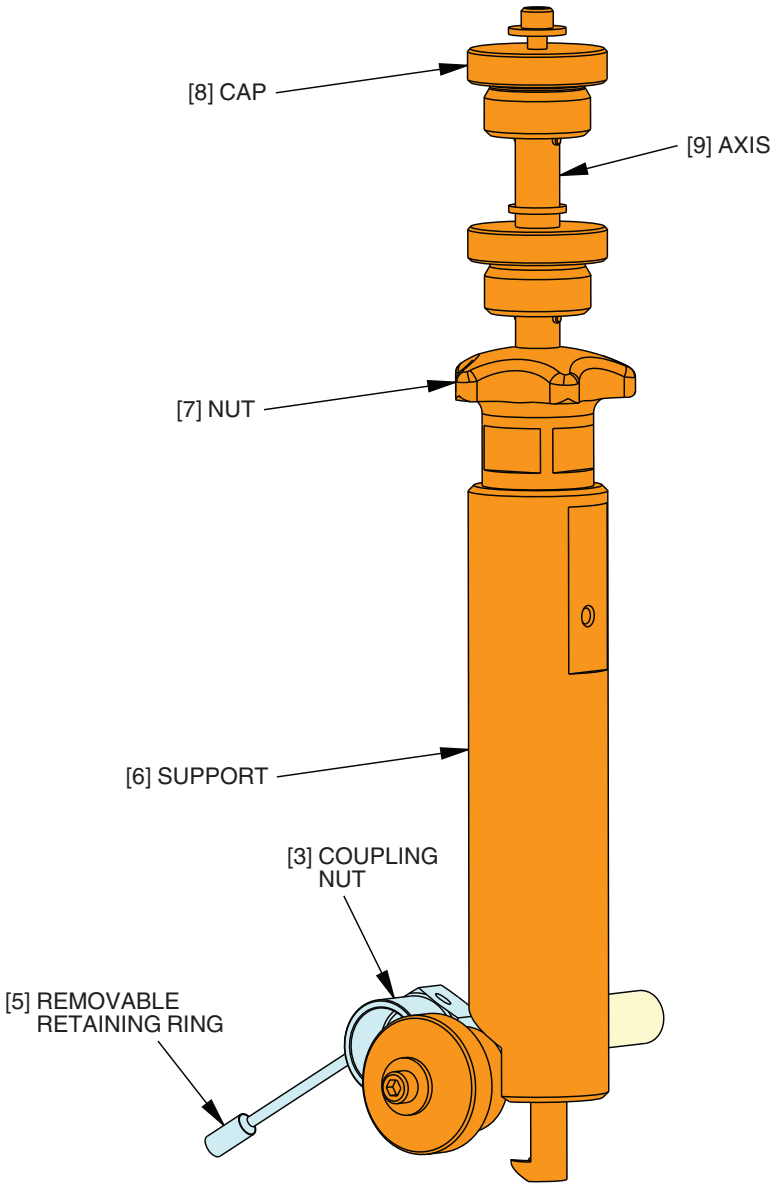
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Internal Exhaust Air Tube Installation
Figure 401/72-59-01-990-801-G00 (Sheet 2 of 2)

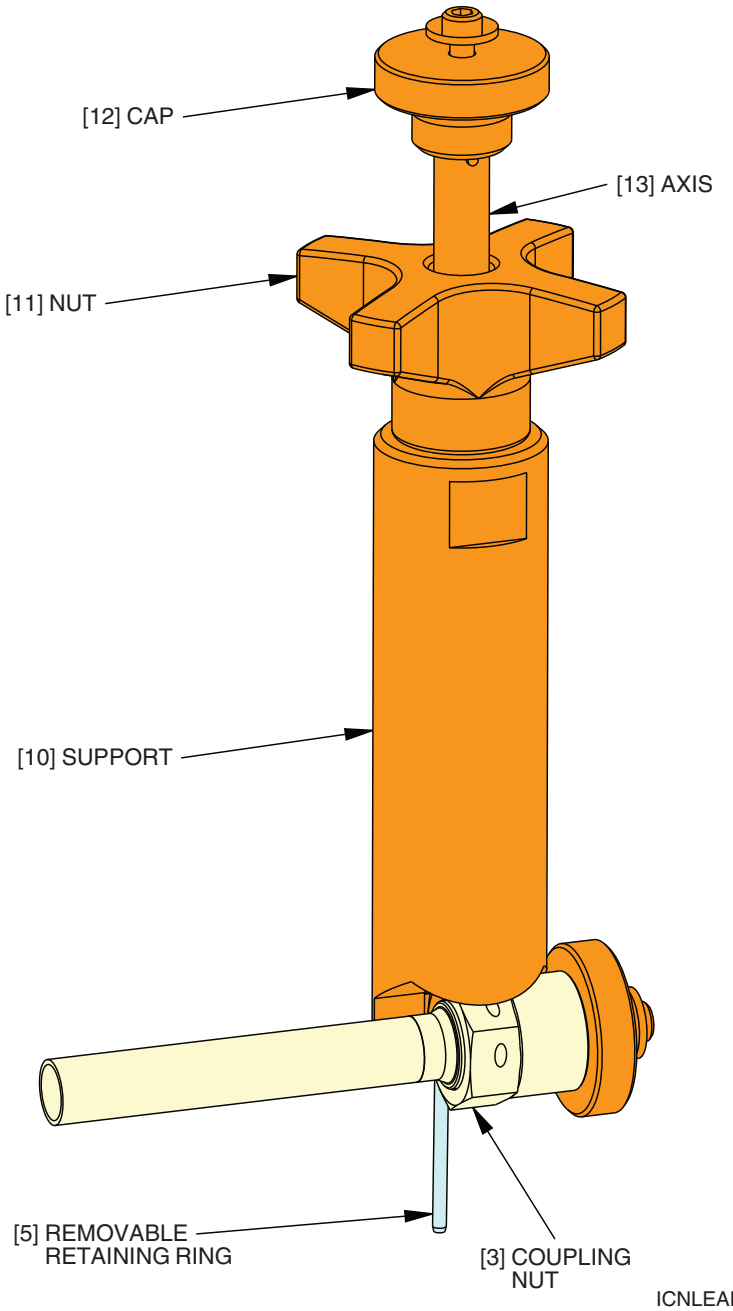
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Removable Retaining Ring Removal
Figure 402/72-59-01-990-802-G00 (Sheet 1 of 2)

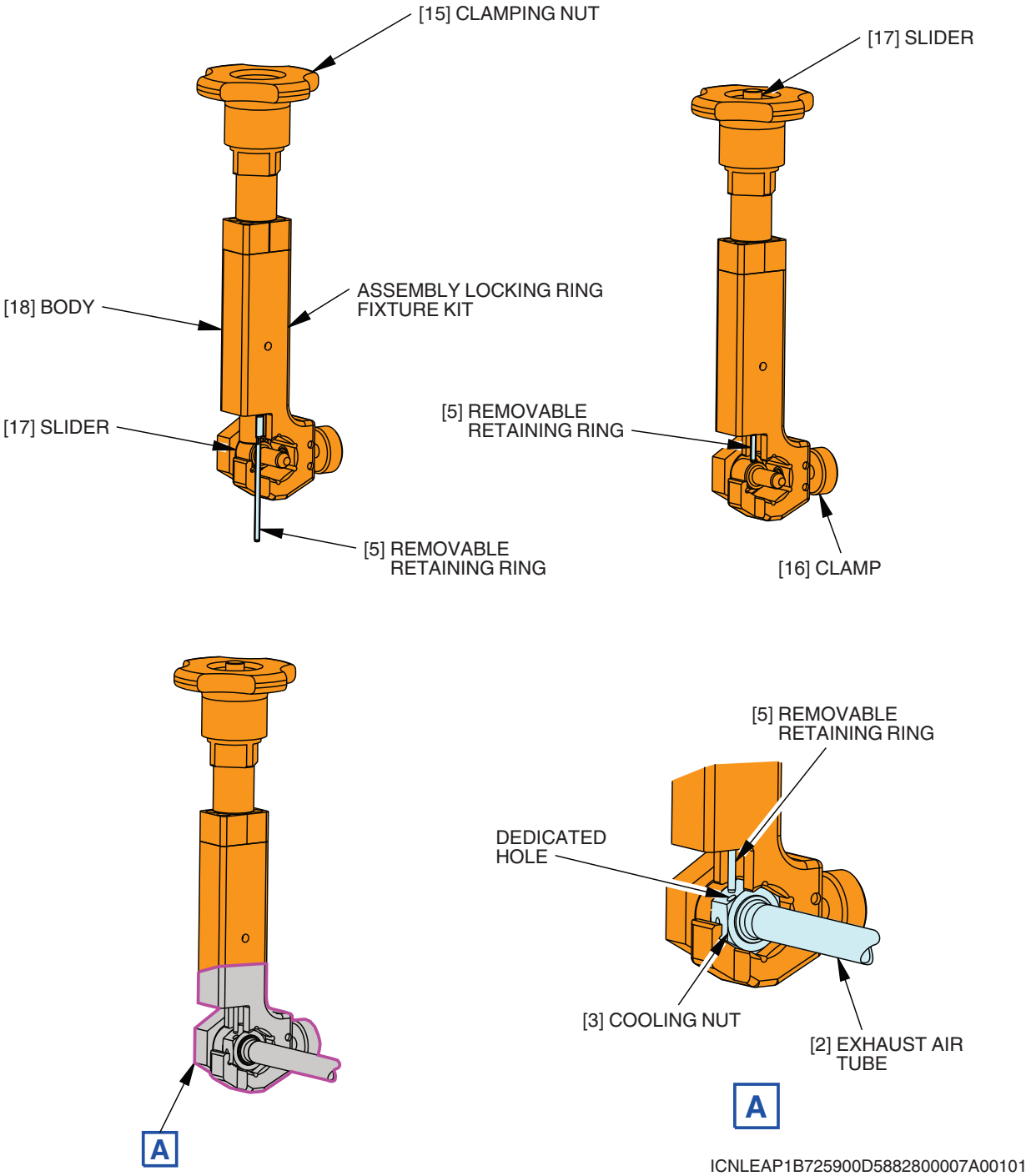
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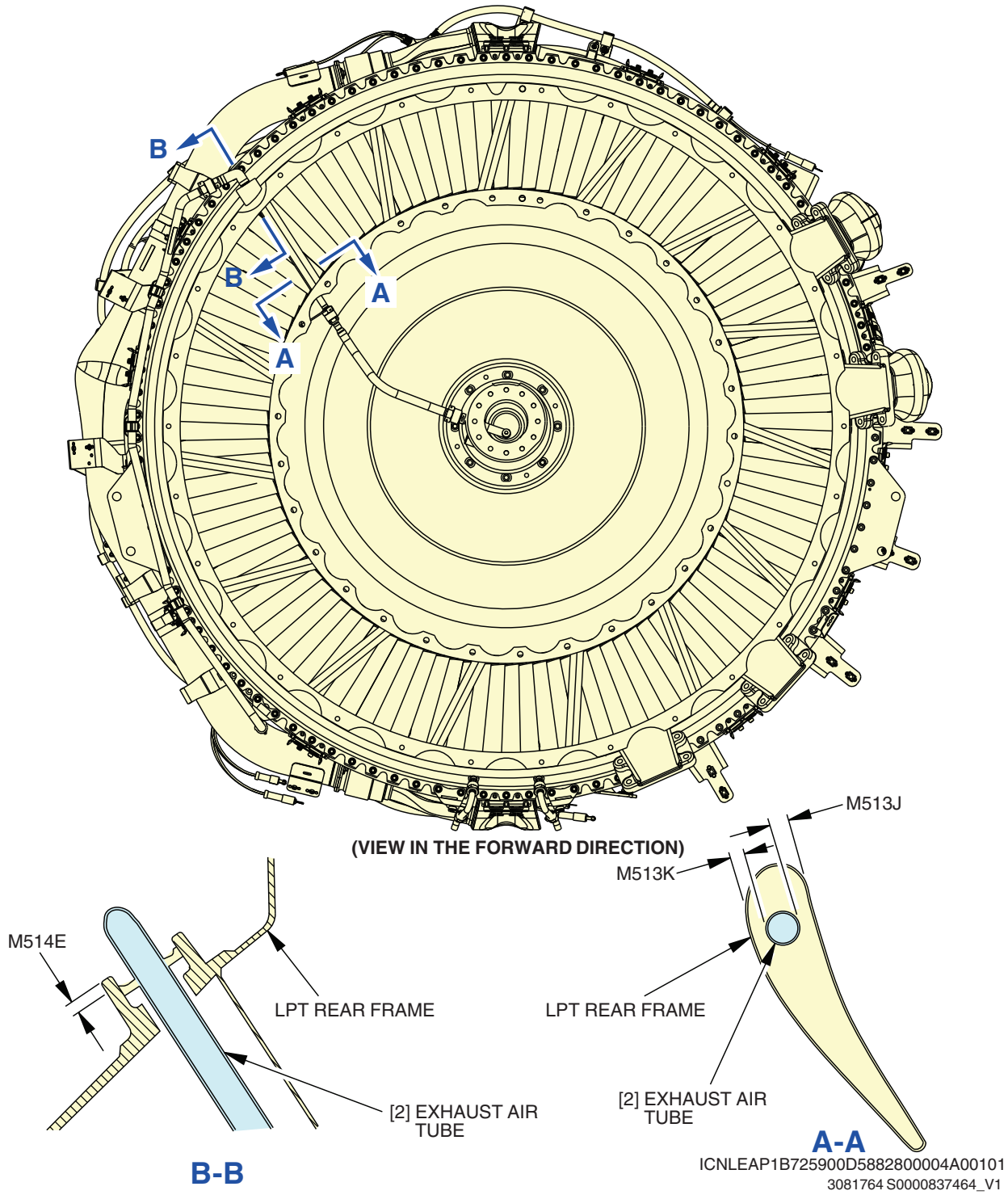
Removable Retaining Ring Removal
Figure 402/72-59-01-990-802-G00 (Sheet 2 of 2)

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Removable Retaining Ring Installation
Figure 403/72-59-01-990-803-G00

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Internal Exhaust Air Tube Installation
Figure 404/72-59-01-990-804-G00

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TASK 72-59-01-400-801-G00

3. Internal Exhaust Air Tube Installation

(Figure 401, Figure 402, Figure 403, and Figure 404)

A. General

(1) This task gives the instructions to install internal exhaust air tube.

(2) The internal exhaust air tube is referred to as the exhaust air tube.

B. References

Reference	Title
72-59-00-400-801-G00	Injection Nozzle Installation (P/B 401)
78-11-01-400-801-G00	Primary Nozzle Assembly Installation (P/B 401)
78-11-02-400-801-G00	Primary Plug Assembly Installation (P/B 401)
78-11-03-400-801-G00	Center Vent Tube Extension Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-25055	Kit Fixture - Assembly Locking Ring Part #: 956A8609G02 Supplier: 58828

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G02495 [CP8002]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS5689
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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G. Exhaust Air Tube Installation

SUBTASK 72-59-01-420-001-G00

(1) Install the exhaust air tube [2] on the Low Pressure Turbine (LPT) rear case as follows (Figure 401):

(a) Remove the caps or plugs from tubes and openings.

(b) Put and hold the exhaust air tube [2] in position.

(c) Make sure that the end of the exhaust air tube [2] is horizontal.

1) If it is necessary, slowly turn the exhaust air tube [2] to put its end in horizontal position.

(d) Apply oil, D00599 [CP2442], on the thread of the nipple of the exhaust air tube [2] and rear flange of the ferrule of the air tube [4].

(e) Connect the nipple of the exhaust air tube [2] to the coupling nut [21] of the air tube [4].

1) Tighten the coupling nut [21] by hand.

(f) Install the coupling nut [3] on the exhaust air tube [2].

1) Use the body [18] of the locking ring fixture, SPL-25055, to install the new removable retaining ring [5].

2) Fully tighten the clamping nut [15] to extend the slider [17].

3) Install the new removable retaining ring [5] on the hook of the slider [17].

4) Fully loosen the clamping nut [15] to retract the slider [17].

5) Install the coupling nut [3] in the insertion equipment of the locking ring fixture, SPL-25055.

6) Make sure that the dedicated hole of the coupling nut [3] is aligned with the removable retaining ring [5].

7) Tighten the clamp [16] of the locking ring fixture, SPL-25055, to lock the coupling nut [3].

8) Install the coupling nut [3] with the locking ring fixture, SPL-25055, on the exhaust air tube [2].

9) Tighten the clamping nut [15] to install the removable retaining ring [5].

10) Loosen the clamp [16] from the locking ring fixture, SPL-25055.

11) Remove the locking ring fixture, SPL-25055, from the coupling nut [3].

(g) Install the exhaust air tube [1] (TASK 72-59-00-400-801-G00).

CAUTION

USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(h) Tighten the coupling nut [21] of the air tube [4] to 450.0 in-lb (50.8 N·m) - 550.0 in-lb (62.1 N·m) (Figure 401).

SUBTASK 72-59-01-860-002-G00

(2) Make sure that the exhaust air tube [2] is installed correctly (Figure 404).

(a) Do the check of the dimension M514E as follows:

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1) Measure and record dimension M514E between the exhaust air tube [2] and LPT rear frame.

2) Make sure that the value of M514E is between 0.0906 in. (2.30 mm) and 0.2244 in. (5.70 mm).

3) If the value is out of tolerance, find the cause.

(b) Do the check of the dimensions M513J and M513K as follows:

1) Measure and record dimensions M513J and M513K between the exhaust air tube [2] and LPT rear frame at the most restrictive section on arm (weld bead) with a feeler gage.

2) Make sure that the values of M513J and M513K are more than 0.04 in. (1.0 mm).

3) If the values are out of tolerance, find the cause.

SUBTASK 72-59-01-420-002-G00

(3) Install safety cable, G50065 [CP8006], or safety wire, G02345 [CP8001], or safety wire, G02495 [CP8002], to the coupling nut [21] of the air tube [4] and exhaust air tube [2].

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-59-01-420-003-G00

(1) Install the center vent tube extension (TASK 78-11-03-400-801-G00).

SUBTASK 72-59-01-420-004-G00

(2) Install the primary plug assembly (TASK 78-11-02-400-801-G00).

SUBTASK 72-59-01-420-005-G00

(3) Install the primary nozzle assembly (TASK 78-11-01-400-801-G00).

SUBTASK 72-59-01-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS WHEN YOU CLOSE THE THRUST REVERSERS, INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Close applicable thrust reversers:

(TASK 78-31-00-010-802-G00)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Close the applicable fan cowl panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

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SUBTASK 72-59-01-020-002-G00

(5) Remove the DO NOT OPERATE tags, from the applicable ENGINE START switch and START LEVER switch.

(a) Remove the DO NOT OPERATE tag, from the applicable ENGINE START switch on the P5 overhead panel.

(b) Remove the DO NOT OPERATE tag, from the from the applicable START LEVER switch on the P5 overhead panel.

——— END OF TASK ———

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ACCESSORY DRIVES - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Accessory Drives Inspection

(2) Accessory Drives Damper Bearings Inspection.

TASK 72-60-00-200-801-G00

2. Accessory Drives Inspection

(Figure 601 and Figure 602)

A. General

(1) This task gives the instructions to examine the accessory drives.

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-62-00-300-802-G00	Blending and Spot Touch-Up of the Transfer Gearbox Module (P/B 801)
72-62-00-300-804-G00	Transfer Gearbox Assembly Corrosion Removal (P/B 801)
72-63-00-300-802-G00	Blending and Spot Touch-Up of the Accessory Gearbox Module (P/B 801)
72-63-03-300-802-G00	Blending and Spot Touch-Up of the Manual Control Cover (P/B 801)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

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(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 72-60-00-860-002-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-60-00-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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G. Accessory Drives (AGB Assembly) Inspection

SUBTASK 72-60-00-230-001-G00

(1) Make sure that the AGB assembly [1] you will examine is clean.

SUBTASK 72-60-00-220-001-G00

(2) Examine all areas of the AGB assembly [1] that you can see for these conditions (Figure 601 (Sheet 3)):

(a) Cracks

1) Not permitted.

2) Not repairable.

(b) Corrosion

1) Not permitted.

2) Not repairable.

(c) Lack of protection on all non-functional surfaces of the AGB assembly [1] that you can see.

NOTE: A non-functional surface is not an interface area or a sealing area.

1) Permitted up to 5 percent of the surface.

(d) Lack of protection on all other surfaces that you can see.

1) Not permitted.

SUBTASK 72-60-00-210-001-G00

(3) Examine seal grooves of the AGB assembly [1] that you can see for:

(a) Scratches, nicks, dents and score

1) Not permitted.

SUBTASK 72-60-00-210-002-G00

(4) Examine all areas of the AGB damper bearings (TASK 72-60-00-200-803-G00).

SUBTASK 72-60-00-210-004-G00

(5) Examine all areas of the AGB central damper [37] that you can see for these conditions (Figure 601 (Sheet 6)):

(a) Cracks

1) Not permitted.

2) Not repairable.

(b) Elastomer ruptured or sheared

1) Not permitted.

2) Not repairable.

(c) Corroded, pitted or galled

1) Not permitted.

2) Not repairable.

(d) Axial and radial play

1) Not permitted.

2) Not repairable.

(e) Loose or missing

1) Not permitted.

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2) Not repairable.

SUBTASK 72-60-00-210-005-G00

(6) Examine all areas of AGB axial lug [24] that you can see for this condition (Figure 601 (Sheet 2)):

(a) Missing protective coating

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-006-G00

(7) Examine the damper bearings of AGB axial lug [26] for these conditions (Figure 601 (Sheet 8)):

(a) Loose or missing

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-008-G00

(8) Examine location 2 and location 3 of the AGB housing lug [14] and lug [25] for these conditions (Figure 601 (Sheet 14)):

(a) Scratches, nicks, dents, and score

1) Not permitted.

2) Not repairable.

SUBTASK 72-60-00-210-011-G00

(9) Examine the AGB inserts [39] for these conditions (Figure 601 (Sheet 15) and Figure 601 (Sheet 16)):

(a) Wear of internal threads

1) Permitted if total cumulated length is less than 1/2 of one full thread.

2) Not repairable.

(b) Looseness

1) Not permitted.

2) Not repairable.

(c) Missing insert or key

1) Not permitted.

(d) Self-locking quality (running torque)

1) 0.2500–28 UNJF-3B

a) Permitted if the running torque is between 3.540 in-lb (0.40 N·m) and 30.004 in-lb (3.39 N·m).

b) Not repairable.

2) 0.3750–24 UNJF-3B

a) Permitted if the running torque is between 9.470 in-lb (1.07 N·m) and 80.011 in-lb (9.04 N·m).

b) Not repairable.

SUBTASK 72-60-00-210-012-G00

(10) Examine the AGB key-locked studs [40] for these conditions (Figure 601 (Sheet 8)):

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